

Project Data Page:

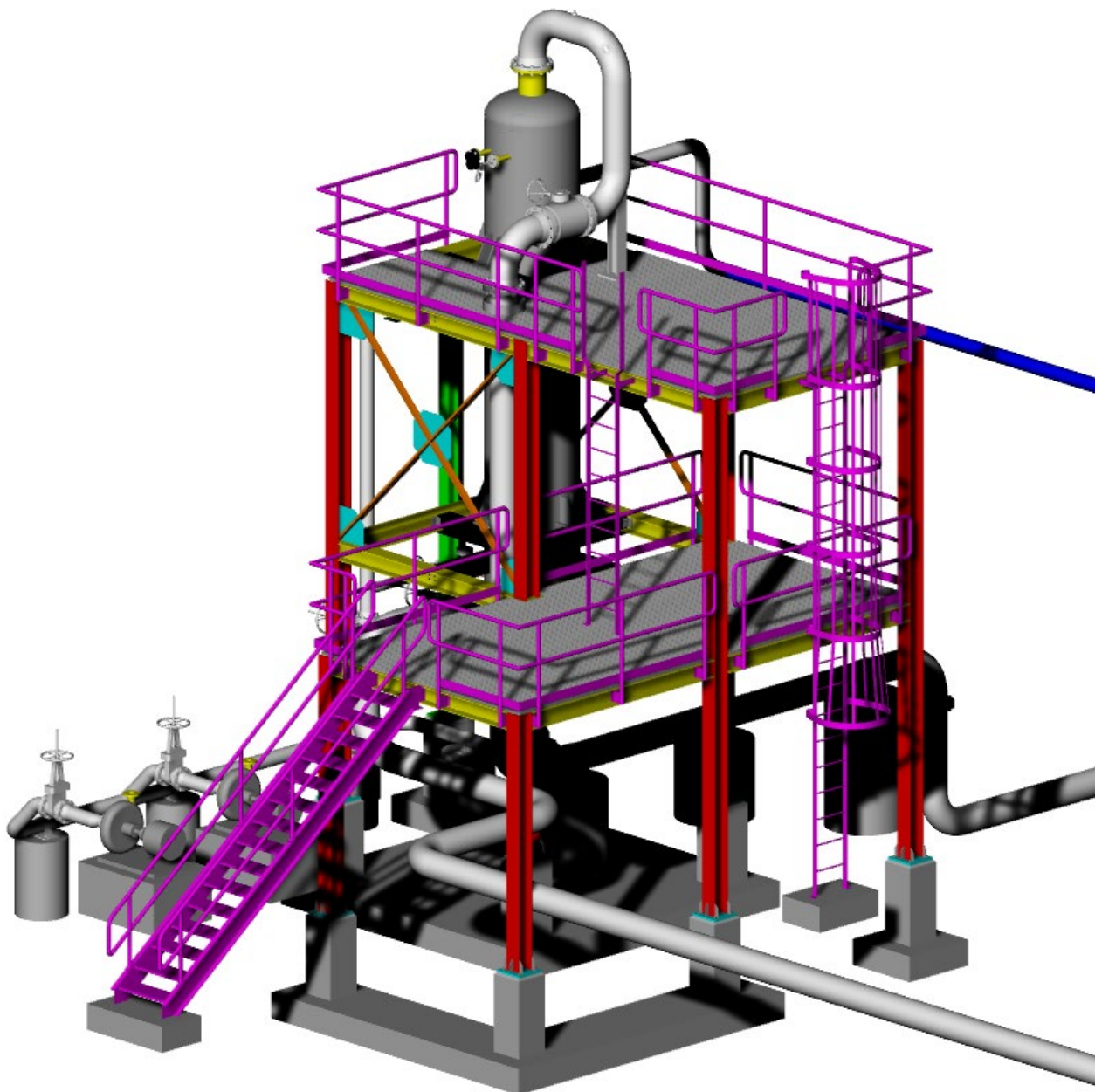




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2021

Analysis Performed by : #

Job File : ...\\174-CBEUDia 1000-AK.pvdb

Date of Analysis : Sep 9, 2026 9:44am

PV Elite 25 SP2, October 2023



PV Elite 25 SP2 Licensee: #
 FileName : 174-C_BEU_Dia 1000-AK
 Vessel Design Summary:

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Vessel Design Summary:

ASME Code, Section VIII Division 1, 2021

Diameter Spec : 1060.000 x 1066.000 mm ID	
Vessel Design Length, Tangent to Tangent	9054.70 mm
Specified Datum Line Distance	50.00 mm
Shell Side Design Temperature	200 °C
Channel Side Design Temperature	82 °C
Shell Side Design Pressure	6100.000 KPa.
Channel Side Design Pressure	1000.000 KPa.
Shop Shell Side Test Pressure	8569.472 KPa.
Field Shell Side Test Pressure	7930.000 KPa.
Shop Channel Side Test Pressure	2067.008 KPa.
Field Channel Side Test Pressure	1300.000 KPa.
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Normal ized	Impact Tested
Shell	SA-516 70	...	...	K02700	No	No
Head	SA-516 70	...	...	K02700	No	No
Flange	SA-266 2	...	...	K03506	No	No
Nozzle	SA-106 B	...	...	K03006	No	No
Re-Pad	SA-516 70	...	...	K02700	No	No
Nozzle Flg	SA-182 F304L	...	<= 130	S30403	No	No
Nozzle Flg	SA-182 F11	...	...	K11572	No	No
Nozzle Flg	SA-105	...	...	K03504	No	No
Tubes	SA-213 TP304L	...	...	S30403	No	No
Tubesheet	SA-965 F304L	...	...	S30403	No	No
Flg Bolting	SA-193 B7	...	<= 64	G41400	No	No
Hrz Bolting	SA-307 A	...	6 <= t <= 100		No	No
Saddles	SA-516 70	...	...	K02700	No	No

- Normalized is determined based on the UCS-66 material curve selection and Figure UCS-66.
- Impact Tested is based on material selection and material data properties.

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Channel head	1010.422	101.30	1032.000	3.2000	No	No
Channel shell	1010.422	103.00	2751.000	3.2000	N/A	No
LHS Ch Flange	1000.000	103.00	3588.000	0.0000	N/A	No
Shell Flange	6100.000	103.00	6600.000	0.0000	N/A	No
Shell	6110.468	103.00	6506.000	2.0000	N/A	No
Shell head	6110.370	103.00	6265.000	2.0000	No	No



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Liquid Level: 1066.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	0.0	50.0	14.0	7.2	7.1	6.0	1.00	1.00
Cylinder	800.0	800.0	14.0	14.0	7.1	6.5	1.00	1.00
Body Flg	1050.5	241.0	212.0	204.0	199.7	199.7	1.00	1.00
Body Flg	1554.7	226.0	194.0	186.0	181.4	181.4	1.00	1.00
Cylinder	8954.7	7400.0	28.0	28.0	26.3	9.9	1.00	1.00
Ellipse	9004.7	50.0	34.0	26.5	25.9	4.9	1.00	1.00

Saddle Parameters:

Saddle Width	405.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1000.000	mm
Wear Pad Width	600.000	mm
Wear Pad Thickness	10.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	1500.000	mm
Baseplate Length	1000.000	mm
Baseplate Thickness	25.000	mm
Baseplate Width	430.000	mm
Number of Ribs (including outside ribs)	3	
Rib Thickness	10.000	mm
Web Thickness	10.000	mm
Height of Center Web	480.000	mm
Number of Bolts in Baseplate	4	

Baseplate Sketch

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|----- 1000.000 mm -----|
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Baseplate Plan View

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Baseplate Side View

Maximum Tensile Bolt Load 277. Kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	26930.90	Kgf
Maximum Transverse Saddle Shear Load	942.09	Kgf
Maximum Longitudinal Saddle Shear Load	1884.18	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	24373.28	Kgf
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Maximum Transverse Saddle Shear Load

146.12 Kgf

Maximum Longitudinal Saddle Shear Load

61.17 Kgf

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Tube inlet	WRC-297	0.974	Nozzle	Passed
Tube outlet	WRC-297	0.976	Nozzle	Passed
Shell outlet	WRC-297	0.928	Shell	Passed
Shell Inlet	WRC-297	0.928	Shell	Passed

Weights:

Fabricated - Bare W/O Removable Internals	24194.4	kg.
Shop Test - Fabricated + Water (Full)	31352.6	kg.
Shipping - Fab. + Rem. Intls.+ Shipping App.	24194.4	kg.
Erected - Fab. + Rem. Intls.+ Insul. (etc)	24694.4	kg.
Empty - Fab. + Intls. + Details + Wghts.	25229.1	kg.
Operating - Empty + Operating Liquid (No CA)	32043.8	kg.
Field Test - Empty Weight + Water (Full)	31887.4	kg.

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PV Elite 25 SP2 Licensee: #
 FileName : 174-C_BEU_Dia 1000-AK
 Nozzle Summary:

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Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
Tube inlet	2751	OK	...	OK	7.08	OK	Passed
Tube outlet	2751	OK	...	OK	7.12	OK	Passed
Shell outlet	6405	OK	...	OK	11.10	OK	Passed
Shell Inlet	6415	OK	...	OK	11.10	OK	Passed
Vent	6506	...	...	OK	6.42	OK	No Calc[*]
Drain	6506	...	...	OK	6.42	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 6405 Nozzle : Shell outlet [Shellside]
 Minimum MAWP Nozzles : 2751 Nozzle : Tube inlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the Spatial Relationship between the Nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	Tube inlet	450.000	0.000	509.016
20	Tube outlet	450.000	180.000	509.016
50	Shell outlet	8104.700	180.000	505.293
50	Shell Inlet	2604.700	0.000	505.293
50	Vent	3604.700	0.000	112.325
50	Drain	4604.700	180.000	112.325

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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 Nozzle Schedule:

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
Vent	2.000 in	XXS	WNF	2.375	11.074	200.32	16.00	278.85	600
Drain	2.000 in	XXS	WNF	2.375	11.074	200.32	16.00	278.85	600
Tube inlet	10.000 in	40	WNF	10.750	9.271	453.05	12.00	281.89	150
Tube outlet	10.000 in	40	WNF	10.750	9.271	453.05	12.00	281.89	150
Shell outlet	10.000 in	80	WNF	10.750	15.088	473.05	28.00	345.78	600
Shell Inlet	10.000 in	80	WNF	10.750	15.088	473.05	28.00	345.78	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Vent	SA-106 B	14.000	20.000	12.000	16.000	...
Drain	SA-106 B	14.000	20.000	15.000	16.000	...
Tube inlet	SA-106 B	9.000	15.000	10.000	12.000	...
Tube outlet	SA-106 B	9.000	15.000	10.000	12.000	...
Shell outle	SA-106 B	15.088	10.000	22.000	28.000	...
Shell Inlet	SA-106 B	15.088	10.000	22.000	28.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Vent	3554.700	0.0	250.00	0.00	Shell
Drain	4554.700	180.0	250.00	0.00	Shell
Tube inlet	400.000	0.0	250.00	0.00	Channel shell
Tube outlet	400.000	180.0	250.00	0.00	Channel shell
Shell outlet	8054.700	180.0	300.00	0.00	Shell
Shell Inlet	2554.700	0.0	300.00	0.00	Shell



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Bill of Material:

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 14.0mm THK X 1060.0mm ID X 50.0mm	SA-516 70
1	BODY FLANGE: 212.0mm THK X 1380.0mm OD	SA-266 2
1	BODY FLANGE: 194.0mm THK X 1380.0mm OD	SA-266 2
1	ELLIPTICAL HEAD: 2.0 X 1, 34.0mm THK X 1066.0mm ID X 50.0mm	SA-516 70
20	BAFFLES: 1040.0mm DIA X 7.0mm THK	SA-516 70
8	TIE RODS: 25mm DIA X 4000mm	SA-193 B7
1	TUBESHEET: 1380.0mm X 275.0mm THK	SA-965 F304L
682	TUBES: 7100.0mm X 25.4mm DIA X 2.8mm THK	SA-213 TP304L
1	INSULATION: 115mm X 50mm THK	Insulation
1	CLADDING: 237.000mm X 3.000mm THK	...
2	SADDLE: 405.0mm X 120 DEG	SA-516 70
1	INSULATION: 7400mm X 50mm THK	Hot
1	CLADDING: 7400.000mm X 3.000mm THK	...
1	INSULATION: 312mm X 50mm THK	Hot
1	CLADDING: 314.000mm X 3.000mm THK	...
1	GASKET: 1118.0mm OD X 1092.0mm ID	...
36	BODY FLANGE BOLTS: 47.6mm DIA	SA-193 B7
72	NUTS FOR BODY FLANGE BOLTS: 47.6mm DIA	...
2	CYLINDRICAL SEGMENT 14.0mm THK X 1322.8mm X 800.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 684.5mm X 800.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 1293.0mm X 7400.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 1283.7mm X 7400.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 772.1mm X 7400.0mm	SA-516 70
1	NAMEPLATE	...



PV Elite 25 SP2 Licensee: #
 FileName : 174-C_BEU_Dia 1000-AK
 MDMT Summary:

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Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20 (f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Shell Flange [11]		B	0	-3		0.949	26.000	1.00	No
Shell [8]		B	2	0		0.960	28.000	1.00	No
Shell head [10]		B	1	0		0.991	26.500	1.00	No
Shell head [7]		B	7	-5		0.780	34.000	1.00	No
Left Saddle [24]		B	-29	-31	-29	0.960	10.000	1.00	No
New Saddle [24]		B	-29	-31	-29	0.960	10.000	1.00	No
Shell outlet [1]		B	2	-1		0.936	28.000	1.00	No
Nozzle Flg [5]		A	-18	-40					
Shell Inlet [1]		B	2	-1		0.934	28.000	1.00	No
Nozzle Flg [5]		A	-18	-40					
Vent [1]		B	-14	-29	-29	0.934	16.000	1.00	No
Nozzle Flg [5]		A	-18	-40					
Drain [1]		B	-14	-29	-29	0.936	16.000	1.00	No
Nozzle Flg [5]		A	-18	-40					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No
Warmest MDMT:			7	0					
LHS Ch Flange[11]		B	-18	-104	-29	0.288	14.000	1.00	No
Channel head [10]		B	-29	-29	-29	0.990	7.200	1.00	No
Channel head [7]		B	-18	-48	-29	0.371	14.000	1.00	No
Channel shell [8]		B	-18	-48	-29	0.371	14.000	1.00	No
Tube inlet [1]		B	-23	-48	-29	0.359	12.000	1.00	No
Nozzle Flg [5]		B	-18	-38					
Tube outlet [1]		B	-23	-48	-29	0.363	12.000	1.00	No
Nozzle Flg [5]		B	-18	-38					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No
Bolting [21]			-48						
Warmest MDMT:			-18	-29					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	0.0	-12.0	Fail
Channel/Tube	-29.0	-12.0	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material



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MDMT Summary:

Step:

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- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning: LHS Ch Flange and Shell Flange

There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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 Input Echo:

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Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	6100	KPa.
Channel Side Design Pressure	999.99	KPa.
Shell Side Design Temperature	200.0	°C
Channel Side Design Temperature	82.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	



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 Input Echo:

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Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	97.2	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment H or Hh	0	mm
Distance Upwind of Crest Lh	0	mm
Distance from Crest to the Vessel x	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000



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Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element From Node	10	
Element To Node	20	
Element Type	Elliptical	
Description	Channel head	
Distance "FROM" to "TO"	50	mm
Inside Diameter	1060	mm
Element Thickness	7.2	mm
Internal Corrosion Allowance	3.2	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	101.3	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	242.6	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	



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Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1060 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Channel shell
Distance "FROM" to "TO"	800 mm
Inside Diameter	1060 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1060 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Nozzle
Detail ID	Tube inlet
Dist. from "FROM" Node / Offset dist	400 mm
Nozzle Diameter	10 in.
Nozzle Schedule	40
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	50.1 Kgf
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-106 B

Element From Node	20
Detail Type	Nozzle
Detail ID	Tube outlet
Dist. from "FROM" Node / Offset dist	400 mm
Nozzle Diameter	10 in.
Nozzle Schedule	40
Nozzle Class	150



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Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	50.1	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Weight	
Detail ID	Partition plate	
Dist. from "FROM" Node / Offset dist	400	mm
Miscellaneous Weight	100	Kgf
Offset from Element Centerline	0	mm

Element From Node	30	
Element To Node	40	
Element Type	Flange	
Description	LHS Ch Flange	
Distance "FROM" to "TO"	241	mm
Flange Inside Diameter	1060	mm
Element Thickness	204	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	212	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	103	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	40	
Element To Node	50	
Element Type	Flange	
Description	Shell Flange	
Distance "FROM" to "TO"	226	mm
Flange Inside Diameter	1056	mm
Element Thickness	186	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	194	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	6100	KPa.
Design Temperature Internal Pressure	200	°C
Design External Pressure	103	KPa.
Design Temperature External Pressure	200	°C



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Effective Diameter Multiplier 1.2
 Material Name SA-266 2
 Perform Flange Stress Calculation (Y/N) Y
 Weld is pre-Heated No

Element From Node 40
 Detail Type Insulation
 Detail ID Insulation
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 115 mm
 Thickness of Insulation 50 mm
 Density 0.000352 kg./cm³

Element From Node 40
 Detail Type Lining
 Detail ID SS 304L Clad
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Lining 237 mm
 Thickness of Lining 3 mm
 Density 0.00785 kg./cm³

Element From Node 50
 Element To Node 60
 Element Type Cylinder
 Description Shell
 Distance "FROM" to "TO" 7400 mm
 Inside Diameter 1066 mm
 Element Thickness 28 mm
 Internal Corrosion Allowance 2 mm
 Nominal Thickness 28 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 6100 KPa.
 Design Temperature Internal Pressure 275 °C
 Design External Pressure 103 KPa.
 Design Temperature External Pressure 200 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Allowable Stress, Ambient 138 N./mm²
 Allowable Stress, Operating 138 N./mm²
 Allowable Stress, Hydrotest 234 N./mm²
 Material Density 0.00775 kg./cm³
 P Number Thickness 32.004 mm
 Yield Stress, Operating 262 N./mm²
 UCS-66 Chart Curve Designation B
 External Pressure Chart Name CS-2
 UNS Number K02700
 Product Form Plate
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Saddle
 Detail ID Left Saddle
 Dist. from "FROM" Node / Offset dist 1500 mm
 Width of Saddle 405 mm
 Height of Saddle at Bottom 1000 mm
 Saddle Contact Angle 120.0



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Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	600	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	50	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	5900	mm
Width of Saddle	405	mm
Height of Saddle at Bottom	1000	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	600	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1066	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	7400	mm
Thickness of Insulation	50	mm
Density	0.000352	kg./cm ³

Element From Node	50	
Detail Type	Lining	
Detail ID	SS 304L Claddin	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	7400	mm
Thickness of Lining	3	mm
Density	0.00785	kg./cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Shell outlet	
Dist. from "FROM" Node / Offset dist	6500	mm
Nozzle Diameter	10	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	180.0	



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Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	209.21	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Shell Inlet	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	10	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	209.21	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Vent	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	XXS	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	12.17	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Drain	
Dist. from "FROM" Node / Offset dist	3000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	XXS	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	12.17	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Weight	
Detail ID	Fire proofing	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	250	Kgf
Offset from Element Centerline	1000	mm

Element From Node	50	
Detail Type	Weight	
Detail ID	Fire proofing	
Dist. from "FROM" Node / Offset dist	5900	mm
Miscellaneous Weight	250	Kgf
Offset from Element Centerline	1000	mm



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Element From Node	60
Element To Node	70
Element Type	Elliptical
Description	Shell head
Distance "FROM" to "TO"	50 mm
Inside Diameter	1066 mm
Element Thickness	26.5 mm
Internal Corrosion Allowance	2 mm
Nominal Thickness	34 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6100 KPa.
Design Temperature Internal Pressure	275 °C
Design External Pressure	103 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1056 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	312.5 mm
Thickness of Insulation	50 mm
Density	0.000352 kg./cm ³

Element From Node	60
Detail Type	Lining
Detail ID	SS304L cladding
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	314 mm
Thickness of Lining	3 mm
Density	0.00785 kg./cm ³



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XY Coordinate Calculations:

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Channel head		50	...	50	...
Channel shell		850	...	800	...
LHS Ch Flange		1100.52	...	241	...
Shell Flange		1604.7	...	226	...
Shell		9004.7	...	7400	...
Shell head		9054.7	...	50	...

PV Elite includes an 1/8 inch (3.175mm) raised face and gasket thicknesses for girth flanges and tubesheet thicknesses where applicable in the Tangent to Tangent length calculation. The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact.

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 Flg Calc [Int P]: New Flange

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Warnings for Flange : New Flange :

Note: Flange Hub Taper 0.5946 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by
 ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values

Description: New Flange :

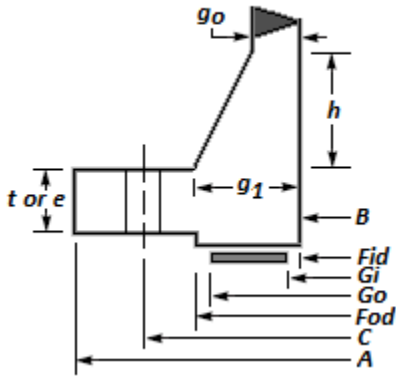
LHS Ch Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	1000.00	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1060.000	mm
Flange Outside Diameter	A	1380.000	mm
Flange Thickness	t	204.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	g1	36.0000	mm
Length of Hub	h	37.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1270.000	mm
Nominal Bolt Diameter	a	47.6250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	1143.000	mm
Flange Face Inside Diameter	Fid	1060.000	mm
Flange Facing Sketch	1, Code Sketch 1a		
Gasket Outside Diameter	Go	1118.000	mm
Gasket Inside Diameter	Gi	1092.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Column for Gasket Seating	2, Code Column II		
Gasket Thickness	tg	6.3500	mm
Length of Partition Gasket	lp	1050.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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 Flg Calc [Int P]: New Flange

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1000.0 \cdot (1060.0/2 + 0.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 1000.0) + C_a$$

$$= 3.8576 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (530.0 + 0.6 \cdot 14.0)$$

$$= 3588.137 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (530.0 + 0.6 \cdot 14.0)$$

$$= 3588.137 \text{ KPa.}$$

Corroded Flange ID,	Bcor = B + 2 * ci	1060.000	mm
Corroded Large Hub,	g1Cor = g1 - ci	36.000	mm
Corroded Small Hub,	g0Cor = go - ci	14.000	mm
Code R Dimension,	R = (C - B) / 2 - g1	69.000	mm
Gasket Contact Width,	N = (Go - Gi) / 2	13.000	mm
Basic Gasket Width,	bo = N / 2	6.500	mm
Effective Gasket Width,	b = Cb * sqrt(bo)	6.425	mm
Gasket Reaction Diameter,	G = Go - 2 * b	1105.151	mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$= \pi/4 \cdot G^2 \cdot P_{eq}$$

$$= \pi/4 \cdot 1105.1509^2 \cdot 1000.0$$

$$= 97820.422 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 1105.1509 \cdot 3.0 \cdot 1000.0$$

$$+ 2 \cdot 1050.0 \cdot 5.0 \cdot 3.0 \cdot 1000.0$$

$$= 16860.027 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1060.0^2 \cdot 1000.0$$



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= 89990.805 Kgf

Pressure Force on Flange Face [Ht]:

= H - Hd
 = 97820 - 89991
 = 7829.618 Kgf

Operating Bolt Load [Wm1]:

= max(H + Hp + H'p, 0)
 = max(97820 + 16860 + 0, 0)
 = 114680.453 Kgf
 = 699550.750 Kgf, Mating Flange Load Governs

Gasket Seating Bolt Load [Wm2]:

= y • b • π • G + yPart • bPart • lp
 = 68.95 • 6.4246 • π • 1105.151 + 68.95 • 5.0 • 1050.0
 = 193738.562 Kgf

Required Bolt Area [Am]:

= max(Wm1/Sb, Wm2/Sa)
 = max(699551/172, 193739/172)
 = 398.866 cm²

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:

= 2a + 6t/(m + 0.5)
 = 2 • 47.625 + 6 • 204.0/(3.0 + 0.5)
 = 444.964 mm

Actual Circumferential Bolt Spacing [Bs]:

= C • sin(π / n)
 = 1270.0 • sin(π/36)
 = 110.688 mm

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

= max(√(Bs/(2a + t)), 1)
 = max(√(110.688/(2 • 47.625 + 204.0)), 1)
 = 1.0000

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	398.866	535.122	
Radial Distance between Hub and Bolts:	60.325	69.000	
Radial Distance between Bolts and Edge:	47.625	55.000	
Circ. Spacing between the Bolts:	101.600	110.688	444.964

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

= Ab • Sa/(y • π(Go + Gi))
 = 535.122 • 172.0/(68.95 • π(1118.0 + 1092.0))
 = 19.226 mm[Note: Exceeds actual gasket width, 13.000]

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :

= 32.000 mm[Note: Exceeds actual gasket width, 13.000]

Flange Design Bolt Load, Gasket Seating [W]:

= Sa • Ab
 = 172.0 • 535.1215
 = 938521.75 Kgf



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 Flg Calc [Int P]: New Flange

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Gasket Load for the Operating Condition [HG]:
 = Wm1 - H
 = 699551 - 97820
 = 601730.31 Kgf

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 = (C - G) / 2
 = (1270.0 - 1105.1509) / 2
 = 82.4246 mm

Distance to Face Pressure Reaction [ht]:
 = (R + g1 + hg) / 2
 = (69.0 + 36.0 + 82.4246) / 2
 = 93.7123 mm

Distance to End Pressure Reaction [hd]:
 = R + (g1 / 2)
 = 69.0 + (36.0 / 2.0)
 = 87.0000 mm

Summary of Moments for Internal Pressure: (Kg-m.)

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	89991.	87.0000	1.0000	7829.
Face Pressure,	Mt	7830.	93.7123	1.0000	734.
Gasket Load,	Mg	601730.	82.4246	1.0000	49598.
Gasket Seating,	Matm	938522.	82.4246	1.0000	77359.

Total Moment for Operation, Mop 58161. Kg-m.
 Total Moment for Gasket seating, Matm 77359. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 121.820 mm
 Hub Ratio, h/h0 = HL / H0 0.304
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 2.571

Flange Factors for Integral Flange:

Factor F 0.868
 Factor V 0.262
 Factor f 3.427
 Factors from Figure 2-7.1 K = 1.302
 T = 1.797 U = 8.249
 Y = 7.506 Z = 3.878
 e = 0.0071 1/mm d = 0.75054E+06 mm³
 Stress Factors Alpha = 2.454
 Beta = 2.939 Gamma = 1.366
 Delta = 11.311 Lambda = 12.678

Longitudinal Hub Stress, Operating [SHo]:
 = (f • Mop / Bcor) / (L • g1²)
 = (3.4269 • 58161 / 1060.0) / (12.6775 • 36.0²)
 = 112.23 N./mm²

Longitudinal Hub Stress, Seating [SHa]:
 = (f • Matm / Bcor) / (L • g1²)
 = (3.4269 • 77359 / 1060.0) / (12.6775 • 36.0²)
 = 149.27 N./mm²



PV Elite 25 SP2 Licensee: #
 FileName : 174-C_BEU_Dia 1000-AK
 Flg Calc [Int P]: New Flange

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Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (2.939 \cdot 58161 / 1060.0) / (12.6775 \cdot 204.0^2)$
 $= 3.00 \text{ N./mm}^2$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (2.939 \cdot 77359 / 1060.0) / (12.6775 \cdot 204.0^2)$
 $= 3.99 \text{ N./mm}^2$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
 $= (7.5062 \cdot 58161 / (204.0^2 \cdot 1060.0)) - 3.8781 \cdot 3$
 $= 85.43 \text{ N./mm}^2$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (7.5062 \cdot 77359 / (204.0^2 \cdot 1060.0)) - 3.8781 \cdot 4$
 $= 113.63 \text{ N./mm}^2$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (112 + \max(3, 85)) / 2$
 $= 98.83 \text{ N./mm}^2$

Average Flange Stress, Seating [SAa]:
 $= (SHA + \max(SRa, STa)) / 2$
 $= (149 + \max(4, 114)) / 2$
 $= 131.45 \text{ N./mm}^2$

Bolt Stress, Operating [BSo]:
 $= Wm1 / Ab$
 $= 699551 / 535.1215$
 $= 128.20 \text{ N./mm}^2$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (193739 / 535.1215)$
 $= 35.51 \text{ N./mm}^2$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	112.23	207.00	149.27	207.00
Radial Flange	3.00	138.00	3.99	138.00
Tangential Flange	85.43	138.00	113.63	138.00
Maximum Average	98.83	138.00	131.45	138.00
Bolting	128.20	172.00	35.51	172.00

Minimum Required Flange Thickness 199.746 mm
 Estimated M.A.W.P. (Operating) 3588 KPa.
 Estimated Finished Weight of Flange at given Thk. 994.0 kg.
 Estimated Unfinished Weight of Forging at given Thk 1145.5 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):



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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	491.97	500.02	0.984	491.97	500.02	0.984
Tangential Flange	St	375.01	375.01	1.000	375.01	375.01	1.000
Tangential Flange	Sto	120.05	250.01	0.480	120.05	250.01	0.480
Combined	Sh/f + St	518.78	750.03	0.692	518.78	750.03	0.692
Combined	Sr + St	388.17	750.03	0.518	388.17	750.03	0.518
Combined	Sh + Sto	612.02	750.03	0.816	612.02	750.03	0.816
Radial Flange	Sr	13.16	500.02	0.026	13.16	500.02	0.026
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:
 = max(1, F1num / F1den)
 = max(1, 3.9425/1.1521)
 = 3.4221

Hub Stress Correction Factor [f2]:
 = max(0.25, fn2) if h/h0 < 0.35, else max (0.25, fn3)
 = max(0.25, 0.7989/0.9567) if h/h0 < 0.35, else max (0.25, 0.6453/0.7412)
 = 0.8351

In this case, the hub ratio h/h0, 0.3037 was < 0.35.



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PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	567.7 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	275.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.260
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	14.8645 cm ²
Max. permissible single flange rotation for gasket	Thetamax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.558 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot lp \cdot bPart)$$

$$= \pi/4 (1118.0^2 - 1092.0^2) + (0.5 \cdot 1050.0 \cdot 5.0)$$

$$= 477.5 \text{ cm}^2$$

Target Bolt Stress [Sbsel]:

$$= SGT \cdot Ag / (nb \cdot Ab) \quad (O-1)$$

$$= 208.0 \cdot 477.539 / (36 \cdot 14.864)$$

$$= 185.6 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]:

$$= \min(Sbsel, Sbmax) \quad (O-4)$$

$$= \min(185.618, 507.5)$$

$$= 185.6 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]:

$$= \max(Sbsel, Sbmin) \quad (O-5)$$

$$= \max(185.618, 290.0)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]:

$$= \min(Sbsel, Sfmax) \quad (O-6)$$

$$= \min(290.0, 567.662)$$

$$= 290.0 \text{ N./mm}^2$$

Gasket Assembly Stress [GAS]:

$$= Sgmin-S (Ag / (nb \cdot Abs)) \quad (O-7)$$

$$= 140.0 (477.539 / (36 \cdot 14.864))$$

$$= 124.9 \text{ N./mm}^2$$

Since Sbsel >= GAS, check passes.

Gasket Operating Stress [GOS]:

$$= (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb) \quad (O-8)$$



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$$= (97.0 \cdot 477.539 + \pi/4 \cdot 1000.0 \cdot 1092.0^2) / (0.7 \cdot 14.864 \cdot 36)$$

$$= 148.664 \text{ N./mm}^2$$

Since $Sbsel \geq GOS$, check passes.

Gasket Maximum Stress [GMS]:

$$= Sgmax(Ag / (nb \cdot Abs)) \quad (O-9)$$

$$= 275.0(477.539 / (36 \cdot 14.864))$$

$$= 245.4 \text{ N./mm}^2$$

Since $Sbsel > GMS$, check fails.

Flange Rotation Limit [FRL]:

$$= Sfmax(thetagmax / thetafmax) \quad (O-10)$$

$$= 567.662(1.0 / 0.558)$$

$$= 1016.8 \text{ N./mm}^2$$

Since $Sbsel \leq FRL$, check passes.

Assembly Bolt Torque [Tb]:

$$= Sbsel \cdot K \cdot Abs \cdot \phi b / 12 \quad (O-2)$$

$$= 290.0 \cdot 0.26 \cdot 14.864 \cdot 47.625 / 12$$

$$= 544.3 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange Rigidity Based on Required Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 77358.8 / 1.0 \cdot 9806.637 \cdot 0.262 / (11.964 \cdot 202713 \cdot 14.0^2 \cdot 121.82 \cdot 0.3)$$

$$= 0.597 \quad (\text{should be } \leq 1)$$

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 58161.4 / 1.0 \cdot 9806.637 \cdot 0.262 / (11.964 \cdot 199226 \cdot 14.0^2 \cdot 121.82 \cdot 0.3)$$

$$= 0.457 \quad (\text{should be } \leq 1)$$

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 77358.8 / 1.0 \cdot 9806.637 \cdot 0.262 / (12.678 \cdot 202713 \cdot 14.0^2 \cdot 121.82 \cdot 0.3)$$

$$= 0.563 \quad (\text{should be } \leq 1)$$

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 58161.4 / 1.0 \cdot 9806.637 \cdot 0.262 / (12.678 \cdot 199226 \cdot 14.0^2 \cdot 121.82 \cdot 0.3)$$

$$= 0.431 \quad (\text{should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.288, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20 (f)	-29 °C



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Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

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Flange Input Data Values

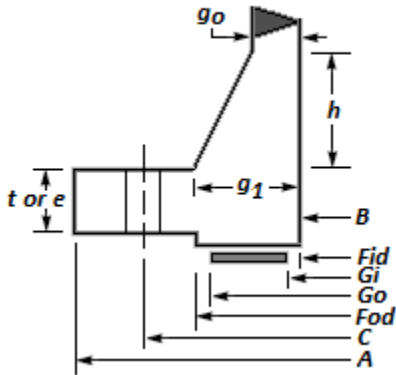
Description: Shell Flange :

Shell Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	6100.00	KPa.
Design Temperature		200	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1056.000	mm
Flange Outside Diameter	A	1380.000	mm
Flange Thickness	t	186.0000	mm
Thickness of Hub at Small End	go	26.0000	mm
Thickness of Hub at Large End	gl	36.0000	mm
Length of Hub	h	40.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1270.000	mm
Nominal Bolt Diameter	a	47.6250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	1143.000	mm
Flange Face Inside Diameter	Fid	1056.000	mm
Flange Facing Sketch	1, Code Sketch 1a		
Gasket Outside Diameter	Go	1118.000	mm
Gasket Inside Diameter	Gi	1092.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Column for Gasket Seating	2, Code Column II		
Gasket Thickness	tg	6.3500	mm
Length of Partition Gasket	lp	1050.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²

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Hub Small End Required Thickness due to Internal Pressure:

$$\begin{aligned}
 &= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (6100.0 \cdot (1056.0/2 + 0.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 6100.0) + C_a \\
 &= 23.9769 \text{ mm}
 \end{aligned}$$

Small End Hub MAWP:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 26.0) / (528.0 + 0.6 \cdot 26.0) \\
 &= 6599.939 \text{ KPa.}
 \end{aligned}$$

Hub Small End Hub MAPnc:

$$\begin{aligned}
 &= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 26.0) / (528.0 + 0.6 \cdot 26.0) \\
 &= 6599.939 \text{ KPa.}
 \end{aligned}$$

Corroded Flange ID,	Bcor = B + 2 * ci	1056.000	mm
Corroded Large Hub,	g1Cor = g1 - ci	36.000	mm
Corroded Small Hub,	g0Cor = go - ci	26.000	mm
Code R Dimension,	R = (C - B) / 2 - g1	71.000	mm

Gasket Contact Width,	N = (Go - Gi) / 2	13.000	mm
Basic Gasket Width,	bo = N / 2	6.500	mm
Effective Gasket Width,	b = Cb * sqrt(bo)	6.425	mm
Gasket Reaction Diameter,	G = Go - 2 * b	1105.151	mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$\begin{aligned}
 &= \pi/4 \cdot G^2 \cdot P_{eq} \\
 &= \pi/4 \cdot 1105.1509^2 \cdot 6100.0 \\
 &= 596704.562 \text{ Kgf}
 \end{aligned}$$

Contact Load on Gasket Surfaces [Hp]:

$$\begin{aligned}
 &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P \\
 &= 2 \cdot 6.4246 \cdot \pi \cdot 1105.1509 \cdot 3.0 \cdot 6100.0 \\
 &\quad + 2 \cdot 1050.0 \cdot 5.0 \cdot 3.0 \cdot 6100.0 \\
 &= 102846.172 \text{ Kgf}
 \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned}
 &= \pi/4 \cdot B_{cor}^2 \cdot P \\
 &= \pi/4 \cdot 1056.0^2 \cdot 6100.0
 \end{aligned}$$



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= 544808.812 Kgf

Pressure Force on Flange Face [Ht]:

= H - Hd
 = 596705 - 544809
 = 51895.750 Kgf

Operating Bolt Load [Wm1]:

= max(H + Hp + H'p, 0)
 = max(596705 + 102846 + 0, 0)
 = 699550.750 Kgf

Gasket Seating Bolt Load [Wm2]:

= y • b • π • G + yPart • bPart • lp
 = 68.95 • 6.4246 • π • 1105.151 + 68.95 • 5.0 • 1050.0
 = 193738.562 Kgf

Required Bolt Area [Am]:

= max(Wm1/Sb, Wm2/Sa)
 = max(699551/172, 193739/172)
 = 398.866 cm²

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:

= 2a + 6t/(m + 0.5)
 = 2 • 47.625 + 6 • 186.0/(3.0 + 0.5)
 = 414.107 mm

Actual Circumferential Bolt Spacing [Bs]:

= C • sin(π / n)
 = 1270.0 • sin(π/36)
 = 110.688 mm

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

= max(√(Bs/(2a + t)), 1)
 = max(√(110.688/(2 • 47.625 + 186.0)), 1)
 = 1.0000

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	398.866	535.122	
Radial Distance between Hub and Bolts:	60.325	71.000	
Radial Distance between Bolts and Edge:	47.625	55.000	
Circ. Spacing between the Bolts:	101.600	110.688	414.107

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

= Ab • Sa/(y • π(Go + Gi))
 = 535.122 • 172.0/(68.95 • π(1118.0 + 1092.0))
 = 19.226 mm[Note: Exceeds actual gasket width, 13.000]

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :

= 32.000 mm[Note: Exceeds actual gasket width, 13.000]

Flange Design Bolt Load, Gasket Seating [W]:

= Sa • Ab
 = 172.0 • 535.1215
 = 938521.75 Kgf



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Gasket Load for the Operating Condition [HG]:
 $= Wm1 - H$
 $= 699551 - 596705$
 $= 102846.20 \text{ Kgf}$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (1270.0 - 1105.1509) / 2$
 $= 82.4246 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= (R + g1 + hg) / 2$
 $= (71.0 + 36.0 + 82.4246) / 2$
 $= 94.7123 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= R + (g1 / 2)$
 $= 71.0 + (36.0 / 2.0)$
 $= 89.0000 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	544809.	89.0000	1.0000	48489.
Face Pressure,	Mt	51896.	94.7123	1.0000	4915.
Gasket Load,	Mg	102846.	82.4246	1.0000	8477.
Gasket Seating,	Matm	938522.	82.4246	1.0000	77359.

Total Moment for Operation, Mop 61881. Kg-m.
 Total Moment for Gasket seating, Matm 77359. Kg-m.

Effective Hub Length, $h_o = \sqrt{Bcor \cdot goCor}$ 165.699 mm
 Hub Ratio, $h/h_o = HL / H_o$ 0.241
 Thickness Ratio, $g1/g_o = (g1Cor/goCor)$ 1.385

Flange Factors for Integral Flange:

Factor F 0.891
 Factor V 0.417
 Factor f 1.136
 Factors from Figure 2-7.1 K = 1.307
 $T = 1.794$ U = 8.136
 $Y = 7.404$ Z = 3.826
 $e = 0.0054 \text{ 1/mm}$ d = 0.21861E+07 mm³
 Stress Factors Alpha = 2.000
 $Beta = 2.333$ Gamma = 1.114
 $Delta = 2.943$ Lambda = 4.058

Longitudinal Hub Stress, Operating [SHo]:
 $= (f \cdot Mop / Bcor) / (L \cdot g1^2)$
 $= (1.136 \cdot 61881 / 1056.0) / (4.0579 \cdot 36.0^2)$
 $= 124.13 \text{ N./mm}^2$

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot Matm / Bcor) / (L \cdot g1^2)$
 $= (1.136 \cdot 77359 / 1056.0) / (4.0579 \cdot 36.0^2)$
 $= 155.18 \text{ N./mm}^2$



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Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (2.333 \cdot 61881 / 1056.0) / (4.0579 \cdot 186.0^2)$
 $= 9.55 \text{ N./mm}^2$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (2.333 \cdot 77359 / 1056.0) / (4.0579 \cdot 186.0^2)$
 $= 11.94 \text{ N./mm}^2$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
 $= (7.4036 \cdot 61881 / (186.0^2 \cdot 1056.0)) - 3.8258 \cdot 10$
 $= 86.44 \text{ N./mm}^2$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (7.4036 \cdot 77359 / (186.0^2 \cdot 1056.0)) - 3.8258 \cdot 12$
 $= 108.06 \text{ N./mm}^2$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (124 + \max(10, 86)) / 2$
 $= 105.29 \text{ N./mm}^2$

Average Flange Stress, Seating [SAa]:
 $= (SHA + \max(SRa, STa)) / 2$
 $= (155 + \max(12, 108)) / 2$
 $= 131.62 \text{ N./mm}^2$

Bolt Stress, Operating [BSo]:
 $= Wm1 / Ab$
 $= 699551 / 535.1215$
 $= 128.20 \text{ N./mm}^2$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (193739 / 535.1215)$
 $= 35.51 \text{ N./mm}^2$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	124.13	207.00	155.18	207.00
Radial Flange	9.55	138.00	11.94	138.00
Tangential Flange	86.44	138.00	108.06	138.00
Maximum Average	105.29	138.00	131.62	138.00
Bolting	128.20	172.00	35.51	172.00

Minimum Required Flange Thickness 181.356 mm
 Estimated M.A.W.P. (Operating) 6600 KPa.
 Estimated Finished Weight of Flange at given Thk. 926.4 kg.
 Estimated Unfinished Weight of Forging at given Thk 1085.8 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):



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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	464.94	486.85	0.955	464.94	500.02	0.930
Tangential Flange	St	322.55	365.14	0.883	322.55	375.01	0.860
Tangential Flange	Sto	109.26	243.43	0.449	109.26	250.01	0.437
Combined	Sh/f + St	730.28	730.28	1.000	730.28	750.03	0.974
Combined	Sr + St	358.18	730.28	0.490	358.18	750.03	0.478
Combined	Sh + Sto	574.20	730.28	0.786	574.20	750.03	0.766
Radial Flange	Sr	35.63	486.85	0.073	35.63	500.02	0.071
Maximum Stress Ratio				1.000			0.974

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:
 = max(1, F1num / F1den)
 = max(1, 1.3194/1.1571)
 = 1.1403

Hub Stress Correction Factor [f2]:
 = max(0.25, fn2) if h/ho < 0.35, else max (0.25, fn3)
 = max(0.25, 0.2882/1.0754) if h/ho < 0.35, else max (0.25, 0.237/0.7871)
 = 0.2680

In this case, the hub ratio h/ho, 0.2414 was < 0.35.



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PV Elite iterates on the assembly bolt load stress, S_{fmax}, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	S _{bmax}	507.5 N./mm ²
Minimum permissible bolt stress	S _{bmin}	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	S _{fmax}	513.4 N./mm ²
Target assembly gasket stress	S _{gT}	208.0 N./mm ²
Maximum permissible gasket stress	S _{gmax}	275.0 N./mm ²
Minimum gasket operating stress	S _{gmin-O}	97.0 N./mm ²
Minimum gasket seating stress	S _{gmin-S}	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	14.8645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	S _{ya}	250.0 N./mm ²
Flange Material Operating Yield Stress	S _{yo}	213.0 N./mm ²
Single flange rotation at S _{fmax}	Theta _{fmax}	0.538 deg

Gasket Contact Area [A_g]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$$

$$= \pi/4 (1118.0^2 - 1092.0^2) + (0.5 \cdot 1050.0 \cdot 5.0)$$

$$= 477.5 \text{ cm}^2$$

Target Bolt Stress [S_{bse}]:

$$= SGT \cdot A_g / (nb \cdot A_b) \quad (O-1)$$

$$= 208.0 \cdot 477.539 / (36 \cdot 14.864)$$

$$= 185.6 \text{ N./mm}^2$$

Upper Bolt Stress Limit [S_{bse}]:

$$= \min(S_{bse}, S_{bmax}) \quad (O-4)$$

$$= \min(185.618, 507.5)$$

$$= 185.6 \text{ N./mm}^2$$

Lower Bolt Stress Limit [S_{bse}]:

$$= \max(S_{bse}, S_{bmin}) \quad (O-5)$$

$$= \max(185.618, 290.0)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [S_{bse}]:

$$= \min(S_{bse}, S_{fmax}) \quad (O-6)$$

$$= \min(290.0, 513.374)$$

$$= 290.0 \text{ N./mm}^2$$

Gasket Assembly Stress [GAS]:

$$= S_{gmin-S} (A_g / (nb \cdot A_b)) \quad (O-7)$$

$$= 140.0 (477.539 / (36 \cdot 14.864))$$

$$= 124.9 \text{ N./mm}^2$$

Since S_{bse} >= GAS, check passes.

Gasket Operating Stress [GOS]:

$$= (S_{gmin-O} \cdot A_g + \pi/4 \cdot P_{max} \cdot GI.D.^2) / (\phi G \cdot A_b \cdot nb) \quad (O-8)$$

$$= (97.0 \cdot 477.539 + \pi/4 \cdot 6100.0 \cdot 1092.0^2) / (0.7 \cdot 14.864 \cdot 36)$$



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= 276.185 N./mm²

Since Sbsel >= GOS, check passes.

Gasket Maximum Stress [GMS]:
 = Sgmax(Ag / (nb • Abs)) (O-9)
 = 275.0(477.539/(36 • 14.864))
 = 245.4 N./mm²

Since Sbsel > GMS, check fails.

Flange Rotation Limit [FRL]:
 = Sfmax(thetagmax / thetafmax) (O-10)
 = 513.374(1.0/0.538)
 = 954.6 N./mm²

Since Sbsel <= FRL, check passes.

Assembly Bolt Torque [Tb]:
 = Sbsel • K • Abs • φb/12 (O-2)
 = 290.0 • 0.2 • 14.864 • 47.625/12
 = 418.7 Kg-m.

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 95 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange Rigidity Based on Required Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•77358.8/1.0•9806.637•0.417/(3.829•201334•26.0²•165.699•0.3)
 = 0.636 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•61881.4/1.0•9806.637•0.417/(3.829•191213•26.0²•165.699•0.3)
 = 0.536 (should be ≤ 1)

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•77358.8/1.0•9806.637•0.417/(4.058•201334•26.0²•165.699•0.3)
 = 0.600 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•61881.4/1.0•9806.637•0.417/(4.058•191213•26.0²•165.699•0.3)
 = 0.505 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.949, Temperature Reduction per Fig. UCS 66.1 = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-3 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.



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Flg Calc [Int P]: Shell Flange

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 Internal Pressure Calculations:

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress (SE) N./mm ²
Channel head		1010.4	14	3.2	1060	138
Channel shell		1010.4	14	3.2	1060	138
LHS Ch Flange		1000	212	...	1060	138
Shell Flange		6100	194	...	1056	138
Shell		6110.5	28	2	1066	138
Shell head		6110.4	34	2	1066	137

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm
Channel head		1000	1032.25	1872.03	7.2	7.07621
Channel shell		1000	2751.01	3588.14	14	7.12156
LHS Ch Flange		1000	3588.14	3588.14	204	199.746
Shell Flange		6100	6599.94	6599.94	186	181.356
Shell		6100	6505.56	7027.48	28	26.3376
Shell head		6100	6265.39	6826.7	26.5	25.8514

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 6265 KPa.
 Shell Side MAPnc = 6600 KPa.
 Channel Side MAWP = 1032 KPa.
 Channel Side MAPnc = 1590 KPa.

Elements Suitable for Design Internal Pressure.

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 Element: Channel head

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Channel head

Elliptical Head Design Information:					
Design Pressure	P	1010.422 KPa.	Int. Design Temperature	82 °C	
Material		SA-516 70	Ext. Design Temperature	82 °C	
Length		50.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.2000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1060.000 mm	Actual Stress	133.7 N./mm ²	
Spec. Min. Thk t or e		7.2000 mm	Spec. Nominal Thk.	14.000 mm	
Surface Area		14540.6 cm ²	SG of Contents	1.000	
Element Volume		200063.2 cm ³	Weight of Contents	851.8 Kgf	
Empty Weight		3153.6 Kgf	Operating Weight	4005.5 Kgf	
			Comp. All. Ope. Stress	99.7 N./mm ²	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.625 mm

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)} \\
 &= (1010.422 \cdot 1066.4 \cdot 0.992) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 1010.422) \\
 &= 3.8762 + 3.2000 = 7.0762 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.422 KPa.

$$\begin{aligned}
 &= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)} \\
 &= (2 \cdot 138.0 \cdot 1.0 \cdot 4.0) / (0.992 \cdot 1066.4 + 0.2 \cdot 4.0) \\
 &= 1043 - 10 = 1032 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)} \\
 &= (2 \cdot 138.0 \cdot 1.0 \cdot 7.2) / (1.0 \cdot 1060.0 + 0.2 \cdot 7.2) \\
 &= 1872 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t) \\
 &= (1010.422 \cdot (0.992 \cdot 1066.4 + 0.2 \cdot 4.0)) / (2 \cdot 1.0 \cdot 4.0) \\
 &= 133.730 \text{ N./mm}^2
 \end{aligned}$$



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Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (1010.422 \cdot 533.2) / (138.0 \cdot 1.0 - 0.6 \cdot 1010.422) + 3.2$$

$$= 7.122 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 10.422 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 10.8) / (533.2 + 0.6 \cdot 10.8)$$

$$= 2761 - 10 = 2751 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (1066.4 / (2 \cdot 268.2))^2) / 6$$

$$= 0.992069$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.609 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 7.2, tr = 3.96, c = 3.2 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 14.0, tr = 4.007, c = 3.2 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.371$, Temp. Reduction = 62 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1032.245 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	7.200 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.990	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-29 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.625 mm

External Pressure Results for: Channel head

Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.



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Element: Channel head

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Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
4.000	1074.40	268.60	0.0005171	51.70

$MAEP = B / (K0 * Do/t) = 51.6968 / (0.9 * 268.6) = 213.8406 \text{ KPa.}$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.753	1074.40	390.24	0.0003559	35.58

$MAEP = B / (K0 * Do/t) = 35.5829 / (0.9 * 390.2374) = 101.3081 \text{ KPa.}$

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 Element: Channel shell

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Channel shell

Cylindrical Shell Design Information:			
Design Pressure P	1010.422 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	800.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1060.000 mm	Actual Stress	50.5 N./mm ²
Spec. Min. Thk t or e	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	27344.4 cm ²	SG of Contents	1.000
Element Volume	706105.2 cm ³	Weight of Contents	851.8 Kgf
Empty Weight	3153.6 Kgf	Operating Weight	4005.5 Kgf
		Comp. All. Ope. Stress	130.0 N./mm ²

Radiography Information:	
User Defined	User Defined
Circ. Joint Efficiency	100 %
	Long. Joint Efficiency E 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1010.422 \cdot 533.2) / (138.0 \cdot 1.0 - 0.6 \cdot 1010.422) \\
 &= 3.9216 + 3.2000 = 7.1216 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.422 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 10.8) / (533.2 + 0.6 \cdot 10.8) \\
 &= 2761 - 10 = 2751 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 14.0) / (530.0 + 0.6 \cdot 14.0) \\
 &= 3588 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1010.422 \cdot (533.2 + 0.6 \cdot 10.8)) / (1.0 \cdot 10.8) \\
 &= 50.494 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 1.304 %

Minimum Design Metal Temperature Results:



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Govrn. thk, tg = 14.0, tr = 4.007, c = 3.2 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.371$, Temp. Reduction = 62 °C

Min Metal Temp. w/o impact per UCS-66, Curve B
Min Metal Temp. at Required thickness (UCS 66.1)
Min Metal Temp. w/o impact per UG-20(f)

-18 °C
-48 °C
-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1032.245 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.371	Temperature Difference	30 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

External Pressure Results for: Channel shell

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.800	1088.00	938.33	100.74	0.8624	0.0015725	98.00

$$MAEP = (4 * B) / (3 * (Do/t)) = (4 * 98.0) / (3 * 100.7407) = 1296.9819 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.277	1088.00	938.33	331.98	0.8624	0.0002565	25.65

$$MAEP = (4 * B) / (3 * (Do/t)) = (4 * 25.6487) / (3 * 331.981) = 103.0069 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.800	1088.00	20377.67	100.74	18.7295	0.0001084	10.84

$$MAEP = (4 * B) / (3 * (Do/t)) = (4 * 10.8364) / (3 * 100.7407) = 143.4145 \text{ KPa.}$$



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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	6110.468 KPa.	Int. Design Temperature	275 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	7400.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	2.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1066.000 mm	Actual Stress	129.4 N./mm ²
Spec. Min. Thk t or e	28.0000 mm	Spec. Nominal Thk.	28.000 mm
Surface Area	260840.2 cm ²	SG of Contents	1.000
Element Volume	4027503.8 cm ³	Weight of Contents	851.8 Kgf
Empty Weight	3153.6 Kgf	Operating Weight	4005.5 Kgf
		Comp. All. Ope. Stress	110.7 N./mm ²

Radiography Information:	
User Defined	User Defined
Circ. Joint Efficiency 100 %	Long. Joint Efficiency E 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (6110.468 \cdot 535.0) / (138.0 \cdot 1.0 - 0.6 \cdot 6110.468) \\
 &= 24.3376 + 2.0000 = 26.3376 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.469 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 26.0) / (535.0 + 0.6 \cdot 26.0) \\
 &= 6516 - 10 = 6506 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 28.0) / (533.0 + 0.6 \cdot 28.0) \\
 &= 7027 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (6110.468 \cdot (535.0 + 0.6 \cdot 26.0)) / (1.0 \cdot 26.0) \\
 &= 129.408 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 2.559 %

Minimum Design Metal Temperature Results:



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Element: Shell

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Govrn. thk, tg = 28.0, tr = 24.972, c = 2.0 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.96$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B

2 °C

Min Metal Temp. at Required thickness (UCS 66.1)

0 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6265.392 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	28.000 mm	Basic MDMT	2 °C
Stress/Thk. Ratio	0.960	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	0 °C
		Joint Efficiency E*	100 %

External Pressure Results for: Shell

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
26.000	1122.00	7538.83	43.15	6.7191	0.0006895	66.75

$$MAEP = (4*B) / (3*(Do/t)) = (4*66.7513) / (3*43.1538) = 2062.3081 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.856	1122.00	7538.83	142.83	6.7191	0.0001140	11.03

$$MAEP = (4*B) / (3*(Do/t)) = (4*11.0348) / (3*142.827) = 103.007 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
26.000	1122.00	578601.00	43.15	50.0000	0.0005907	57.18

$$MAEP = (4*B) / (3*(Do/t)) = (4*57.1821) / (3*43.1538) = 1766.6644 \text{ KPa.}$$

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 Element: Shell head

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell head

Elliptical Head Design Information:					
Design Pressure	P	6110.370 KPa.	Int. Design Temperature		275 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm	Allowable Stress (ope)	S	137.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1066.000 mm	Actual Stress		133.4 N./mm ²
Spec. Min. Thk t or e		26.5000 mm	Spec. Nominal Thk.		34.000 mm
Surface Area		15720.9 cm ²	SG of Contents		1.000
Element Volume		203227.1 cm ³	Weight of Contents		851.8 Kgf
Empty Weight		3153.6 Kgf	Operating Weight		4005.5 Kgf
			Comp. All. Ope. Stress		110.7 N./mm ²
			Aspect Ratio		2.00

Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Efficiency	E	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.325 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.663 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (6110.37 \cdot 1070.0 \cdot 0.995) / (2 \cdot 137.0 \cdot 1.0 - 0.2 \cdot 6110.37)$$

$$= 23.8514 + 2.0000 = 25.8514 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.371 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 137.0 \cdot 1.0 \cdot 24.5) / (0.995 \cdot 1070.0 + 0.2 \cdot 24.5)$$

$$= 6276 - 10 = 6265 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 26.5) / (1.0 \cdot 1066.0 + 0.2 \cdot 26.5)$$

$$= 6827 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (6110.37 \cdot (0.995 \cdot 1070.0 + 0.2 \cdot 24.5)) / (2 \cdot 1.0 \cdot 24.5)$$

$$= 133.388 \text{ N./mm}^2$$



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 Element: Shell head

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Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (6110.37 \cdot 535.0) / (137.0 \cdot 1.0 - 0.6 \cdot 6110.37) + 2.0$$

$$= 26.520 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 10.371 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.0 \cdot 1.0 \cdot 32.0) / (535.0 + 0.6 \cdot 32.0)$$

$$= 7910 - 10 = 7900 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (1070.0 / (2 \cdot 268.5))^2) / 6$$

$$= 0.995043$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 12.864 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 26.5, tr = 24.281, c = 2.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.991$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	0 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 34.0, tr = 24.972, c = 2.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.78$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-5 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6265.392 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	34.000 mm	Basic MDMT	7 °C
Stress/Thk. Ratio	0.780	Temperature Difference	12 °C
UCS-66(c) red.	No	Computed Min. Temp.	-5 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.325 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.663 mm

External Pressure Results for: Shell head

Elliptical Head From 60 to 70 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.



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Element: Shell head

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Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
24.500	1119.00	45.67	0.0030409	99.75

$$MAEP = B / (K0 * Do / t) = 99.7506 / (0.9 * 45.6735) = 2426.5205 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.938	1119.00	380.82	0.0003647	35.31

$$MAEP = B / (K0 * Do / t) = 35.3062 / (0.9 * 380.8222) = 103.0058 \text{ KPa.}$$

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Pressure Test Results: Shop Test

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Hydrostatic Test Pressure Results:**Exchanger Shell Side Hydrostatic Test Pressures:**

Pressure per UG99b	= 1.30 * M.A.W.P. * Sa/S	8145.009	KPa.
Pressure per UG99b[35]	= 1.30 * Design Pres * Sa/S	7930.000	KPa.
Pressure per UG99c	= 1.30 * M.A.P. - Head(Hyd)	8569.472	KPa.
Pressure per UG100	= 1.10 * M.A.W.P. * Sa/S	6891.931	KPa.
Pressure per PED	= max(1.43*DP, 1.25*DP*ratio)	8722.999	KPa.
Pressure per PED&UG99b	= max(1.43*DP, 1.3*DP*ratio)	8722.999	KPa.
Pressure per App 27-4	= M.A.W.P.	6265.392	KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

Pressure per UG99b	= 1.30 * M.A.W.P. * Sa/S	1341.919	KPa.
Pressure per UG99b[35]	= 1.30 * Design Pres * Sa/S	1300.000	KPa.
Pressure per UG99c	= 1.30 * M.A.P. - Head(Hyd)	2067.008	KPa.
Pressure per UG100	= 1.10 * M.A.W.P. * Sa/S	1135.470	KPa.
Pressure per PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000	KPa.
Pressure per PED&UG99b	= max(1.43*DP, 1.3*DP*ratio)	1430.000	KPa.
Pressure per App 27-4	= M.A.W.P.	1032.245	KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Channel head	153.1	234.0	0.654	2077.40
Channel shell	79.9	234.0	0.341	2077.40
Shell	168.5	234.0	0.720	8579.92
Shell head	173.4	234.0	0.741	8579.92

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Tube inlet	Nozzle	118.00	118.00	1.000
Tube inlet	Pad	138.00	138.00	1.000
Tube outlet	Nozzle	118.00	118.00	1.000
Tube outlet	Pad	138.00	138.00	1.000
Shell outlet	Nozzle	118.00	118.00	1.000
Shell outlet	Pad	138.00	137.00	1.007
Shell Inlet	Nozzle	118.00	118.00	1.000
Shell Inlet	Pad	138.00	137.00	1.007
Vent	Nozzle	118.00	118.00	1.000
Vent	Pad	138.00	137.00	1.007
Drain	Nozzle	118.00	118.00	1.000
Drain	Pad	138.00	137.00	1.007

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):



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Pressure Test Results: Shop Test

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Description	Ambient	Operating	Ratio
Channel head	138.00	138.00	1.000
Channel shell	138.00	138.00	1.000
LHS Ch Flange	138.00	138.00	1.000
Shell Flange	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Shell head	138.00	137.00	1.007

Minimum

1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	100.35	1.146
Tubesheet Material	115.00	74.15	1.551

Minimum

1.146

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Tube inlet	29.76	216.00	0.138
Tube outlet	29.76	216.00	0.138
Shell outlet	85.30	216.00	0.395
Shell Inlet	85.30	216.00	0.395
Vent	19.94	216.00	0.092
Drain	19.94	216.00	0.092

Elements Suitable for Test Pressure.

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Pressure Test Results: Field Test

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Hydrostatic Test Pressure Results:**Exchanger Shell Side Hydrostatic Test Pressures:**

Pressure per UG99b	= 1.30 * M.A.W.P. * Sa/S	8145.009	KPa.
Pressure per UG99b[35]	= 1.30 * Design Pres * Sa/S	7930.000	KPa.
Pressure per UG99c	= 1.30 * M.A.P. - Head(Hyd)	8569.472	KPa.
Pressure per UG100	= 1.10 * M.A.W.P. * Sa/S	6891.931	KPa.
Pressure per PED	= max(1.43*DP, 1.25*DP*ratio)	8722.999	KPa.
Pressure per PED&UG99b	= max(1.43*DP, 1.3*DP*ratio)	8722.999	KPa.
Pressure per App 27-4	= M.A.W.P.	6265.392	KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

Pressure per UG99b	= 1.30 * M.A.W.P. * Sa/S	1341.919	KPa.
Pressure per UG99b[35]	= 1.30 * Design Pres * Sa/S	1300.000	KPa.
Pressure per UG99c	= 1.30 * M.A.P. - Head(Hyd)	2067.008	KPa.
Pressure per UG100	= 1.10 * M.A.W.P. * Sa/S	1135.470	KPa.
Pressure per PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000	KPa.
Pressure per PED&UG99b	= max(1.43*DP, 1.3*DP*ratio)	1430.000	KPa.
Pressure per App 27-4	= M.A.W.P.	1032.245	KPa.

UG-99(b) Note 35, Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 6100.0 • 1.0

= 7930.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 1000.0 • 1.0

= 1300.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Channel head	173.4	234.0	0.741	1310.39
Channel shell	65.5	234.0	0.280	1310.39
Shell	168.2	234.0	0.719	7940.45
Shell head	173.3	234.0	0.741	7940.45

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Tube inlet	Nozzle	118.00	118.00	1.000
Tube inlet	Pad	138.00	138.00	1.000
Tube outlet	Nozzle	118.00	118.00	1.000
Tube outlet	Pad	138.00	138.00	1.000
Shell outlet	Nozzle	118.00	118.00	1.000
Shell outlet	Pad	138.00	137.00	1.007
Shell Inlet	Nozzle	118.00	118.00	1.000



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Pressure Test Results: Field Test

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Shell Inlet	Pad	138.00	137.00	1.007
Vent	Nozzle	118.00	118.00	1.000
Vent	Pad	138.00	137.00	1.007
Drain	Nozzle	118.00	118.00	1.000
Drain	Pad	138.00	137.00	1.007

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Channel head	138.00	138.00	1.000
Channel shell	138.00	138.00	1.000
LHS Ch Flange	138.00	138.00	1.000
Shell Flange	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Shell head	138.00	137.00	1.007

Minimum

1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	100.35	1.146
Tubesheet Material	115.00	74.15	1.551

Minimum

1.146

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Tube inlet	18.77	216.00	0.087
Tube outlet	18.77	216.00	0.087
Shell outlet	78.94	216.00	0.365
Shell Inlet	78.94	216.00	0.365
Vent	18.45	216.00	0.085
Drain	18.45	216.00	0.085

Elements Suitable for Test Pressure.

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PV Elite 25 SP2 Licensee: #
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 External Pressure Calculations:

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External Pressure Results Summary:

External Pressure Calculation Results:

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External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	1074.4	4	0.00051708	51.6968
20	30	938.333	1088	10.8	0.0015725	98
30	40	No Calc	...	204	No Calc	No Calc
40	50	No Calc	...	186	No Calc	No Calc
50	60	7538.83	1122	26	0.00068953	66.7513
60	70	No Calc	1119	24.5	0.0030409	99.7506

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	7.2	5.9532	101.3	213.841
20	30	14	6.4773	103	1296.98
30	40	204	199.746	103	No Calc
40	50	186	181.356	103	No Calc
50	60	28	9.85566	103	2062.31
60	70	26.5	4.93838	103	2426.52

Minimum

214

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	938.333	20377.7	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	7538.83	578601	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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PV Elite 25 SP2 Licensee: #
 FileName : 174-C_BEU_Dia 1000-AK
 Flg Calc [Ext P]: New Flange

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Warnings for Flange : New Flange :

Note: Flange Hub Taper 0.5946 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by
 ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values

Description: New Flange :

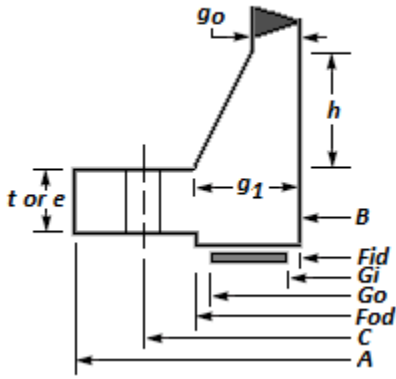
LHS Ch Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	-103.00	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1060.000	mm
Flange Outside Diameter	A	1380.000	mm
Flange Thickness	t	204.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	g1	36.0000	mm
Length of Hub	h	37.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1270.000	mm
Nominal Bolt Diameter	a	47.6250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	1143.000	mm
Flange Face Inside Diameter	Fid	1060.000	mm
Flange Facing Sketch	1, Code Sketch 1a		
Gasket Outside Diameter	Go	1118.000	mm
Gasket Inside Diameter	Gi	1092.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Column for Gasket Seating	2, Code Column II		
Gasket Thickness	tg	6.3500	mm
Length of Partition Gasket	lp	1050.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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 FileName : 174-C_BEU_Dia 1000-AK
 Flg Calc [Ext P]: New Flange

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Bolted Flange Calculation Results:

Corroded Flange ID,	$B_{cor} = B + 2 \cdot c_i$	1060.000	mm
Corroded Large Hub,	$g1_{cor} = g1 - c_i$	36.000	mm
Corroded Small Hub,	$g0_{cor} = g0 - c_i$	14.000	mm
Code R Dimension,	$R = (C - B) / 2 - g1$	69.000	mm
Gasket Contact Width,	$N = (G_o - G_i) / 2$	13.000	mm
Basic Gasket Width,	$b_o = N / 2$	6.500	mm
Effective Gasket Width,	$b = C_b \cdot \sqrt{b_o}$	6.425	mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	1105.151	mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$\begin{aligned}
 &= \pi / 4 \cdot G^2 \cdot P_{eq} \\
 &= \pi / 4 \cdot 1105.1509^2 \cdot -103.0 \\
 &= -10075.504 \text{ Kg}
 \end{aligned}$$

Contact Load on Gasket Surfaces [Hp]:

$$\begin{aligned}
 &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{part} \cdot m_{part} \cdot P \\
 &= 2 \cdot 6.4246 \cdot \pi \cdot 1105.1509 \cdot 3.0 \cdot -103.0 \\
 &\quad + 2 \cdot 1050.0 \cdot 5.0 \cdot 3.0 \cdot -103.0 \\
 &= -1736.583 \text{ Kg}
 \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned}
 &= \pi / 4 \cdot B_{cor}^2 \cdot P \\
 &= \pi / 4 \cdot 1060.0^2 \cdot -103.0 \\
 &= -9269.054 \text{ Kg}
 \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned}
 &= H - H_d \\
 &= -10076 - -9269 \\
 &= -806.450 \text{ Kg}
 \end{aligned}$$

Operating Bolt Load [Wm1]:

$$\begin{aligned}
 &= \max(H + H_p + H'p, 0) \\
 &= \max(-10076 + -1737 + 0, 0) \\
 &= 0.000 \text{ Kg}
 \end{aligned}$$

Gasket Seating Bolt Load [Wm2]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{part} \cdot b_{part} \cdot l_p \\
 &= 68.95 \cdot 6.4246 \cdot \pi \cdot 1105.151 + 68.95 \cdot 5.0 \cdot 1050.0
 \end{aligned}$$



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 Flg Calc [Ext P]: New Flange

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$$= 193738.562 \text{ Kgf}$$

Required Bolt Area [Am]:
 $= \max(Wm1/Sb, Wm2/Sa)$
 $= \max(0/172, 193739/172)$
 $= 110.465 \text{ cm}^2$

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:
 $= 2a + 6t/(m + 0.5)$
 $= 2 \cdot 47.625 + 6 \cdot 204.0/(3.0 + 0.5)$
 $= 444.964 \text{ mm}$

Actual Circumferential Bolt Spacing [Bs]:
 $= C \cdot \sin(\pi / n)$
 $= 1270.0 \cdot \sin(\pi/36)$
 $= 110.688 \text{ mm}$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:
 $= \max(\sqrt{ Bs/(2a + t) }, 1)$
 $= \max(\sqrt{ 110.688/(2 \cdot 47.625 + 204.0) }, 1)$
 $= 1.0000$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	110.465	535.122	
Radial Distance between Hub and Bolts:	60.325	69.000	
Radial Distance between Bolts and Edge:	47.625	55.000	
Circ. Spacing between the Bolts:	101.600	110.688	444.964

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
 $= Ab \cdot Sa/(y \cdot \pi(Go + Gi))$
 $= 535.122 \cdot 172.0/(68.95 \cdot \pi(1118.0 + 1092.0))$
 $= 19.226 \text{ mm}$ [Note: Exceeds actual gasket width, 13.000]

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :
 = 32.000 mm [Note: Exceeds actual gasket width, 13.000]

Flange Design Bolt Load, Gasket Seating [W]:
 $= Sa \cdot Ab$
 $= 172.0 \cdot 535.1215$
 $= 938521.75 \text{ Kgf}$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (1270.0 - 1105.1509) / 2$
 $= 82.4246 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= (R + g1 + hg) / 2$
 $= (69.0 + 36.0 + 82.4246) / 2$
 $= 93.7123 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= R + (g1 / 2)$
 $= 69.0 + (36.0/2.0)$



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 Flg Calc [Ext P]: New Flange

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= 87.0000 mm

Summary of Moments for External Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	-9269.	4.5754	1.0000	-42.
Face Pressure, Mt	-806.	11.2877	1.0000	-9.
Gasket Seating, Matm	938522.	82.4246	1.0000	77359.

Total Moment for Operation, Mop -52. Kg-m.
 Total Moment for Gasket seating, Matm 77359. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 121.820 mm
 Hub Ratio, h/h0 = HL / H0 0.304
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 2.571

Flange Factors for Integral Flange:

Factor F 0.868
 Factor V 0.262
 Factor f 3.427
 Factors from Figure 2-7.1 K = 1.302
 T = 1.797 U = 8.249
 Y = 7.506 Z = 3.878
 e = 0.0071 1/mm d = 0.75054E+06 mm³
 Stress Factors Alpha = 2.454
 Beta = 2.939 Gamma = 1.366
 Delta = 11.311 Lambda = 12.678

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
 = (3.4269•-52/1060.0)/(12.6775•36.0²)
 = 0.10 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)
 = (3.4269•77359/1060.0)/(12.6775•36.0²)
 = 149.27 N./mm²

Radial Flange Stress, Operating [SRo]:

= (β • Mop / Bcor) / (L • t²)
 = (2.939•-52/1060.0)/(12.6775•204.0²)
 = 0.00 N./mm²

Radial Flange Stress, Seating [SRa]:

= (β • Matm/Bcor) / (L • t²)
 = (2.939•77359/1060.0)/(12.6775•204.0²)
 = 3.99 N./mm²

Tangential Flange Stress, Operating [STo]:

= (Y • Mo / (t² • Bcor)) - Z • SRo
 = (7.5062•-52/(204.0²•1060.0))+3.8781•0.0
 = 0.08 N./mm²

Tangential Flange Stress, Seating [STa]:

= (y • Matm / (t² • Bcor)) - Z • SRa
 = (7.5062•77359/(204.0²•1060.0))-3.8781•4
 = 113.63 N./mm²

Average Flange Stress, Operating [SAo]:



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$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (0.1 + \max(0.0, 0.1)) / 2$$

$$= 0.09 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (149 + \max(4, 114)) / 2$$

$$= 131.45 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 0 / 535.1215$$

$$= 0.00 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (193739 / 535.1215)$$

$$= 35.51 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	0.10	207.00	149.27	207.00
Radial Flange	0.00	138.00	3.99	138.00
Tangential Flange	0.08	138.00	113.63	138.00
Maximum Average	0.09	138.00	131.45	138.00
Bolting	0.00	172.00	35.51	172.00

Minimum Required Flange Thickness 199.746 mm
 Stress Check: OK for External Pressure -103.0 KPa.
 Estimated Finished Weight of Flange at given Thk. 994.0 kg.
 Estimated Unfinished Weight of Forging at given Thk 1145.5 kg.

Flange Rigidity Based on Required Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 77358.8 / 1.0 \cdot 9806.637 \cdot 0.262 / (11.964 \cdot 202713 \cdot 14.0^2 \cdot 121.82 \cdot 0.3)$$

$$= 0.597 \text{ (should be } \leq 1)$$

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 51.5 / 1.0 \cdot 9806.637 \cdot 0.262 / (11.964 \cdot 199226 \cdot 14.0^2 \cdot 121.82 \cdot 0.3)$$

$$= 0.000 \text{ (should be } \leq 1)$$

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 77358.8 / 1.0 \cdot 9806.637 \cdot 0.262 / (12.678 \cdot 202713 \cdot 14.0^2 \cdot 121.82 \cdot 0.3)$$

$$= 0.563 \text{ (should be } \leq 1)$$

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 51.5 / 1.0 \cdot 9806.637 \cdot 0.262 / (12.678 \cdot 199226 \cdot 14.0^2 \cdot 121.82 \cdot 0.3)$$

$$= 0.000 \text{ (should be } \leq 1)$$



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 Flg Calc [Ext P]: Shell Flange

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Flange Input Data Values

Description: Shell Flange :

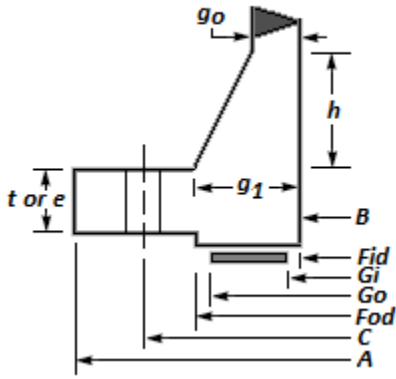
Shell Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	-103.00	KPa.
Design Temperature		200	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1056.000	mm
Flange Outside Diameter	A	1380.000	mm
Flange Thickness	t	186.0000	mm
Thickness of Hub at Small End	go	26.0000	mm
Thickness of Hub at Large End	gl	36.0000	mm
Length of Hub	h	40.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1270.000	mm
Nominal Bolt Diameter	a	47.6250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	1143.000	mm
Flange Face Inside Diameter	Fid	1056.000	mm
Flange Facing Sketch	1, Code Sketch 1a		
Gasket Outside Diameter	Go	1118.000	mm
Gasket Inside Diameter	Gi	1092.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Column for Gasket Seating	2, Code Column II		
Gasket Thickness	tg	6.3500	mm
Length of Partition Gasket	lp	1050.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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 Flg Calc [Ext P]: Shell Flange

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Bolted Flange Calculation Results:

Corroded Flange ID,	$B_{cor} = B + 2 \cdot c_i$	1056.000	mm
Corroded Large Hub,	$g1_{cor} = g1 - c_i$	36.000	mm
Corroded Small Hub,	$g0_{cor} = g0 - c_i$	26.000	mm
Code R Dimension,	$R = (C - B) / 2 - g1$	71.000	mm
Gasket Contact Width,	$N = (G_o - G_i) / 2$	13.000	mm
Basic Gasket Width,	$b_o = N / 2$	6.500	mm
Effective Gasket Width,	$b = C_b \cdot \sqrt{b_o}$	6.425	mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	1105.151	mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$\begin{aligned}
 &= \pi / 4 \cdot G^2 \cdot P_{eq} \\
 &= \pi / 4 \cdot 1105.1509^2 \cdot -103.0 \\
 &= -10075.504 \text{ Kg}
 \end{aligned}$$

Contact Load on Gasket Surfaces [Hp]:

$$\begin{aligned}
 &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P \\
 &= 2 \cdot 6.4246 \cdot \pi \cdot 1105.1509 \cdot 3.0 \cdot -103.0 \\
 &\quad + 2 \cdot 1050.0 \cdot 5.0 \cdot 3.0 \cdot -103.0 \\
 &= -1736.583 \text{ Kg}
 \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned}
 &= \pi / 4 \cdot B_{cor}^2 \cdot P \\
 &= \pi / 4 \cdot 1056.0^2 \cdot -103.0 \\
 &= -9199.230 \text{ Kg}
 \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned}
 &= H - H_d \\
 &= -10076 - -9199 \\
 &= -876.274 \text{ Kg}
 \end{aligned}$$

Operating Bolt Load [Wm1]:

$$\begin{aligned}
 &= \max(H + H_p + H'p, 0) \\
 &= \max(-10076 + -1737 + 0, 0) \\
 &= 0.000 \text{ Kg}
 \end{aligned}$$

Gasket Seating Bolt Load [Wm2]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 68.95 \cdot 6.4246 \cdot \pi \cdot 1105.151 + 68.95 \cdot 5.0 \cdot 1050.0
 \end{aligned}$$



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 Flg Calc [Ext P]: Shell Flange

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= 193738.562 Kgf

Required Bolt Area [Am]:
 = max(Wm1/Sb, Wm2/Sa)
 = max(0/172, 193739/172)
 = 110.465 cm²

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:
 = 2a + 6t/(m + 0.5)
 = 2 • 47.625 + 6 • 186.0/(3.0 + 0.5)
 = 414.107 mm

Actual Circumferential Bolt Spacing [Bs]:
 = C • sin(π / n)
 = 1270.0 • sin(π/36)
 = 110.688 mm

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:
 = max(√(Bs/(2a + t)), 1)
 = max(√(110.688/(2 • 47.625 + 186.0)), 1)
 = 1.0000

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	110.465	535.122	
Radial Distance between Hub and Bolts:	60.325	71.000	
Radial Distance between Bolts and Edge:	47.625	55.000	
Circ. Spacing between the Bolts:	101.600	110.688	414.107

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
 = Ab • Sa/(y • π(Go + Gi))
 = 535.122 • 172.0/(68.95 • π(1118.0 + 1092.0))
 = 19.226 mm[Note: Exceeds actual gasket width, 13.000]

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :
 = 32.000 mm[Note: Exceeds actual gasket width, 13.000]

Flange Design Bolt Load, Gasket Seating [W]:
 = Sa • Ab
 = 172.0 • 535.1215
 = 938521.75 Kgf

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 = (C - G) / 2
 = (1270.0 - 1105.1509)/2
 = 82.4246 mm

Distance to Face Pressure Reaction [ht]:
 = (R + g1 + hg) / 2
 = (71.0 + 36.0 + 82.4246)/2
 = 94.7123 mm

Distance to End Pressure Reaction [hd]:
 = R + (g1 / 2)
 = 71.0 + (36.0/2.0)



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= 89.0000 mm

Summary of Moments for External Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	-9199.	6.5754	1.0000	-60.
Face Pressure, Mt	-876.	12.2877	1.0000	-11.
Gasket Seating, Matm	938522.	82.4246	1.0000	77359.

Total Moment for Operation, Mop -71. Kg-m.
 Total Moment for Gasket seating, Matm 77359. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 165.699 mm
 Hub Ratio, h/h0 = HL / H0 0.241
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.385

Flange Factors for Integral Flange:

Factor F 0.891
 Factor V 0.417
 Factor f 1.136
 Factors from Figure 2-7.1 K = 1.307
 T = 1.794 U = 8.136
 Y = 7.404 Z = 3.826
 e = 0.0054 1/mm d = 0.21861E+07 mm³
 Stress Factors Alpha = 2.000
 Beta = 2.333 Gamma = 1.114
 Delta = 2.943 Lambda = 4.058

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
 = (1.136•71/1056.0)/(4.0579•36.0²)
 = 0.14 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)
 = (1.136•77359/1056.0)/(4.0579•36.0²)
 = 155.18 N./mm²

Radial Flange Stress, Operating [SRo]:

= (β • Mop / Bcor) / (L • t²)
 = (2.333•71/1056.0)/(4.0579•186.0²)
 = 0.01 N./mm²

Radial Flange Stress, Seating [SRa]:

= (β • Matm/Bcor) / (L • t²)
 = (2.333•77359/1056.0)/(4.0579•186.0²)
 = 11.94 N./mm²

Tangential Flange Stress, Operating [STo]:

= (Y • Mo / (t² • Bcor)) - Z • SRo
 = (7.4036•71/(186.0²•1056.0))+3.8258•0.0
 = 0.10 N./mm²

Tangential Flange Stress, Seating [STa]:

= (y • Matm / (t² • Bcor)) - Z • SRA
 = (7.4036•77359/(186.0²•1056.0))-3.8258•12
 = 108.06 N./mm²

Average Flange Stress, Operating [SAo]:



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$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (0.1 + \max(0.0, 0.1)) / 2$$

$$= 0.12 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (155 + \max(12, 108)) / 2$$

$$= 131.62 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 0 / 535.1215$$

$$= 0.00 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (193739 / 535.1215)$$

$$= 35.51 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	0.14	207.00	155.18	207.00
Radial Flange	0.01	138.00	11.94	138.00
Tangential Flange	0.10	138.00	108.06	138.00
Maximum Average	0.12	138.00	131.62	138.00
Bolting	0.00	172.00	35.51	172.00

Minimum Required Flange Thickness 181.356 mm
 Stress Check: OK for External Pressure -103.0 KPa.
 Estimated Finished Weight of Flange at given Thk. 926.4 kg.
 Estimated Unfinished Weight of Forging at given Thk 1085.8 kg.

Flange Rigidity Based on Required Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 77358.8 / 1.0 \cdot 9806.637 \cdot 0.417 / (3.829 \cdot 201334 \cdot 26.0^2 \cdot 165.699 \cdot 0.3)$$

$$= 0.636 \text{ (should be } \leq 1)$$

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 71.3 / 1.0 \cdot 9806.637 \cdot 0.417 / (3.829 \cdot 191213 \cdot 26.0^2 \cdot 165.699 \cdot 0.3)$$

$$= 0.000 \text{ (should be } \leq 1)$$

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 77358.8 / 1.0 \cdot 9806.637 \cdot 0.417 / (4.058 \cdot 201334 \cdot 26.0^2 \cdot 165.699 \cdot 0.3)$$

$$= 0.600 \text{ (should be } \leq 1)$$

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 71.3 / 1.0 \cdot 9806.637 \cdot 0.417 / (4.058 \cdot 191213 \cdot 26.0^2 \cdot 165.699 \cdot 0.3)$$

$$= 0.000 \text{ (should be } \leq 1)$$



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 Element and Detail Weights:

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	168.104	200063	129.68	203439	16.8104
20	30	292.889	706105	226.617	714658	29.2889
30	40	1032.05	219775	1032.05	219775	103.205
40	50	964.879	204980	964.879	204980	96.4879
50	60	5519.33	4027503	5134.46	4077170	551.934
60	70	427.156	203227	402.029	205355	42.7156
Total		8404	5561655.00	7889	5625377.50	840



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	199.941	-88.3333	...	...	Liquid: 10
20	Liqd	705.674	400	...	...	Liquid: 20
20	Nozl	55.1103	400	666.525	...	Tube inlet
20	Nozl	55.1103	400	-666.525	...	Tube outlet
20	Wght	100	400	...	...	Partition plate
40	Insl	15.4382	57.5	...	...	Insulation
40	Lini	18.4853	118.5	...	...	SS 304L Clad
50	Sadl	273.607	1500	747.5	...	Left Saddle
50	Sadl	273.607	5900	747.5	...	New Saddle
50	Liqd	4025.05	3700	...	...	Liquid: 40
50	Insl	480.099	3700	...	...	Ins: 40
50	Lini	582.658	3700	...	...	SS 304L Claddin
50	Nozl	230.13	6500	-669.525	...	Shell outlet
50	Nozl	230.13	1000	669.525	...	Shell Inlet
50	Nozl	13.3874	2000	552.088	...	Vent
50	Nozl	13.3874	3000	-552.088	...	Drain
50	Wght	250	1500	...	-1000	Fire proofing
50	Wght	250	5900	...	-1000	Fire proofing
60	Liqd	203.05	138.833	5	...	Liquid: 50
60	Insl	39.2125	156.25	...	...	Ins: 40
60	Lini	35.8233	157	...	...	SS304L cladding
30	FTsh	2792.83	385.85	...	...	Tubesheet
30	Tube	9256.26	3798.35	...	...	Tubes
40	Bafl	50.9	327	...	...	Baffles
40	Bafl	50.9	654	...	...	Baffles
40	Bafl	50.9	981	...	...	Baffles
40	Bafl	50.9	1308	...	...	Baffles
40	Bafl	50.9	1635	...	...	Baffles
40	Bafl	50.9	1962	...	...	Baffles
40	Bafl	50.9	2289	...	...	Baffles
40	Bafl	50.9	2616	...	...	Baffles
40	Bafl	50.9	2943	...	...	Baffles
40	Bafl	50.9	3270	...	...	Baffles



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 Element and Detail Weights:

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40	Bafl	50.9	3597	...	...	Baffles
40	Bafl	50.9	3924	...	...	Baffles
40	Bafl	50.9	4251	...	...	Baffles
40	Bafl	50.9	4578	...	...	Baffles
40	Bafl	50.9	4905	...	...	Baffles
40	Bafl	50.9	5232	...	...	Baffles
40	Bafl	50.9	5559	...	...	Baffles
40	Bafl	50.9	5886	...	...	Baffles
40	Bafl	50.9	6213	...	...	Baffles
40	Bafl	51.8979	6540	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	547.2
Liquid	5133.7
Insulation	534.7
Cladding & Overlay	637.0
Nozzles	597.3
Weights	600.0
Exchanger Components	13068.1
Liquid in Tubes	1681.0

Sum of the Detail Weights 22799.0 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	9244.9	9244.9	9244.9	9244.9	9244.9	9244.9
Saddles	547.2	547.2	547.2	547.2	547.2	547.2
Cladding	637.0	637.0	637.0	637.0	637.0	637.0
Nozzles	597.3	597.3	597.3	597.3	597.3	597.3
Wld Weights	100.0	100.0	100.0	100.0	100.0	100.0
Exchanger	13068.1	13068.1	13068.1	13068.1	13068.1	13068.1
Insulation	...	...	...	...	534.7	534.7
Empty Weights	...	...	...	500.0	500.0	...
Ope Weights	...	...	...	...	...	500.0
Ope. Liquid	...	...	...	...	...	5133.7
Tube Ope Lqd	...	...	...	...	...	1681.0
Test Liquid	...	5477.2	...	...	...	...
Tube Tst Lqd	...	1681.0	...	...	...	...
Totals	24194.4	31352.6	24194.4	24694.4	25229.1	32043.8

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare Weight without Removable Internals	24194.4 kg.
Shop Test Wt.	- Fabricated Weight + Water (Full)	31352.6 kg.
Shipping Wt.	- Fab. Weight + removable Intls.+ Shipping App.	24194.4 kg.
Erected Wt.	- Fab. Wt + or - loose items (trays,platforms etc.)	24694.4 kg.



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Element and Detail Weights:

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Ope. Wt. no Liq - Fab. Weight + Internals. + Details + Weights	25229.1 kg.
Operating Wt. - Empty Weight + Operating Liq. Uncorroded	32043.8 kg.
Oper. Wt. + CA - Corr Wt. + Operating Liquid	31477.6 kg.
Field Test Wt. - Empty Weight + Water (Full)	31887.4 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	1682002.2 cm ³
Weight of Ope Liq in tubes :	1681.0 kg.
Weight of Water in tubes :	1681.0 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	14540.6	13844.7
20	30	27344.4	26640.7
30	40	16639.3	8291.92
40	50	16027	7763
50	60	260840	247821
60	70	15720.9	13992.4

Total	351112.4	318354.2 cm ²
Total	377.9	342.7 ft ²

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Nozzle Flange MAWP:

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Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Design Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44 (b)	50%	DNV
Tube inlet	1402.0	1590.0	82	150	GR 2.3	2539.5	544.9	1402.0	-436.5
Tube outlet	1402.0	1590.0	82	150	GR 2.3	2539.5	544.9	1402.0	-436.5
Shell outlet	8175.0	10210.0	275	600	GR 1.1	2700.7	8175.0	8175.0	8175.0
Shell Inlet	8175.0	10210.0	275	600	GR 1.1	2700.7	8175.0	8175.0	8175.0
Vent	8175.0	10210.0	275	600	GR 1.1	...	...	...	...
Drain	8175.0	10210.0	275	600	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope) [ShellSide]: 8175.003 KPa.
Lowest Flange Pressure Rating was (Amb) [ShellSide]: 10210.003 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope) [TubeSide]: 1402.006 KPa.
Lowest Flange Pressure Rating was (Amb) [TubeSide]: 1590.006 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: None Selected

It appears that the ASME UG-44(b) method of pressure de-rating was not selected. If external loads (forces and bending moments) are also required to be evaluated on standard flanged joints with welding neck flanges as mentioned in UG-44(a)(2), (a)(9) and (a)(10) for vessels constructed per the ASME VIII-1 2019 Code Edition onward, they may be evaluated per the requirements specified in UG-44(b).

You can set the year of construction in Tools->Enter U1 Form Information.

Pressure Ratings are per ASME B16.5 2017 Metric Edition

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 FileName : 174-C_BEU_Dia 1000-AK
 Wind Load Calculation:

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	97.2	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm
Percent Wind for Hydrotect	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment H or Hh	0	mm
Distance Upwind of Crest Lh	0	mm
Distance from Crest to the Vessel x	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.983) / (1 + 1.7 \cdot 3.4 \cdot 0.228)))$
 $= \min(0.85, 0.916)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz
User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.606
Structure Height to Diameter ratio	8.131

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Z_g = 274320.0$ mm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1000.0 + 300.0 = 1300.0$ mm
 $= 4.265$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (4.265 ft.) < 15 ft.
 $= 2.01(15 / Z_g)^{2/\alpha}$



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$$= 2.01(15/900.0)^{2/9.5}$$

$$= 0.849$$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
 = 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:
 $= (1 + K1 \cdot K2 \cdot K3)^2$
 $= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$
 = 1.0

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 $= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 $= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$
 = 16.0 psf [78.12] Kgs/m²

Force on the first element [F]:
 $= qz \cdot G \cdot Cf \cdot \text{WindArea}$
 $= 16.0 \cdot 0.85 \cdot 0.606 \cdot 3.166$
 = 26.1 lbs. [11.8] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Channel head	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
Channel shell	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS Ch Flange	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell Flange	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell head	1300.0	0.000	0.000	0.000	0.849	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Partition pla	0.00	1300.00	78.12	0.00
Fire proofing	0.00	1300.00	78.12	0.00
Fire proofing	0.00	1300.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1300	1289.28	2941.55	78.12	13.0265
20	30	1300	1305.6	10444.8	78.12	46.2541
30	40	1300	1272	3065.52	78.12	13.5754
40	50	1300	1328.26	3001.87	78.12	13.2936
50	60	1300	1466.4	10851.4	78.12	480.545
60	70	1300	1462.8	4015.28	78.12	17.7814



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Wind Load Calculation:

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Note:

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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 Earthquake Load Calculation:

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\
 SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(5.02950.75)$
 = 0.067 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.067$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.157 \cdot 32044 \cdot 1.25$
 = 1884.177 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.157 \cdot 32044$
 = 1004.9 Kgf

Earthquake Load Calculation:



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Earthquake Load Calculation:

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From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	530	4005.48	234.968
20	30	530	4005.48	234.968
30	40	530	4005.48	234.968
40	50	528	4005.48	234.081
50	Sadl	533	4005.48	236.298
Sadl	60	533	4005.48	236.298
50	60	533	4005.48	236.298
60	70	533	4005.48	236.298

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Center of Gravity Calculation:

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Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.
Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5304.700 mm
Center of Gravity of the Liquid	4579.171 mm
Center of Gravity of the Insulation	5475.790 mm
Center of Gravity of the Lining	5411.123 mm
Center of Gravity of the Nozzles	4393.522 mm
Center of Gravity of the Added Weights (Operating)	4495.583 mm
Center of Gravity of the Added Weights (Empty)	4495.583 mm
Center of Gravity of the Tubesheet(s)	1245.375 mm
Center of Gravity of the Tubes	4657.875 mm
Center of Gravity of the Baffles	4815.243 mm
Center of Gravity of Bare Shell New and Cold	4254.425 mm
Center of Gravity of Bare Shell Corroded	4240.785 mm
Vessel CG in the Operating Condition	4264.638 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4150.814 mm
Vessel CG in the Test Condition	4207.479 mm

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 Saddle Calcs: Bundle Pull New:

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, New Condition

Input and Calculated Values:

Vessel Mean Radius	Rm	547.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	7676.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1500.00	mm
Saddle Width	b1 or b	405.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		13100.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	
Saddle Force Q, + Bundle Pull, Force		18549.61	Kgf
Pressure used in Analysis	P	0.000	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-3.03	-234.01	
Long. Stress at Top of Midspan	3.03	138.00	
Long. Stress at Bottom of Midspan	3.03	138.00	
Long. Stress at Top of Saddles	17.74	138.00	
Long. Stress at Bottom of Saddles	-9.83	-234.01	
Long. Stress at Bottom of Saddles	9.83	138.00	
Tangential Shear in Shell	8.47	110.40	
Circ. Stress at Horn of Saddle	11.99	207.00	
Circ. Compressive Stress in Shell	0.61	138.00	

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 13100 • 1.5 • 1000.0/4400.0
 = 4465.9 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)



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 Saddle Calcs: Bundle Pull New:

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= 14084 + max(4466, 0, 0, 0)
 = 18549.6 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	18823.21	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	19650.00	Kgf

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(1066.0/2 + 28.0 + 10.0)sin(120.0/2), 2•547.0)
 = 989.001 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm²)/3L)]
 = -18550•1500.0[1-(1-1500.0/7676.0+(547.0²-0.0²)/
 (2•1500.0•7676.0))/(1+(4•0.0)/(3•7676.0))]
 = -5075.9 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4(1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L
 = 18550•7676/4(1+2(547²-0²)/(7676²))/(1+(4•0)/
 (3•7676))-4•1500/7676
 = 8134.0 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)
 = 0.0 • 547.0/(2•28.0) - 8134.0/(π •547.0²•28.0)
 = -3.03 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 0.0 • 547.0/(2 • 28.0) + 8134.0/(π • 547.0² • 28.0)
 = 3.03 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 0.0•547.0/(2•28.0)-5075.9/(0.1066• π •547.0²•28.0)
 = 17.74 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 0.0•547.0/(2•28.0)+5075.9/(0.1923• π •547.0²•28.0)
 = -9.83 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:



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 Saddle Calcs: Bundle Pull New:

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$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 18550(7676.0 - 2 \cdot 1500.0)/(7676.0 + (4 \cdot 0.0/3))$$

$$= 11299.9 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ 2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 11299.89 / (547.0 \cdot 28.0)$$

$$= 8.47 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(547.0 \cdot 28.0)}$$

$$= 96.531 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(405.0 + 1.56 \cdot \sqrt{(547.0 \cdot 28.0)}, 2 \cdot 1500.0)$$

$$= 598.06 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, bundle pull case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 18550 \cdot 0.1 / (598.062(28.0 + 1.0 \cdot 10.0))$$

$$= -0.61 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q/(4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q/(2(t + \eta \cdot tr)^2)$$

$$= -18550/(4(28.0 + 1.0 \cdot 10.0)598.062) -$$

$$3 \cdot 0.053 \cdot 18550/(2(28.0 + 1.0 \cdot 10.0)^2)$$

$$= -11.99 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	430.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw, c	480.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(533.0 \cdot 28.0)} + 600.0)28.0^3)/12$$

$$= 144.623 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 144.623 + 221.361(148.24 - 14.0)^2$$

$$= 40034.879 \text{ cm}^4$$

B	D	Y	A	AY	I + AD ⁽²⁾
mm	mm	mm	cm ²	mm ³	cm ⁴



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Shell	790.6	28.0	14.0	221.4	309905.6	0.400E+05
Wearplate	600.0	10.0	33.0	60.0	198000.0	0.797E+04
Web	10.0	404.0	240.0	40.4	969599.8	0.890E+04
BasePlate	430.0	25.0	454.5	107.5	4885874.5	0.101E+06
Totals	...	...	...	429.3	6363380.0	0.158E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6363380 / 429.261$$

$$= 148.240 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 18549.605$$

$$= 3775.2468 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.7808 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000.0 - 533.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 559.213 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2111.2070 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.9451 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(18550 + 274)430.0 / (4 \cdot 1000.0 \cdot 174.663)}$$

$$= 18.462 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (533.0 + 28.0 + 10.0)$$

$$= 989.001 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 989.001 / (3 - 1)$$

$$= 494.501 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 494.5005 \cdot 430.0/2$$

$$= 1063.176 \text{ cm}^2$$



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Bearing Pressure [Bp]:
 $= Q / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 18549.605 / (1000.0 \cdot 430.0)$
 $= 4.314 \text{ Kgf/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 1063.2 \cdot 4.31$
 $= 4586.395 \text{ Kgf}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 39.5 + 24.725$
 $= 64.225 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 4586.4 / 64.225$
 $= 7.003 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	I _o
Rib+Web	10.0	405.0	...	40.5	...	0.554E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 405.0 - 10.0$
 $= 395.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0 / 64.225$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 405.0 / 2$
 $= 202.500 \text{ mm}$

Radius of Gyration [r]:
 $= \sqrt{ \text{Total Inertia} / \text{Total Area} }$
 $= \sqrt{ 5535.8 / 64.225 }$
 $= 92.841 \text{ mm}$

Length of Outer Rib [L]:
 $= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 1000.0 - \cos(120.0 / 2) (533.0 + 28.0 + 10.0) - 25.0$
 $= 689.500 \text{ mm}$

Intermediate Term [Cc]:
 $= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} }$
 $= \sqrt{ 2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 262.0 }$
 $= 122.739$

Slenderness ratio [KL/r]:



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$$= KL/r$$

$$= 1 \cdot 689.5/92.841$$

$$= 7.427$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 19650.0 / (2 \cdot 1000.0) \cdot 494.501 \cdot 689.5 / 2$$

$$= 1674.990 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (7.4267 < 122.7394) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.43)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (7.43) / (8 \cdot 122.74) - (7.43^3) / (8 \cdot 122.74^3))$$

$$= 154.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 7.0/154.8 + (1674.99 \cdot 202.5/55358440) / 174.66$$

$$= 0.389$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	I _o
Rib	10.0	395.0	0.0	39.5	0.0	0.554E+04
Web	494.5	10.0	0.0	49.5	0.0	4.12
Totals	...	...	...	89.0	...	0.554E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/88.95$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 405.0/2$$

$$= 202.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1000.0 - \cos(571.0/2) (25.0 + 0.0 + 0.0) - 0.0$$

$$= 404.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (5539.9/88.95) }$$

$$= 78.918 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 404.0/78.918$$

$$= 5.119$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 19650.0 / (2 \cdot 1000.0)$$

$$= 9.825 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:



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$$= \text{Unit Force} \cdot e \cdot L$$

$$= 9.825 \cdot 494.501 \cdot 404.0$$

$$= 1962.860 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1962.86 / 273574.375$$

$$= 70.361 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (5.1192 < 122.7394) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.12)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (5.12) / (8 \cdot 122.74) - (5.12)^3 / (8 \cdot 122.74^3))$$

$$= 155.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 10.12 / 155.6 + (1962.86 \cdot 202.5 / 5539.876) / 174.66$$

$$= 0.468$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-307 A	
Bolt Allowable Stress	Stba	138.00	N./mm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm
User Defined Area in a Single Bolt	UserBoltArea	8.6100	cm ²
BasePlate Allowable Stress	S	138.00	N./mm ²
Area Available in a Single Bolt	BlArea	7.1203	cm ²
Saddle Load QO (Weight)	QO	18823.2	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	19650.0	Kgf
Maximum Transverse Force	Ft	0.0	Kgf
Maximum Longitudinal Force	Fl	19650.0	Kgf
Saddle Bolted to Steel Foundation	No		

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 19650 / (7.12 \cdot 4)$$

$$= 67.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0 / (7.12 \cdot 4)$$

$$= 0.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= (Q_L - Q_O) / (Stba \cdot Rnbolts)$$

$$= (19650.0 - 18823.21) / (138.0 \cdot 4.0)$$

$$= 0.1469 \text{ cm}^2$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 19650.0 / (82.8 \cdot 4.0)$$

$$= 5.8184 \text{ cm}^2$$

Bolt Area due to Transverse Load:



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Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$

= $1000.0 \cdot 0.0 + 0.0$

= 0.00 Kg-m.

Eccentricity (e):

= $Rmom / QO$

= $0.0 / 18823.21$

= 0.00 mm < $B_{plen}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$

= $\max[0.1469, 5.8184, 0.0]$

= 5.8184 cm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, Corroded Condition

Input and Calculated Values:

Vessel Mean Radius	Rm	548.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	7676.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1500.00	mm
Saddle Width	b1 or b	405.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		13100.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	
Saddle Force Q, + Bundle Pull, Force		18266.52	Kgf
Pressure used in Analysis	P	0.000	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-3.20	-234.01
Long. Stress at Top of Midspan	3.20	138.00
Long. Stress at Bottom of Midspan	3.20	138.00
Long. Stress at Top of Saddles	18.74	138.00
Long. Stress at Bottom of Saddles	-10.39	-234.01
Long. Stress at Bottom of Saddles	10.39	138.00
Tangential Shear in Shell	8.97	110.40
Circ. Stress at Horn of Saddle	13.06	207.00
Circ. Compressive Stress in Shell	0.64	138.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 13100 • 1.5 • 1000.0/4400.0
 = 4465.9 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)



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= 13801 + max(4466, 0, 0, 0)
 = 18266.5 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	18540.13	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	19650.00	Kgf

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(1066.0/2 + 28.0 + 10.0)sin(120.0/2), 2•548.0)
 = 989.001 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm²)/3L)]
 = -18267•1500.0[1-(1-1500.0/7676.0+(548.0²-0.0²)/
 (2•1500.0•7676.0))/(1+(4•0.0)/(3•7676.0))]
 = -4997.1 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4(1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L
 = 18267•7676/4(1+2(548²-0²)/(7676²))/(1+(4•0)/
 (3•7676))-4•1500/7676
 = 8011.1 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)
 = 0.0 • 548.0/(2•26.0) - 8011.1/(π •548.0²•26.0)
 = -3.20 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 0.0 • 548.0/(2 • 26.0) + 8011.1/(π • 548.0² • 26.0)
 = 3.20 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 0.0•548.0/(2•26.0)-4997.1/(0.1066• π •548.0²•26.0)
 = 18.74 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 0.0•548.0/(2•26.0)+4997.1/(0.1923• π •548.0²•26.0)
 = -10.39 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:



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$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 18267(7676.0 - 2 \cdot 1500.0)/(7676.0 + (4 \cdot 0.0/3))$$

$$= 11127.4 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ 2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 11127.44 / (548.0 \cdot 26.0)$$

$$= 8.97 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(548.0 \cdot 26.0)}$$

$$= 93.105 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(405.0 + 1.56 \cdot \sqrt{(548.0 \cdot 26.0)}, 2 \cdot 1500.0)$$

$$= 591.21 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, bundle pull case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 18267 \cdot 0.1 / (591.209(26.0 + 1.0 \cdot 10.0))$$

$$= -0.64 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q/(4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q/(2(t + \eta \cdot tr)^2)$$

$$= -18267/(4(26.0 + 1.0 \cdot 10.0)591.209) -$$

$$3 \cdot 0.053 \cdot 18267/(2(26.0 + 1.0 \cdot 10.0)^2)$$

$$= -13.06 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	430.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw, c	480.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(535.0 \cdot 26.0)} + 600.0)26.0^3)/12$$

$$= 114.828 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 114.828 + 203.837(152.449 - 13.0)^2$$

$$= 39752.898 \text{ cm}^4$$

B	D	Y	A	AY	I + AD ⁽²⁾
mm	mm	mm	cm ²	mm ³	cm ⁴



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Shell	784.0	26.0	13.0	203.8	264987.8	0.398E+05
Wearplate	600.0	10.0	31.0	60.0	186000.0	0.885E+04
Web	10.0	404.0	238.0	40.4	961520.0	0.845E+04
BasePlate	430.0	25.0	452.5	107.5	4864375.5	0.968E+05
Totals	...	...	...	411.7	6276883.0	0.154E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6276883 / 411.737$$

$$= 152.449 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 18266.521$$

$$= 3717.6333 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.7536 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1000.0 - 535.0 \cdot \sin(120.0 / 2) / 1.0472$$

$$= 557.558 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2072.8389 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.0136 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(18267 + 274)430.0 / (4 \cdot 1000.0 \cdot 174.663)}$$

$$= 18.322 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0 / 2) (533.0 + 28.0 + 10.0)$$

$$= 989.001 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 989.001 / (3 - 1)$$

$$= 494.501 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 494.5005 \cdot 430.0 / 2$$

$$= 1063.176 \text{ cm}^2$$



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Bearing Pressure [Bp]:
 $= Q / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 18266.521 / (1000.0 \cdot 430.0)$
 $= 4.248 \text{ Kgf/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 1063.2 \cdot 4.25$
 $= 4516.402 \text{ Kgf}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 39.5 + 24.725$
 $= 64.225 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 4516.4 / 64.225$
 $= 6.896 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	I _o
Rib+Web	10.0	405.0	...	40.5	...	0.554E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 405.0 - 10.0$
 $= 395.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0 / 64.225$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 405.0 / 2$
 $= 202.500 \text{ mm}$

Radius of Gyration [r]:
 $= \sqrt{ \text{Total Inertia} / \text{Total Area} }$
 $= \sqrt{ 5535.8 / 64.225 }$
 $= 92.841 \text{ mm}$

Length of Outer Rib [L]:
 $= \text{Saddle Height} - \cos(\theta / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 1000.0 - \cos(120.0 / 2) (533.0 + 28.0 + 10.0) - 25.0$
 $= 689.500 \text{ mm}$

Intermediate Term [Cc]:
 $= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} }$
 $= \sqrt{ 2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 262.0 }$
 $= 122.739$

Slenderness ratio [KL/r]:



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$$= KL/r$$

$$= 1 \cdot 689.5/92.841$$

$$= 7.427$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 19650.0 / (2 \cdot 1000.0) \cdot 494.501 \cdot 689.5 / 2$$

$$= 1674.990 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (7.4267 < 122.7394) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.43)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (7.43) / (8 \cdot 122.74) - (7.43^3) / (8 \cdot 122.74^3))$$

$$= 154.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 6.9/154.8 + (1674.99 \cdot 202.5/55358440) / 174.66$$

$$= 0.389$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	I _o
Rib	10.0	395.0	0.0	39.5	0.0	0.554E+04
Web	494.5	10.0	0.0	49.5	0.0	4.12
Totals	...	...	...	89.0	...	0.554E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/88.95$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 405.0/2$$

$$= 202.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1000.0 - \cos(571.0/2) (25.0 + 0.0 + 0.0) - 0.0$$

$$= 404.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (5539.9/88.95) }$$

$$= 78.918 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 404.0/78.918$$

$$= 5.119$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 19650.0 / (2 \cdot 1000.0)$$

$$= 9.825 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:



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$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 9.825 \cdot 494.501 \cdot 404.0 \\
 &= 1962.860 \text{ Kg-m.}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 1962.86 / 273574.375 \\
 &= 70.361 \text{ N./mm}^2
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ (5.1192 < 122.7394) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (5.12)^2 / (2 \cdot 122.74^2)) 262 / \\
 &\quad (5/3 + 3 \cdot (5.12) / (8 \cdot 122.74) - (5.12^3) / (8 \cdot 122.74^3)) \\
 &= 155.6 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba} \\
 &= 9.96 / 155.6 + (1962.86 \cdot 202.5 / 5539.876) / 174.66 \\
 &= 0.467
 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-307 A	
Bolt Allowable Stress	Stba	138.00	N./mm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm
User Defined Area in a Single Bolt	UserBoltArea	8.6100	cm ²
BasePlate Allowable Stress	S	138.00	N./mm ²
Area Available in a Single Bolt	BltArea	7.1203	cm ²
Saddle Load QO (Weight)	QO	18540.1	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	19650.0	Kgf
Maximum Transverse Force	Ft	0.0	Kgf
Maximum Longitudinal Force	Fl	19650.0	Kgf
Saddle Bolted to Steel Foundation	No		

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$\begin{aligned}
 &= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 19650 / (7.12 \cdot 4) \\
 &= 67.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.
 \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$\begin{aligned}
 &= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 0 / (7.12 \cdot 4) \\
 &= 0.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.
 \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$\begin{aligned}
 &= (Q_L - Q_O) / (Stba \cdot Rnbolts) \\
 &= (19650.0 - 18540.13) / (138.0 \cdot 4.0) \\
 &= 0.1972 \text{ cm}^2
 \end{aligned}$$

Bolt Area due to Shear Load [Bltarears]:

$$\begin{aligned}
 &= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts}) \\
 &= 19650.0 / (82.8 \cdot 4.0) \\
 &= 5.8184 \text{ cm}^2
 \end{aligned}$$

Bolt Area due to Transverse Load:



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Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$

= $1000.0 \cdot 0.0 + 0.0$

= 0.00 Kg-m.

Eccentricity (e):

= $Rmom / QO$

= $0.0/18540.13$

= 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$

= $\max[0.1972, 5.8184, 0.0]$

= 5.8184 cm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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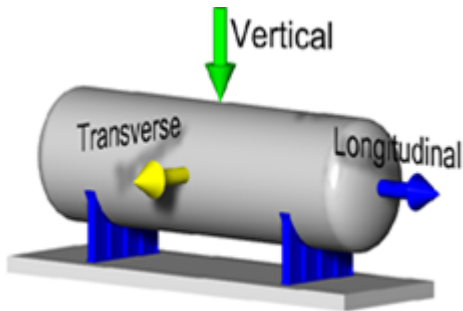
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	547.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	7676.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1500.00	mm
Saddle Width	b1 or b	405.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	
Saddle Force Q for this Transportation Case		19764.13	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	-3.23	-234.01
Long. Stress at Top	of Midspan	3.23	138.00
Long. Stress at Bottom	of Midspan	3.23	138.00
Long. Stress at Top	of Saddles	18.90	138.00
Long. Stress at Bottom	of Saddles	-10.48	-234.01
Long. Stress at Bottom	of Saddles	10.48	138.00
Tangential Shear in Shell		9.02	110.40



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Circ. Stress at Horn of Saddle	12.77	207.00
Circ. Compressive Stress in Shell	0.65	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf$$

$$= 142.06 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0$$

$$= 1206.7 \text{ lbf}$$

$$= 547.375 \text{ Kgf}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$$

$$= \pi \cdot 3.681^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \cdot 1.2$$

$$= 108.5 \text{ lbf}$$

$$= 49.2 \text{ Kgf}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$= gz \cdot \text{Saddle Load} \cdot gw + Ftw$$

$$= 0.1 \cdot 18549.61 \cdot 1.0 + 547.38$$

$$= 2402.3 \text{ Kgf}$$

Longitudinal Force due to Axial Acceleration [Fl]:

$$= gx \cdot \text{Saddle Load} \cdot gw + Flw$$

$$= 0.1 \cdot 18549.61 \cdot 1.0 + 49.21$$

$$= 1904.2 \text{ Kgf}$$

Saddle Reaction Force due to Transverse Load [Fwt]:

$$= Ft/\text{Num of Saddles} \cdot B / E$$

$$= 2402.3/2 \cdot 1000.0/989.001$$

$$= 1214.5 \text{ Kgf}$$

Saddle Reaction Force due to Longitudinal Load [Fwl]:

$$= Fl \cdot B / Ls$$

$$= 1904 \cdot 1000.0/4400.0$$

$$= 432.8 \text{ Kgf}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [Fsl]:

$$= \text{Saddle Load} \cdot gy$$

$$= 18549.6 \cdot 0.05$$

$$= 927.480 \text{ Kgf}$$

Load Combination Results due to Transport Loading [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl)$$

$$= 18550 + \max(433, 1215, 927)$$

$$= 19764.1 \text{ Kgf}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	20037.74	Kgf
Transverse Shear Load	1214.53	Kgf
Longitudinal Shear Load	1904.17	Kgf

Formulas and Substitutions for Horizontal Vessel Analysis:**Note: Wear Plate is Welded to the Shell, k = 0.1**

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(1066.0/2 + 28.0 + 10.0)\sin(120.0/2), 2 \cdot 547.0)$$



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= 989.001 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm^2)/3L)] \\
 &= -19764 \cdot 1500.0 [1 - (1 - 1500.0/7676.0 + (547.0^2 - 0.0^2) / (2 \cdot 1500.0 \cdot 7676.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 7676.0))] \\
 &= -5408.2 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L / 4 (1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L \\
 &= 19764 \cdot 7676 / 4 (1 + 2(547^2 - 0^2)/(7676^2)) / (1 + (4 \cdot 0) / (3 \cdot 7676)) - 4 \cdot 1500 / 7676 \\
 &= 8666.5 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 0.0 \cdot 547.0 / (2 \cdot 28.0) - 8666.5 / (\pi \cdot 547.0^2 \cdot 28.0) \\
 &= -3.23 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 0.0 \cdot 547.0 / (2 \cdot 28.0) + 8666.5 / (\pi \cdot 547.0^2 \cdot 28.0) \\
 &= 3.23 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 0.0 \cdot 547.0 / (2 \cdot 28.0) - 5408.2 / (0.1066 \cdot \pi \cdot 547.0^2 \cdot 28.0) \\
 &= 18.90 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1* \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 0.0 \cdot 547.0 / (2 \cdot 28.0) + 5408.2 / (0.1923 \cdot \pi \cdot 547.0^2 \cdot 28.0) \\
 &= -10.48 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 19764 (7676.0 - 2 \cdot 1500.0) / (7676.0 + (4 \cdot 0.0 / 3)) \\
 &= 12039.7 \text{ Kg}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.1707 \cdot 12039.74 / (547.0 \cdot 28.0) \\
 &= 9.02 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x1,x2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(547.0 \cdot 28.0)} \\
 &= 96.531 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:



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$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(405.0 + 1.56 \cdot \sqrt{547.0 \cdot 28.0}, 2 \cdot 1500.0)$$

$$= 598.06 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:
 = 1.0000 Materials are the same, transport case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:
 $= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$
 $= -0.7603 \cdot 19764 \cdot 0.1 / (598.062(28.0 + 1.0 \cdot 10.0))$
 $= -0.65 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:
 $= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$
 $= -19764 / (4(28.0 + 1.0 \cdot 10.0)598.062) -$
 $3 \cdot 0.053 \cdot 19764 / (2(28.0 + 1.0 \cdot 10.0)^2)$
 $= -12.77 \text{ N./mm}^2$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	430.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw, c	480.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{R \cdot t}) + \text{WearPlateWidth})t^3 / 12$
 $= ((1.56 \cdot \sqrt{533.0 \cdot 28.0}) + 600.0)28.0^3 / 12$
 $= 144.623 \text{ cm}^4$

Resolved Inertia for the Shell [I, shell]:
 $= \text{ShellInertia} + A_{\text{shell}}(C1 - Y)^2$
 $= 144.623 + 221.361(148.24 - 14.0)^2$
 $= 40034.879 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	790.6	28.0	14.0	221.4	309905.6	0.400E+05
Wearplate	600.0	10.0	33.0	60.0	198000.0	0.797E+04
Web	10.0	404.0	240.0	40.4	969599.8	0.890E+04
BasePlate	430.0	25.0	454.5	107.5	4885874.5	0.101E+06
Totals	...	...	...	429.3	6363380.0	0.158E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 6363380 / 429.261$
 $= 148.240 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 120.0/2$
 $= 120.0$



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Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 19764.131$$

$$= 4022.4294 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.8974 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000.0 - 533.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 559.213 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2249.4375 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.0724 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3 \cdot (19764 + 274) \cdot 430.0) / (4 \cdot 1000.0 \cdot 174.663)}$$

$$= 19.048 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) \cdot (533.0 + 28.0 + 10.0)$$

$$= 989.001 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 989.001 / (3 - 1)$$

$$= 494.501 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 494.5005 \cdot 430.0 / 2$$

$$= 1063.176 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= Q / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 19764.131 / (1000.0 \cdot 430.0)$$

$$= 4.596 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1063.2 \cdot 4.6$$

$$= 4886.687 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 39.5 + 24.725$$

$$= 64.225 \text{ cm}^2$$



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Compressive Stress [Sc]:

= P/A_r
 = 4886.7/64.225
 = 7.462 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	I _o
Rib+Web	10.0	405.0	...	40.5	...	0.554E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 405.0 - 10.0
 = 395.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/64.225
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 405.0/2
 = 202.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{5535.8/64.225}$
 = 92.841 mm

Length of Outer Rib [L]:

= Saddle Height - $\cos(\theta/2)$ (radius + shlthk + wpdthk) - bpthk
 = 1000.0 - $\cos(120.0/2)$ (533.0 + 28.0 + 10.0) - 25.0
 = 689.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/262.0}$
 = 122.739

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 689.5/92.841
 = 7.427

Bending Moment [Rm]:

= $Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$
 = 1904.2/(2 • 1000.0) • 494.501 • 689.5/2
 = 162.314 Kg-m.

Compressive Allowable, $KL/r < C_c$ (7.4267 < 122.7394) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2/(2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r)/(8 \cdot C_c) - (KL/r)^3/(8 \cdot C_c^3))$
 = $(1 - (7.43)^2/(2 \cdot 122.74^2))^{262} /$
 $(5/3 + 3 \cdot (7.43)/(8 \cdot 122.74) - (7.43)^3/(8 \cdot 122.74^3))$
 = 154.8 N./mm²



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AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 7.46/154.8 + (162.31 \cdot 202.5/55358440) / 174.66$$

$$= 0.082$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	I _o
Rib	10.0	395.0	0.0	39.5	0.0	0.554E+04
Web	494.5	10.0	0.0	49.5	0.0	4.12
Totals	...	...	...	89.0	...	0.554E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/88.95$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 405.0/2$$

$$= 202.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000.0 - \cos(571.0/2) (25.0 + 0.0 + 0.0) - 0.0$$

$$= 404.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(5539.9/88.95)}$$

$$= 78.918 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 404.0/78.918$$

$$= 5.119$$

Unit Force [Force,u]:

$$= F1 / (2 \cdot \text{Baseplate Length})$$

$$= 1904.167 / (2 \cdot 1000.0)$$

$$= 0.952 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.952 \cdot 494.501 \cdot 404.0$$

$$= 190.209 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σB,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 190.209/273574.375$$

$$= 6.818 \text{ N./mm}^2$$

Compressive Allowable, KL/r < Cc (5.1192 < 122.7394) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (5.12)^2 / (2 \cdot 122.74^2)) 262 / (5/3 + 3 \cdot (5.12) / (8 \cdot 122.74) - (5.12)^3 / (8 \cdot 122.74^3))$$

$$= 155.6 \text{ N./mm}^2$$



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AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 10.78/155.6 + (190.21 \cdot 202.5/5539.876)/174.66$$

$$= 0.108$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	547.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	7676.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1500.00	mm
Saddle Width	b1 or b	405.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	279.75	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		137.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	
Saddle Force Q for this Transportation Case		6157.29	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	-0.79	-234.01
Long. Stress at Top	of Midspan	0.79	138.00
Long. Stress at Bottom	of Midspan	0.79	138.00
Long. Stress at Top	of Saddles	7.22	138.00
Long. Stress at Bottom	of Saddles	-4.00	-234.01
Long. Stress at Bottom	of Saddles	4.00	138.00
Tangential Shear in Shell		2.68	110.40
Circ. Stress at Horn of Saddle		3.98	207.00
Circ. Compressive Stress in Shell		0.20	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:
 = Transverse Area • 0.00256 • wv² • wlsf



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$$= 142.06 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0$$

$$= 1206.7 \text{ lbf}$$

$$= 547.375 \text{ Kgf}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$$

$$= \pi \cdot 3.681^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \cdot 1.2$$

$$= 108.5 \text{ lbf}$$

$$= 49.2 \text{ Kgf}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$= gz \cdot \text{Saddle Load} \cdot gw + Flw$$

$$= 0.1 \cdot 5597.57 \cdot 1.0 + 547.38$$

$$= 1107.1 \text{ Kgf}$$

Longitudinal Force due to Axial Acceleration [FI]:

$$= gx \cdot \text{Saddle Load} \cdot gw + Flw$$

$$= 0.1 \cdot 5597.57 \cdot 1.0 + 49.21$$

$$= 609.0 \text{ Kgf}$$

Saddle Reaction Force due to Transverse Load [Fwt]:

$$= Ft/\text{Num of Saddles} \cdot B / E$$

$$= 1107.1/2 \cdot 1000.0/989.001$$

$$= 559.7 \text{ Kgf}$$

Saddle Reaction Force due to Longitudinal Load [Fwl]:

$$= Fl \cdot B / Ls$$

$$= 609 \cdot 1000.0/4400.0$$

$$= 138.4 \text{ Kgf}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [Fsl]:

$$= \text{Saddle Load} \cdot gy$$

$$= 5597.6 \cdot 0.05$$

$$= 279.878 \text{ Kgf}$$

Load Combination Results due to Transport Loading [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl)$$

$$= 5598 + \max(138, 560, 280)$$

$$= 6157.3 \text{ Kgf}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	6430.90	Kgf
Transverse Shear Load	559.72	Kgf
Longitudinal Shear Load	608.96	Kgf

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(1066.0/2 + 28.0 + 10.0) \sin(120.0/2), 2 \cdot 547.0)$$

$$= 989.001 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			



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The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm^2)/3L)] \\
 &= -6157 \cdot 1500.0 [1 - (1 - 1500.0/7676.0 + (547.0^2 - 279.75^2)/(2 \cdot 1500.0 \cdot 7676.0))/(1 + (4 \cdot 279.75)/(3 \cdot 7676.0))] \\
 &= -2064.7 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 (1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L \\
 &= 6157 \cdot 7676/4 (1 + 2(547^2 - 280^2)/(7676^2))/(1 + (4 \cdot 280)/(3 \cdot 7676)) - 4 \cdot 1500/7676 \\
 &= 2116.9 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t) \\
 &= 0.0 \cdot 547.0/(2 \cdot 28.0) - 2116.9/(\pi \cdot 547.0^2 \cdot 28.0) \\
 &= -0.79 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\
 &= 0.0 \cdot 547.0/(2 \cdot 28.0) + 2116.9/(\pi \cdot 547.0^2 \cdot 28.0) \\
 &= 0.79 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 0.0 \cdot 547.0/(2 \cdot 28.0) - 2064.7/(0.1066 \cdot \pi \cdot 547.0^2 \cdot 28.0) \\
 &= 7.22 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 0.0 \cdot 547.0/(2 \cdot 28.0) + 2064.7/(0.1923 \cdot \pi \cdot 547.0^2 \cdot 28.0) \\
 &= -4.00 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot hm/3)) \\
 &= 6157(7676.0 - 2 \cdot 1500.0)/(7676.0 + (4 \cdot 279.75/3)) \\
 &= 3577.0 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (\pi \cdot Rm \cdot t) \\
 &= 1.1707 \cdot 3577.03 / (\pi \cdot 547.0 \cdot 28.0) \\
 &= 2.68 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x1,x2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(\pi \cdot Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(\pi \cdot 547.0 \cdot 28.0)} \\
 &= 96.531 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(\pi \cdot Rm \cdot t)}, 2a) \\
 &= \min(405.0 + 1.56 \cdot \sqrt{(\pi \cdot 547.0 \cdot 28.0)}, 2 \cdot 1500.0) \\
 &= 598.06 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:



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$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 6157 \cdot 0.1 / (598.062(28.0 + 1.0 \cdot 10.0))$$

$$= -0.20 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) $[\sigma_7, r]$:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -6157 / (4(28.0 + 1.0 \cdot 10.0)598.062) -$$

$$3 \cdot 0.053 \cdot 6157 / (2(28.0 + 1.0 \cdot 10.0)^2)$$

$$= -3.98 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	430.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw, c	480.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(533.0 \cdot 28.0)} + 600.0)28.0^3) / 12$$

$$= 144.623 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}}(C1 - Y)^2$$

$$= 144.623 + 221.361(148.24 - 14.0)^2$$

$$= 40034.879 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	790.6	28.0	14.0	221.4	309905.6	0.400E+05
Wearplate	600.0	10.0	33.0	60.0	198000.0	0.797E+04
Web	10.0	404.0	240.0	40.4	969599.8	0.890E+04
BasePlate	430.0	25.0	454.5	107.5	4885874.5	0.101E+06
Totals	...	...	...	429.3	6363380.0	0.158E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6363380 / 429.261$$

$$= 148.240 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:



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$$= K1 \cdot Q$$

$$= 0.204 \cdot 6157.289$$

$$= 1253.1418 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.5911 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000.0 - 533.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 559.213 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 700.7864 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.6456 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(6157 + 274)430.0 / (4 \cdot 1000.0 \cdot 174.663)}$$

$$= 10.791 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (533.0 + 28.0 + 10.0)$$

$$= 989.001 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 989.001 / (3 - 1)$$

$$= 494.501 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 494.5005 \cdot 430.0 / 2$$

$$= 1063.176 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= Q / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 6157.29 / (1000.0 \cdot 430.0)$$

$$= 1.432 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1063.2 \cdot 1.43$$

$$= 1522.391 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 39.5 + 24.725$$

$$= 64.225 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 1522.4 / 64.225$$

$$= 2.325 \text{ N./mm}^2$$

Check of Outside Ribs:



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Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	I _o
Rib+Web	10.0	405.0	...	40.5	...	0.554E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 405.0 - 10.0

= 395.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/64.225

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 405.0/2

= 202.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{5535.8/64.225}$

= 92.841 mm

Length of Outer Rib [L]:

= Saddle Height - $\cos(\theta/2)$ (radius + shlthk + wpdthk) - bpthk

= 1000.0 - $\cos(120.0/2)$ (533.0 + 28.0 + 10.0) - 25.0

= 689.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/262.0}$

= 122.739

Slenderness ratio [KL/r]:

= KL/r

= 1 • 689.5/92.841

= 7.427

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 609.0 / (2 • 1000.0) • 494.501 • 689.5/2

= 51.909 Kg-m.

Compressive Allowable, KL/r < Cc (7.4267 < 122.7394) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (7.43)^2 / (2 \cdot 122.74^2)) 262 /$

$(5/3 + 3 \cdot (7.43) / (8 \cdot 122.74) - (7.43^3) / (8 \cdot 122.74^3))$

= 154.8 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba

= 2.32/154.8 + (51.91 • 202.5/55358440)/174.66

= 0.026

Check of Inside Ribs:



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Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	I _o
Rib	10.0	395.0	0.0	39.5	0.0	0.554E+04
Web	494.5	10.0	0.0	49.5	0.0	4.12
Totals	...	...	...	89.0	...	0.554E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/88.95$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 405.0/2$$

$$= 202.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000.0 - \cos(571.0/2) (25.0 + 0.0 + 0.0) - 0.0$$

$$= 404.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(5539.9/88.95)}$$

$$= 78.918 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 404.0/78.918$$

$$= 5.119$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 608.964 / (2 \cdot 1000.0)$$

$$= 0.304 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.304 \cdot 494.501 \cdot 404.0$$

$$= 60.830 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 60.83/273574.375$$

$$= 2.181 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (5.1192 < 122.7394) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.12)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (5.12) / (8 \cdot 122.74) - (5.12^3) / (8 \cdot 122.74^3))$$

$$= 155.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 3.36/155.6 + (60.83 \cdot 202.5/5539.876) / 174.66$$

$$= 0.034$$

Review notes about nozzle loadings and supports in the Warnings report.



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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	548.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	7676.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1500.00	mm
Saddle Width	b1 or b	405.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	
Saddle Force Q, Operating Case		26657.29	Kgf
Pressure used in Analysis	P	6105.224	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	59.67	138.00
Long. Stress at Bottom	of Midspan	69.02	138.00
Long. Stress at Top	of Saddles	91.69	138.00
Long. Stress at Bottom	of Saddles	49.19	138.00
Tangential Shear in Shell		13.09	110.40
Circ. Stress at Horn of Saddle		19.06	172.50
Circ. Compressive Stress in Shell		0.93	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (584.5/2 + 0) \cdot 1000.0/989.001$
 $= 886.5 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_l, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(122.35, 0) \cdot 1000.0/4400.0$
 $= 27.8 \text{ Kgf}$

Saddle Reaction Force due to Earthquake FI [Fsl]:
 $= \max(F_l, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(1884.18, 0) \cdot 1000.0/4400.0$



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= 428.2 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3.0(1884/2 + 0) • 1000.0/989.001

= 2857.7 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 23800 + max(28, 886, 428, 2858)

= 26657.3 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))

= 1.1 • 766.08(0.0 + $\pi/4$ (1.122 • 1.2)²)

= 1199.798 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)

26930.90 Kgf

Transverse Shear Load

942.09 Kgf

Longitudinal Shear Load

1884.18 Kgf

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)

= min(2(1066.0/2 + 28.0 + 10.0)sin(120.0/2), 2•548.0)

= 989.001 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066

K2 = 1.1707

K3 = 0.8799

K4 = 0.4011

K5 = 0.7603

K6 = 0.0529

K7 = 0.0529

K8 = 0.3405

K9 = 0.2711

K10 = 0.0581

K1* = 0.1923

K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm²)/3L)]

= -26657•1500.0[1-(1-1500.0/7676.0+(548.0²-0.0²)/

(2•1500.0•7676.0))/(1+(4•0.0)/(3•7676.0))]

= -7292.5 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4(1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L

= 26657•7676/4(1+2(548²-0²)/(7676²))/(1+(4•0)/

(3•7676))-4•1500/7676

= 11691.1 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)

= 6105.224 • 548.0/(2•26.0) - 11691.1/(π •548.0²•26.0)

= 59.67 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)



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$$= 6105.224 \cdot 548.0 / (2 \cdot 26.0) + 11691.1 / (\pi \cdot 548.0^2 \cdot 26.0)$$

$$= 69.02 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 6105.224 \cdot 548.0 / (2 \cdot 26.0) - 7292.5 / (0.1066 \cdot \pi \cdot 548.0^2 \cdot 26.0)$$

$$= 91.69 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 6105.224 \cdot 548.0 / (2 \cdot 26.0) + 7292.5 / (0.1923 \cdot \pi \cdot 548.0^2 \cdot 26.0)$$

$$= 49.19 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 26657(7676.0 - 2 \cdot 1500.0) / (7676.0 + (4 \cdot 0.0 / 3))$$

$$= 16238.9 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 16238.86 / (548.0 \cdot 26.0)$$

$$= 13.09 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(548.0 \cdot 26.0)}$$

$$= 93.105 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(405.0 + 1.56 \cdot \sqrt{(548.0 \cdot 26.0)}, 2 \cdot 1500.0)$$

$$= 591.21 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(137.997 / 137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 26657 \cdot 0.1 / (591.209(26.0 + 1.0 \cdot 10.0))$$

$$= -0.93 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -26657 / (4(26.0 + 1.0 \cdot 10.0)591.209) -$$

$$3 \cdot 0.053 \cdot 26657 / (2(26.0 + 1.0 \cdot 10.0)^2)$$

$$= -19.06 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 4400.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000013 \cdot 4400.0 (275.0 - 21.1)$$

$$= 14.677 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm



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Baseplate Width	Bpwid	430.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw, c	480.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(535.0 \cdot 26.0) + 600.0})^3) / 12$$

$$= 114.828 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 114.828 + 203.837(152.449 - 13.0)^2$$

$$= 39752.898 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	784.0	26.0	13.0	203.8	264987.8	0.398E+05
Wearplate	600.0	10.0	31.0	60.0	186000.0	0.885E+04
Web	10.0	404.0	238.0	40.4	961520.0	0.845E+04
BasePlate	430.0	25.0	452.5	107.5	4864375.5	0.968E+05
Totals	...	...	...	411.7	6276883.0	0.154E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6276883 / 411.737$$

$$= 152.449 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 26657.293$$

$$= 5425.3374 \text{ Kg}$$

Tension Stress, St = (Fh/As) = 2.5592 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1000.0 - 535.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 557.558 \text{ mm}$$

Bending Moment, M = Fh * d = 3025.0027 Kg-m.



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Bending Stress, $S_b = (M \cdot C_1 / I) = 2.9386 \text{ N./mm}^2$ Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$ **Minimum Thickness of Baseplate per Moss:**

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(26657 + 274)430.0 / (4 \cdot 1000.0 \cdot 174.663))}$$

$$= 22.083 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (533.0 + 28.0 + 10.0)$$

$$= 989.001 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 989.001 / (3 - 1)$$

$$= 494.501 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 494.5005 \cdot 430.0 / 2$$

$$= 1063.176 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= Q / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 26657.295 / (1000.0 \cdot 430.0)$$

$$= 6.199 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 1063.2 \cdot 6.2$$

$$= 6591.022 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 39.5 + 24.725$$

$$= 64.225 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 6591.0 / 64.225$$

$$= 10.064 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	I _o
Rib+Web	10.0	405.0	...	40.5	...	0.554E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 405.0 - 10.0$$

$$= 395.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$



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= 0.0/64.225
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 405.0/2
 = 202.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{5535.8/64.225}$
 = 92.841 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1000.0 - cos(120.0/2)(533.0 + 28.0 + 10.0) - 25.0
 = 689.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/262.0}$
 = 122.739

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 689.5/92.841
 = 7.427

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 1884.2 / (2 • 1000.0) • 494.501 • 689.5/2
 = 160.609 Kg-m.

Compressive Allowable, KL/r < Cc (7.4267 < 122.7394) per AISC E2-1 [Sca]:

= $(1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$
 = $(1 - (7.43)^2 / (2 \cdot 122.74^2)) 262 /$
 $(5/3 + 3 \cdot (7.43) / (8 \cdot 122.74) - (7.43^3) / (8 \cdot 122.74^3))$
 = 154.8 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 10.06/154.8 + (160.61 • 202.5/55358440) / 174.66
 = 0.098

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	I _o
Rib	10.0	395.0	0.0	39.5	0.0	0.554E+04
Web	494.5	10.0	0.0	49.5	0.0	4.12
Totals	...	...	...	89.0	...	0.554E+04

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/88.95
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2



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= 405.0/2
 = 202.500 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = 1000.0 - cos(571.0/2)(25.0 + 0.0 + 0.0) - 0.0
 = 404.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{5539.9/88.95}$
 = 78.918 mm

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 404.0/78.918
 = 5.119

Unit Force [Force,u]:
 $= F_l / (2 \cdot \text{Baseplate Length})$
 $= 1884.177 / (2 \cdot 1000.0)$
 = 0.942 Kg/mm

Moment at base of inner Rib [Mbase,c]:
 = Unit Force • e • L
 = 0.942 • 494.501 • 404.0
 = 188.212 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = 188.212/273574.375
 = 6.747 N./mm²

Compressive Allowable, $KL/r < C_c$ (5.1192 < 122.7394) per AISC E2-1 [Sca]:
 $= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (5.12)^2 / (2 \cdot 122.74^2)) 262 /$
 $(5/3 + 3 \cdot (5.12) / (8 \cdot 122.74) - (5.12^3) / (8 \cdot 122.74^3))$
 = 155.6 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$
 $= 14.54/155.6 + (188.21 \cdot 202.5/5539.876) / 174.66$
 = 0.132

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-307 A	
Bolt Allowable Stress	Stba	138.00	N./mm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm
User Defined Area in a Single Bolt	UserBoltArea	8.6100	cm ²
BasePlate Allowable Stress	S	138.00	N./mm ²
Area Available in a Single Bolt	BltArea	7.1203	cm ²
Saddle Load QO (Weight)	QO	24073.2	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	428.2	Kgf
Maximum Transverse Force	Ft	942.1	Kgf
Maximum Longitudinal Force	Fl	1884.2	Kgf
Saddle Bolted to Steel Foundation	No		



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Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1884 / (7.12 \cdot 4)$$

$$= 6.5 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 942 / (7.12 \cdot 4)$$

$$= 3.2 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 1884.18 / (82.8 \cdot 4.0)$$

$$= 0.5579 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1000.0 \cdot 942.09 + 0.0$$

$$= 942.11 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 942.11 / 24073.21$$

$$= 39.13 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.5579, 0.0]$$

$$= 0.5579 \text{ cm}^2$$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10.0, c = 2.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.96$, Temp. Reduction = 2 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B

-29 °C

Min Metal Temp. at Required thickness (UCS 66.1)

-31 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:**Input and Calculated Values:**

Vessel Mean Radius	Rm	548.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	7676.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1500.00	mm



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Saddle Width	b1 or b	405.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	280.75	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		137.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	
Saddle Force Q, Operating Case		11559.60	Kgf
Pressure used in Analysis	P	6105.224	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	62.76	138.00	
Long. Stress at Bottom of Midspan	65.93	138.00	
Long. Stress at Top of Saddles	78.89	138.00	
Long. Stress at Bottom of Saddles	56.28	138.00	
Tangential Shear in Shell	5.41	110.40	
Circ. Stress at Horn of Saddle	8.27	172.50	
Circ. Compressive Stress in Shell	0.40	138.00	

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(584.5/2 + 0) \cdot 1000.0/989.001$
 $= 886.5\ Kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl, \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(122.35, 0) \cdot 1000.0/4400.0$
 $= 27.8\ Kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl, \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(1884.18, 0) \cdot 1000.0/4400.0$
 $= 428.2\ Kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(1884/2 + 0) \cdot 1000.0/989.001$
 $= 2857.7\ Kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$
 $= 8702 + \max(28, 886, 428, 2858)$



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= 11559.6 Kgf

Longitudinal Wind Force [FI]:
 = WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 766.08(0.0 + $\pi/4$ (1.122 • 1.2)²)
 = 1199.798 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	11833.20	Kgf
Transverse Shear Load	942.09	Kgf
Longitudinal Shear Load	1884.18	Kgf

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 = min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(1066.0/2 + 28.0 + 10.0)sin(120.0/2), 2•548.0)
 = 989.001 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm²)/3L)]
 = -11560•1500.0[1 - (1 - 1500.0/7676.0 + (548.0²-280.75²)/
 (2•1500.0•7676.0))/(1+(4•280.75)/(3•7676.0))]
 = -3878.1 Kg-m.

Moment per Equation 4.15.2 [M2]:
 = Q•L/4(1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L)) - 4a/L
 = 11560•7676/4(1+2(548²-281²)/(7676²))/(1+(4•281)/
 (3•7676)) - 4•1500/7676
 = 3971.1 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 = P • Rm/(2t) - M2/(π •Rm²t)
 = 6105.224 • 548.0/(2•26.0) - 3971.1/(π •548.0²•26.0)
 = 62.76 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 = P • Rm/(2t) + M2/(π • Rm² • t)
 = 6105.224 • 548.0/(2 • 26.0) + 3971.1/(π • 548.0² • 26.0)
 = 65.93 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 = P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 6105.224•548.0/(2•26.0)--3878.1/(0.1066• π •548.0²•26.0)
 = 78.89 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:
 = P • Rm/(2t) + M1/(K1* • π • Rm² • t)



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$$= 6105.224 \cdot 548.0 / (2 \cdot 26.0) + -3878.1 / (0.1923 \cdot \pi \cdot 548.0^2 \cdot 26.0)$$

$$= 56.28 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 11560(7676.0 - 2 \cdot 1500.0) / (7676.0 + (4 \cdot 280.75 / 3))$$

$$= 6714.3 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 6714.34 / (548.0 \cdot 26.0)$$

$$= 5.41 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(548.0 \cdot 26.0)}$$

$$= 93.105 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(405.0 + 1.56 \cdot \sqrt{(548.0 \cdot 26.0)}, 2 \cdot 1500.0)$$

$$= 591.21 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(137.997 / 137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.7603 \cdot 11560 \cdot 0.1 / (591.209(26.0 + 1.0 \cdot 10.0))$$

$$= -0.40 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot t_r) b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot t_r)^2)$$

$$= -11560 / (4(26.0 + 1.0 \cdot 10.0) 591.209) -$$

$$3 \cdot 0.053 \cdot 11560 / (2(26.0 + 1.0 \cdot 10.0)^2)$$

$$= -8.27 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	430.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw, c	480.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(535.0 \cdot 26.0)} + 600.0) 26.0^3) / 12$$

$$= 114.828 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:



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$$\begin{aligned}
 &= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2 \\
 &= 114.828 + 203.837(152.449 - 13.0)^2 \\
 &= 39752.898 \text{ cm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	784.0	26.0	13.0	203.8	264987.8	0.398E+05
Wearplate	600.0	10.0	31.0	60.0	186000.0	0.885E+04
Web	10.0	404.0	238.0	40.4	961520.0	0.845E+04
BasePlate	430.0	25.0	452.5	107.5	4864375.5	0.968E+05
Totals	...	...	...	411.7	6276883.0	0.154E+06

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 6276883 / 411.737 \\
 &= 152.449 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle} / 2 \\
 &= 180 - 120.0 / 2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.204 \cdot 11559.596 \\
 &= 2352.6282 \text{ Kg}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / A_s) = 1.1098 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1000.0 - 535.0 \cdot \sin(120.0/2) / 1.0472 \\
 &= 557.558 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1311.7538 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 1.2743 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{3(11560 + 274)430.0 / (4 \cdot 1000.0 \cdot 174.663)} \\
 &= 14.638 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120.0/2) (533.0 + 28.0 + 10.0) \\
 &= 989.001 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$



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$$= 989.001 / (3 - 1)$$

$$= 494.501 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 494.5005 \cdot 430.0 / 2$$

$$= 1063.176 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= Q / (BasePlateLength \cdot BasePlateWidth)$$

$$= 11559.596 / (1000.0 \cdot 430.0)$$

$$= 2.688 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1063.2 \cdot 2.69$$

$$= 2858.113 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 39.5 + 24.725$$

$$= 64.225 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 2858.1 / 64.225$$

$$= 4.364 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	I _o
Rib+Web	10.0	405.0	...	40.5	...	0.554E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 405.0 - 10.0$$

$$= 395.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 64.225$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 405.0 / 2$$

$$= 202.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(Total \text{ Inertia} / Total \text{ Area})}$$

$$= \sqrt{(5535.8 / 64.225)}$$

$$= 92.841 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta/2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1000.0 - \cos(120.0/2) (533.0 + 28.0 + 10.0) - 25.0$$

$$= 689.500 \text{ mm}$$



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Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 262.0)}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 689.5 / 92.841$$

$$= 7.427$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 1884.2 / (2 \cdot 1000.0) \cdot 494.501 \cdot 689.5 / 2$$

$$= 160.609 \text{ Kg-m.}$$

Compressive Allowable, KL/r < Cc (7.4267 < 122.7394) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (7.43)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (7.43) / (8 \cdot 122.74) - (7.43)^3 / (8 \cdot 122.74^3))$$

$$= 154.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 4.36 / 154.8 + (160.61 \cdot 202.5 / 55358440) / 174.66$$

$$= 0.061$$

Check of Inside Ribs:**Inertia of Saddle, Inner Ribs - Axial Direction**

	B	D	Y	A	AY	I _o
Rib	10.0	395.0	0.0	39.5	0.0	0.554E+04
Web	494.5	10.0	0.0	49.5	0.0	4.12
Totals	...	...	...	89.0	...	0.554E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 88.95$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 405.0 / 2$$

$$= 202.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1000.0 - \cos(571.0/2) (25.0 + 0.0 + 0.0) - 0.0$$

$$= 404.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(5539.9 / 88.95)}$$

$$= 78.918 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 404.0 / 78.918$$

$$= 5.119$$



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Unit Force [Force,u]:
 $= F_l / (2 \cdot \text{Baseplate Length})$
 $= 1884.177 / (2 \cdot 1000.0)$
 $= 0.942 \text{ Kg/mm}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 0.942 \cdot 494.501 \cdot 404.0$
 $= 188.212 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 188.212 / 273574.375$
 $= 6.747 \text{ N./mm}^2$

Compressive Allowable, $KL/r < C_c$ ($5.1192 < 122.7394$) per AISC E2-1 [Sca]:
 $= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (5.12)^2 / (2 \cdot 122.74^2)) 262 /$
 $(5/3 + 3 \cdot (5.12) / (8 \cdot 122.74) - (5.12^3) / (8 \cdot 122.74^3))$
 $= 155.6 \text{ N./mm}^2$

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$
 $= 6.31 / 155.6 + (188.21 \cdot 202.5 / 5539.876) / 174.66$
 $= 0.079$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-307 A	
Bolt Allowable Stress	Stba	138.00	N./mm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm
User Defined Area in a Single Bolt	UserBoltArea	8.6100	cm ²
BasePlate Allowable Stress	S	138.00	N./mm ²
Area Available in a Single Bolt	BltArea	7.1203	cm ²
Saddle Load QO (Weight)	QO	8975.5	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	428.2	Kgf
Maximum Transverse Force	Ft	942.1	Kgf
Maximum Longitudinal Force	Fl	1884.2	Kgf
Saddle Bolted to Steel Foundation	No		

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 $= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 1884 / (7.12 \cdot 4)$
 $= 6.5 \text{ N./mm}^2$. Must be less than 82.8 N./mm^2 .

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 $= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 942 / (7.12 \cdot 4)$
 $= 3.2 \text{ N./mm}^2$. Must be less than 82.8 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:



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= FI / (BoltShearAllowable • Nbolts)
= 1884.18/(82.8 • 4.0)
= 0.5579 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 1000.0 • 942.09 + 0.0
= 942.11 Kg-m.

Eccentricity (e):
= Rmom / QO
= 942.11/8975.5
= 104.96 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.5579, 0.0]
= 0.5579 cm²

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10.0, c = 2.0 mm, E* = 1.0
Thickness Ratio = tr • E*/(tg - c) = 0.96, Temp. Reduction = 2 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	547.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	7676.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1500.00	mm
Saddle Width	b1 or b	405.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.01	N./mm ²
Head Allowable Stress used in Calculation		234.01	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	
Saddle Force Q, Test Case, no Ext. Forces		24099.67	Kgf
Pressure used in Analysis	P	7935.361	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		73.58	234.01
Long. Stress at Bottom of Midspan		81.45	234.01
Long. Stress at Top of Saddles		100.56	234.01
Long. Stress at Bottom of Saddles		64.74	234.01
Tangential Shear in Shell		11.00	138.59
Circ. Stress at Horn of Saddle		15.58	351.01
Circ. Compressive Stress in Shell		0.79	234.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (292.2/2 + 0) \cdot 1000.0/989.001$
 $= 443.2 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_l, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(61.17, 0) \cdot 1000.0/4400.0$
 $= 13.9 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 23656 + \max(14, 443, 0, 0)$



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= 24099.7 Kgf

Longitudinal Wind Force [FI]:
 = WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 766.08(0.0 + $\pi/4$ (1.122 • 1.2)²)
 = 1199.798 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	24373.28	Kgf
Transverse Shear Load	146.12	Kgf
Longitudinal Shear Load	61.17	Kgf

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 = min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(1066.0/2 + 28.0 + 10.0)sin(120.0/2), 2•547.0)
 = 989.001 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = -Q•a [1 - (1- a/L + (Rm²-hm²)/(2a•L))/(1+(4hm²)/3L)]
 = -24100•1500.0[1-(1-1500.0/7676.0+(547.0²-0.0²)/
 (2•1500.0•7676.0))/(1+(4•0.0)/(3•7676.0))]
 = -6594.6 Kg-m.

Moment per Equation 4.15.2 [M2]:
 = Q•L/4(1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L
 = 24100•7676/4(1+2(547²-0²)/(7676²))/(1+(4•0)/
 (3•7676))-4•1500/7676
 = 10567.7 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 = P • Rm/(2t) - M2/(π •Rm²t)
 = 7935.361 • 547.0/(2•28.0) - 10567.7/(π •547.0²•28.0)
 = 73.58 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 = P • Rm/(2t) + M2/(π • Rm² • t)
 = 7935.361 • 547.0/(2 • 28.0) + 10567.7/(π • 547.0² • 28.0)
 = 81.45 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 = P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 7935.361•547.0/(2•28.0)--6594.6/(0.1066• π •547.0²•28.0)
 = 100.56 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:
 = P • Rm/(2t) + M1/(K1* • π • Rm² • t)



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$$= 7935.361 \cdot 547.0 / (2 \cdot 28.0) + 6594.6 / (0.1923 \cdot \pi \cdot 547.0^2 \cdot 28.0)$$

$$= 64.74 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 24100(7676.0 - 2 \cdot 1500.0) / (7676.0 + (4 \cdot 0.0 / 3))$$

$$= 14680.8 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 14680.83 / (547.0 \cdot 28.0)$$

$$= 11.00 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(547.0 \cdot 28.0)}$$

$$= 96.531 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(405.0 + 1.56 \cdot \sqrt{(547.0 \cdot 28.0)}, 2 \cdot 1500.0)$$

$$= 598.06 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 24100 \cdot 0.1 / (598.062(28.0 + 1.0 \cdot 10.0))$$

$$= -0.79 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -24100 / (4(28.0 + 1.0 \cdot 10.0)598.062) -$$

$$3 \cdot 0.053 \cdot 24100 / (2(28.0 + 1.0 \cdot 10.0)^2)$$

$$= -15.58 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	430.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw, c	480.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(533.0 \cdot 28.0)} + 600.0)^3) / 12$$

$$= 144.623 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 144.623 + 221.361(148.24 - 14.0)^2$$

$$= 40034.879 \text{ cm}^4$$



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	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	790.6	28.0	14.0	221.4	309905.6	0.400E+05
Wearplate	600.0	10.0	33.0	60.0	198000.0	0.797E+04
Web	10.0	404.0	240.0	40.4	969599.8	0.890E+04
BasePlate	430.0	25.0	454.5	107.5	4885874.5	0.101E+06
Totals	...	...	...	429.3	6363380.0	0.158E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6363380 / 429.261$$

$$= 148.240 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 24099.668$$

$$= 4904.8052 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.3136 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1000.0 - 533.0 \cdot \sin(120.0 / 2) / 1.0472$$

$$= 559.213 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2742.8826 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.5270 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(24100 + 274)430.0) / (4 \cdot 1000.0 \cdot 174.663)}$$

$$= 21.008 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0 / 2) (533.0 + 28.0 + 10.0)$$

$$= 989.001 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 989.001 / (3 - 1)$$

$$= 494.501 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 494.5005 \cdot 430.0 / 2$$

$$= 1063.176 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= Q / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 24099.668 / (1000.0 \cdot 430.0)$$

$$= 5.605 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1063.2 \cdot 5.6$$

$$= 5958.649 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 39.5 + 24.725$$

$$= 64.225 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 5958.6 / 64.225$$

$$= 9.099 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	I _o
Rib+Web	10.0	405.0	...	40.5	...	0.554E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 405.0 - 10.0$$

$$= 395.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 64.225$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 405.0 / 2$$

$$= 202.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 5535.8 / 64.225 }$$

$$= 92.841 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000.0 - \cos(120.0 / 2) (533.0 + 28.0 + 10.0) - 25.0$$

$$= 689.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} }$$



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$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 262.0)}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 689.5 / 92.841$$

$$= 7.427$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 61.2 / (2 \cdot 1000.0) \cdot 494.501 \cdot 689.5 / 2$$

$$= 5.215 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($7.4267 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.43)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (7.43) / (8 \cdot 122.74) - (7.43)^3 / (8 \cdot 122.74^3))$$

$$= 154.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 9.1 / 154.8 + (5.21 \cdot 202.5 / 55358440) / 174.66$$

$$= 0.060$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	I _o
Rib	10.0	395.0	0.0	39.5	0.0	0.554E+04
Web	494.5	10.0	0.0	49.5	0.0	4.12
Totals	...	...	...	89.0	...	0.554E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 88.95$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 405.0 / 2$$

$$= 202.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1000.0 - \cos(571.0/2) (25.0 + 0.0 + 0.0) - 0.0$$

$$= 404.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(5539.9 / 88.95)}$$

$$= 78.918 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 404.0 / 78.918$$

$$= 5.119$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$



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$$= 61.174 / (2 \cdot 1000.0)$$

$$= 0.031 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.031 \cdot 494.501 \cdot 404.0$$

$$= 6.111 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 6.111 / 273574.375$$

$$= 0.219 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.1192 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.12)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (5.12) / (8 \cdot 122.74) - (5.12)^3 / (8 \cdot 122.74^3))$$

$$= 155.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 13.14 / 155.6 + (6.11 \cdot 202.5 / 5539.876) / 174.66$$

$$= 0.086$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-307 A	
Bolt Allowable Stress	Stba	138.00	N./mm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm
User Defined Area in a Single Bolt	UserBoltArea	8.6100	cm ²
BasePlate Allowable Stress	S	138.00	N./mm ²
Area Available in a Single Bolt	BlArea	7.1203	cm ²
Saddle Load QO (Weight)	QO	23930.0	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	13.9	Kgf
Maximum Transverse Force	Ft	146.1	Kgf
Maximum Longitudinal Force	Fl	61.2	Kgf
Saddle Bolted to Steel Foundation		No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 61 / (7.12 \cdot 4)$$

$$= 0.2 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 146 / (7.12 \cdot 4)$$

$$= 0.5 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 61.17 / (82.8 \cdot 4.0)$$

$$= 0.0181 \text{ cm}^2$$



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Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 1000.0 \cdot 146.12 + 0.0$
 $= 146.12 \text{ Kg-m.}$

Eccentricity (e):
 $= Rmom / QO$
 $= 146.12 / 23930.04$
 $= 6.11 \text{ mm} < Bplen/6 \rightarrow \text{No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 $= 0 \text{ (No Uplift)}$

Required Area of a Single Bolt [Bltarear]:
 $= \max[Bltarearl, Bltarears, Bltareart]$
 $= \max[0.0, 0.0181, 0.0]$
 $= 0.0181 \text{ cm}^2$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	547.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	7676.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1500.00	mm
Saddle Width	b1 or b	405.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	279.75	mm
Shell Allowable Stress used in Calculation		234.01	N./mm ²
Head Allowable Stress used in Calculation		234.01	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	
Saddle Force Q, Test Case, no Ext. Forces		8173.21	Kgf
Pressure used in Analysis	P	7935.361	KPa.

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	76.47	234.01
Long. Stress at Bottom	of Midspan	78.56	234.01
Long. Stress at Top	of Saddles	87.09	234.01
Long. Stress at Bottom	of Saddles	72.21	234.01



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Tangential Shear in Shell	3.56	138.59
Circ. Stress at Horn of Saddle	5.28	351.01
Circ. Compressive Stress in Shell	0.27	234.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr} (\text{Ft} / \text{Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0 (292.2/2 + 0) \cdot 1000.0/989.001$
 $= 443.2 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl}, \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(61.17, 0) \cdot 1000.0/4400.0$
 $= 13.9 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 7730 + \max(14, 443, 0, 0)$
 $= 8173.2 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08 (0.0 + \pi/4 (1.122 \cdot 1.2)^2)$
 $= 1199.798 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	8446.81	Kgf
Transverse Shear Load	146.12	Kgf
Longitudinal Shear Load	61.17	Kgf

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2 (1066.0/2 + 28.0 + 10.0) \sin(120.0/2), 2 \cdot 547.0)$
 $= 989.001 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2) / (2a \cdot L)) / (1 + (4\text{hm}^2) / 3L)]$
 $= -8173 \cdot 1500.0 [1 - (1 - 1500.0/7676.0 + (547.0^2 - 279.75^2) / (2 \cdot 1500.0 \cdot 7676.0)) / (1 + (4 \cdot 279.75) / (3 \cdot 7676.0))]$
 $= -2740.7 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 (1 + 2 (\text{Rm}^2 - \text{hm}^2) / (L^2)) / (1 + (4\text{hm}^2) / (3L)) - 4a/L$



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$$= 8173 \cdot 7676 / 4(1 + 2(547^2 - 280^2) / (7676^2)) / (1 + (4 \cdot 280) / (3 \cdot 7676)) - 4 \cdot 1500 / 7676$$

$$= 2810.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 7935.361 \cdot 547.0 / (2 \cdot 28.0) - 2810.0 / (\pi \cdot 547.0^2 \cdot 28.0)$$

$$= 76.47 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 7935.361 \cdot 547.0 / (2 \cdot 28.0) + 2810.0 / (\pi \cdot 547.0^2 \cdot 28.0)$$

$$= 78.56 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 7935.361 \cdot 547.0 / (2 \cdot 28.0) - 2740.7 / (0.1066 \cdot \pi \cdot 547.0^2 \cdot 28.0)$$

$$= 87.09 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 7935.361 \cdot 547.0 / (2 \cdot 28.0) + 2740.7 / (0.1923 \cdot \pi \cdot 547.0^2 \cdot 28.0)$$

$$= 72.21 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 8173(7676.0 - 2 \cdot 1500.0) / (7676.0 + (4 \cdot 279.75 / 3))$$

$$= 4748.2 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 4748.16 / (547.0 \cdot 28.0)$$

$$= 3.56 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(547.0 \cdot 28.0)}$$

$$= 96.531 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(405.0 + 1.56 \cdot \sqrt{(547.0 \cdot 28.0)}, 2 \cdot 1500.0)$$

$$= 598.06 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 8173 \cdot 0.1 / (598.062(28.0 + 1.0 \cdot 10.0))$$

$$= -0.27 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -8173 / (4(28.0 + 1.0 \cdot 10.0)598.062) -$$

$$3 \cdot 0.053 \cdot 8173 / (2(28.0 + 1.0 \cdot 10.0)^2)$$

$$= -5.28 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:



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Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	430.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw,c	480.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(533.0 \cdot 28.0)} + 600.0)28.0^3)/12$$

$$= 144.623 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 144.623 + 221.361(148.24 - 14.0)^2$$

$$= 40034.879 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	790.6	28.0	14.0	221.4	309905.6	0.400E+05
Wearplate	600.0	10.0	33.0	60.0	198000.0	0.797E+04
Web	10.0	404.0	240.0	40.4	969599.8	0.890E+04
BasePlate	430.0	25.0	454.5	107.5	4885874.5	0.101E+06
Totals	...	...	...	429.3	6363380.0	0.158E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6363380/429.261$$

$$= 148.240 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 8173.206$$

$$= 1663.4246 \text{ Kg}$$

Tension Stress, St = (Fh/As)	=	0.7847	N./mm ²
Allowed Stress, Sa = 0.6 * Yield Str	=	157.1972	N./mm ²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1000.0 - 533.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 559.213 \text{ mm}$$



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Bending Moment, M = Fh * d = 930.2262 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 0.8570 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 174.6635 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(8173 + 274)430.0 / (4 \cdot 1000.0 \cdot 174.663))}$$

= 12.367 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= 2 * cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)

= 2 * cos(90 - 120.0/2)(533.0 + 28.0 + 10.0)

= 989.001 mm

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= 989.001 / (3 - 1)

= 494.501 mm

Baseplate Pressure Area [Ap]:

= e * Bpwid / 2

= 494.5005 * 430.0 / 2

= 1063.176 cm²

Bearing Pressure [Bp]:

= Q / (BasePlateLength * BasePlateWidth)

= 8173.206 / (1000.0 * 430.0)

= 1.901 Kg/cm²

Axial Load [P]:

= Ap * Bp

= 1063.2 * 1.9

= 2020.827 Kg

Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 39.5 + 24.725

= 64.225 cm²

Compressive Stress [Sc]:

= P / Ar

= 2020.8 / 64.225

= 3.086 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	405.0	...	40.5	...	0.554E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 405.0 - 10.0

= 395.000 mm



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Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/64.225$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 405.0/2$$

$$= 202.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(5535.8/64.225)}$$

$$= 92.841 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000.0 - \cos(120.0/2) (533.0 + 28.0 + 10.0) - 25.0$$

$$= 689.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/262.0)}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 689.5/92.841$$

$$= 7.427$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 61.2 / (2 \cdot 1000.0) \cdot 494.501 \cdot 689.5/2$$

$$= 5.215 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($7.4267 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (7.43)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (7.43) / (8 \cdot 122.74) - (7.43)^3 / (8 \cdot 122.74^3))$$

$$= 154.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 3.09/154.8 + (5.21 \cdot 202.5/5535.8440) / 174.66$$

$$= 0.021$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	I _o
Rib	10.0	395.0	0.0	39.5	0.0	0.554E+04
Web	494.5	10.0	0.0	49.5	0.0	4.12
Totals	...	...	...	89.0	...	0.554E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/88.95$$

$$= 0.000 \text{ mm}$$



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Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 405.0 / 2$$

$$= 202.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000.0 - \cos(571.0/2) (25.0 + 0.0 + 0.0) - 0.0$$

$$= 404.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{5539.9/88.95}$$

$$= 78.918 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 404.0/78.918$$

$$= 5.119$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 61.174 / (2 \cdot 1000.0)$$

$$= 0.031 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.031 \cdot 494.501 \cdot 404.0$$

$$= 6.111 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 6.111/273574.375$$

$$= 0.219 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.1192 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.12)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (5.12) / (8 \cdot 122.74) - (5.12^3) / (8 \cdot 122.74^3))$$

$$= 155.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 4.46/155.6 + (6.11 \cdot 202.5/5539.876) / 174.66$$

$$= 0.030$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification	SA-307 A		
Bolt Allowable Stress	Stba	138.00	N./mm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm
User Defined Area in a Single Bolt	UserBoltArea	8.6100	cm ²
BasePlate Allowable Stress	S	138.00	N./mm ²
Area Available in a Single Bolt	BltArea	7.1203	cm ²
Saddle Load QO (Weight)	QO	8003.6	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	13.9	Kgf
Maximum Transverse Force	Ft	146.1	Kgf



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Maximum Longitudinal Force
Saddle Bolted to Steel Foundation

F1

61.2 Kgf

No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $61 / (7.12 \cdot 4)$

= 0.2 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $146 / (7.12 \cdot 4)$

= 0.5 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= $61.17 / (82.8 \cdot 4.0)$

= 0.0181 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$

= $1000.0 \cdot 146.12 + 0.0$

= 146.12 Kg-m.

Eccentricity (e):

= R_{mom} / QO

= $146.12 / 8003.58$

= 18.26 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$

= $\max[0.0, 0.0181, 0.0]$

= 0.0181 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	548.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	7676.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1500.00	mm
Saddle Width	b1 or b	405.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	
Saddle Force Q, External Pressure Case		26657.29	Kgf
Pressure used in Analysis	P	-103.000	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-5.76	-235.80
Long. Stress at Top of Midspan		5.76	138.00
Long. Stress at Bottom of Midspan		3.59	138.00
Long. Stress at Top of Saddles		26.26	138.00
Long. Stress at Bottom of Saddles		-16.24	-235.80
Long. Stress at Bottom of Saddles		16.24	138.00
Tangential Shear in Shell		13.09	110.40
Circ. Stress at Horn of Saddle		19.06	172.50
Circ. Compressive Stress in Shell		0.93	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (584.5 / 2 + 0) \cdot 1000.0 / 989.001$
 $= 886.5 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_l, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(122.35, 0) \cdot 1000.0 / 4400.0$
 $= 27.8 \text{ Kgf}$

Saddle Reaction Force due to Earthquake FI [Fsl]:



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= max(FI, Sum of X Forces) • B / Ls
 = max(1884.18, 0) • 1000.0/4400.0
 = 428.2 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:
 = Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3.0(1884/2 + 0) • 1000.0/989.001
 = 2857.7 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 23800 + max(28, 886, 428, 2858)
 = 26657.3 Kgf

Longitudinal Wind Force [FI]:
 = WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 766.08(0.0 + $\pi/4$ (1.122 • 1.2)²)
 = 1199.798 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	26930.90	Kgf
Transverse Shear Load	942.09	Kgf
Longitudinal Shear Load	1884.18	Kgf

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(1066.0/2 + 28.0 + 10.0)sin(120.0/2), 2•548.0)
 = 989.001 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = -Q•a [1 - (1- a/L + (Rm²-hm²)/(2a•L))/(1+(4hm²)/3L)]
 = -26657•1500.0[1-(1-1500.0/7676.0+(548.0²-0.0²)/
 (2•1500.0•7676.0))/(1+(4•0.0)/(3•7676.0))]]
 = -7292.5 Kg-m.

Moment per Equation 4.15.2 [M2]:
 = Q•L/4(1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L
 = 26657•7676/4(1+2(548²-0²)/(7676²))/(1+(4•0)/
 (3•7676))-4•1500/7676
 = 11691.1 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 = P • Rm/(2t) - M2/(π •Rm²t)
 = -103.0 • 548.0/(2•26.0) - 11691.1/(π •548.0²•26.0)
 = -5.76 N./mm²



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Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -103.0 \cdot 548.0 / (2 \cdot 26.0) + 11691.1 / (\pi \cdot 548.0^2 \cdot 26.0)$$

$$= 3.59 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -103.0 \cdot 548.0 / (2 \cdot 26.0) - 7292.5 / (0.1066 \cdot \pi \cdot 548.0^2 \cdot 26.0)$$

$$= 26.26 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -103.0 \cdot 548.0 / (2 \cdot 26.0) + 7292.5 / (0.1923 \cdot \pi \cdot 548.0^2 \cdot 26.0)$$

$$= -16.24 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 26657(7676.0 - 2 \cdot 1500.0) / (7676.0 + (4 \cdot 0.0 / 3))$$

$$= 16238.9 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 16238.86 / (548.0 \cdot 26.0)$$

$$= 13.09 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{548.0 \cdot 26.0}$$

$$= 93.105 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(405.0 + 1.56 \cdot \sqrt{548.0 \cdot 26.0}, 2 \cdot 1500.0)$$

$$= 591.21 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r / S, 1)$$

$$= \min(137.997 / 137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 26657 \cdot 0.1 / (591.209(26.0 + 1.0 \cdot 10.0))$$

$$= -0.93 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -26657 / (4(26.0 + 1.0 \cdot 10.0)591.209) -$$

$$3 \cdot 0.053 \cdot 26657 / (2(26.0 + 1.0 \cdot 10.0)^2)$$

$$= -19.06 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	430.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./



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Height of Web at Center Hw, c 480.0 mm
 Friction Coefficient mu 0.000

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(535.0 \cdot 26.0)} + 600.0)26.0^3)/12$$

$$= 114.828 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 114.828 + 203.837(152.449 - 13.0)^2$$

$$= 39752.898 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	784.0	26.0	13.0	203.8	264987.8	0.398E+05
Wearplate	600.0	10.0	31.0	60.0	186000.0	0.885E+04
Web	10.0	404.0	238.0	40.4	961520.0	0.845E+04
BasePlate	430.0	25.0	452.5	107.5	4864375.5	0.968E+05
Totals	...	...	...	411.7	6276883.0	0.154E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6276883/411.737$$

$$= 152.449 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 26657.293$$

$$= 5425.3374 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 2.5592 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1000.0 - 535.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 557.558 \text{ mm}$$

Bending Moment, M = Fh * d = 3025.0027 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 2.9386 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 174.6635 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(26657 + 274)430.0) / (4 \cdot 1000.0 \cdot 174.663)}$$



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= 22.083 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= $2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$

= $2 \cdot \cos(90 - 120.0/2) (533.0 + 28.0 + 10.0)$

= 989.001 mm

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= $989.001 / (3 - 1)$

= 494.501 mm

Baseplate Pressure Area [Ap]:

= $e \cdot \text{Bpwid} / 2$

= $494.5005 \cdot 430.0 / 2$

= 1063.176 cm²

Bearing Pressure [Bp]:

= $Q / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

= $26657.295 / (1000.0 \cdot 430.0)$

= 6.199 Kg/cm²

Axial Load [P]:

= $A_p \cdot B_p$

= $1063.2 \cdot 6.2$

= 6591.022 Kg

Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= $39.5 + 24.725$

= 64.225 cm²

Compressive Stress [Sc]:

= P / A_r

= $6591.0 / 64.225$

= 10.064 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	I _o
Rib+Web	10.0	405.0	...	40.5	...	0.554E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= $405.0 - 10.0$

= 395.000 mm

Distance to Centroid from Datum [ytot]:

= A_Y / A

= $0.0 / 64.225$

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= $405.0 / 2$



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= 202.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{5535.8/64.225}$
 = 92.841 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1000.0 - cos(120.0/2)(533.0 + 28.0 + 10.0) - 25.0
 = 689.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/262.0}$
 = 122.739

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 689.5/92.841
 = 7.427

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 1884.2 / (2 • 1000.0) • 494.501 • 689.5/2
 = 160.609 Kg-m.

Compressive Allowable, KL/r < Cc (7.4267 < 122.7394) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (7.43)^2 / (2 \cdot 122.74^2)) 262 /$
 $(5/3 + 3 \cdot (7.43) / (8 \cdot 122.74) - (7.43^3) / (8 \cdot 122.74^3))$
 = 154.8 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba
 = 10.06/154.8 + (160.61 • 202.5/55358440)/174.66
 = 0.098

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	I _o
Rib	10.0	395.0	0.0	39.5	0.0	0.554E+04
Web	494.5	10.0	0.0	49.5	0.0	4.12
Totals	...	...	...	89.0	...	0.554E+04

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/88.95
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 405.0/2
 = 202.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 1000.0 - cos(571.0/2)(25.0 + 0.0 + 0.0) - 0.0



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= 404.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{5539.9/88.95}$
 = 78.918 mm

Slenderness ratio [KL/r]:

= KL/r
 = $1 \cdot 404.0/78.918$
 = 5.119

Unit Force [Force,u]:

= $FI / (2 \cdot \text{Baseplate Length})$
 = $1884.177 / (2 \cdot 1000.0)$
 = 0.942 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force $\cdot e \cdot L$
 = $0.942 \cdot 494.501 \cdot 404.0$
 = 188.212 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = $188.212/273574.375$
 = 6.747 N./mm²

Compressive Allowable, KL/r < Cc (5.1192 < 122.7394) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (5.12)^2 / (2 \cdot 122.74^2)) 262 /$
 $(5/3 + 3 \cdot (5.12) / (8 \cdot 122.74) - (5.12^3) / (8 \cdot 122.74^3))$
 = 155.6 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 = $14.54/155.6 + (188.21 \cdot 202.5/5539.876) / 174.66$
 = 0.132

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	548.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	7676.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1500.00	mm
Saddle Width	b1 or b	405.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	280.75	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		137.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	



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Circumferential Efficiency at Mid-Span 1.00

Distance from Saddle Base to Centerline B 1000.00 mm

Coefficient of Friction mu 0.30

Saddle Force Q, External Pressure Case 11559.60 Kgf

Pressure used in Analysis P -103.000 KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-2.67	-235.80
Long. Stress at Top of Midspan		2.67	138.00
Long. Stress at Bottom of Midspan		0.50	138.00
Long. Stress at Top of Saddles		13.46	138.00
Long. Stress at Bottom of Saddles		-9.15	-235.80
Long. Stress at Bottom of Saddles		9.15	138.00
Tangential Shear in Shell		5.41	110.40
Circ. Stress at Horn of Saddle		8.27	172.50
Circ. Compressive Stress in Shell		0.40	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(584.5/2 + 0) \cdot 1000.0/989.001$
 $= 886.5\ Kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl, \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(122.35, 0) \cdot 1000.0/4400.0$
 $= 27.8\ Kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl, \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(1884.18, 0) \cdot 1000.0/4400.0$
 $= 428.2\ Kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(1884/2 + 0) \cdot 1000.0/989.001$
 $= 2857.7\ Kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$
 $= 8702 + \max(28, 886, 428, 2858)$
 $= 11559.6\ Kgf$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08(0.0 + \pi/4(1.122 \cdot 1.2)^2)$
 $= 1199.798\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	11833.20	Kgf
Transverse Shear Load	942.09	Kgf
Longitudinal Shear Load	1884.18	Kgf



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Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1066.0/2 + 28.0 + 10.0) \sin(120.0/2), 2 \cdot 548.0)$$

$$= 989.001 \text{ mm}$$

The Computed K values from Table 4.15.1:

$K_1 = 0.1066$	$K_2 = 1.1707$	$K_3 = 0.8799$	$K_4 = 0.4011$
$K_5 = 0.7603$	$K_6 = 0.0529$	$K_7 = 0.0529$	$K_8 = 0.3405$
$K_9 = 0.2711$	$K_{10} = 0.0581$	$K_{1*} = 0.1923$	$K_{6p} = 0.0434$
$K_{7p} = 0.0434$			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m^2)/3L)]$$

$$= -11560 \cdot 1500.0 [1 - (1 - 1500.0/7676.0 + (548.0^2 - 280.75^2) / (2 \cdot 1500.0 \cdot 7676.0)) / (1 + (4 \cdot 280.75^2) / (3 \cdot 7676.0))]$$

$$= -3878.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 (1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L$$

$$= 11560 \cdot 7676/4 (1 + 2(548^2 - 281^2)/(7676^2)) / (1 + (4 \cdot 281^2) / (3 \cdot 7676)) - 4 \cdot 1500/7676$$

$$= 3971.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= -103.0 \cdot 548.0 / (2 \cdot 26.0) - 3971.1 / (\pi \cdot 548.0^2 \cdot 26.0)$$

$$= -2.67 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -103.0 \cdot 548.0 / (2 \cdot 26.0) + 3971.1 / (\pi \cdot 548.0^2 \cdot 26.0)$$

$$= 0.50 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= -103.0 \cdot 548.0 / (2 \cdot 26.0) - 3878.1 / (0.1066 \cdot \pi \cdot 548.0^2 \cdot 26.0)$$

$$= 13.46 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_{1*} \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -103.0 \cdot 548.0 / (2 \cdot 26.0) + 3878.1 / (0.1923 \cdot \pi \cdot 548.0^2 \cdot 26.0)$$

$$= -9.15 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m/3))$$

$$= 11560(7676.0 - 2 \cdot 1500.0) / (7676.0 + (4 \cdot 280.75/3))$$

$$= 6714.3 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 6714.34 / (548.0 \cdot 26.0)$$



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$$= 5.41 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(548.0 \cdot 26.0)}$$

$$= 93.105 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(405.0 + 1.56 \cdot \sqrt{(548.0 \cdot 26.0)}, 2 \cdot 1500.0)$$

$$= 591.21 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(137.997/137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 11560 \cdot 0.1 / (591.209(26.0 + 1.0 \cdot 10.0))$$

$$= -0.40 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -11560 / (4(26.0 + 1.0 \cdot 10.0)591.209) -$$

$$3 \cdot 0.053 \cdot 11560 / (2(26.0 + 1.0 \cdot 10.0)^2)$$

$$= -8.27 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	430.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw, c	480.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(535.0 \cdot 26.0)} + 600.0)26.0^3) / 12$$

$$= 114.828 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 114.828 + 203.837(152.449 - 13.0)^2$$

$$= 39752.898 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	784.0	26.0	13.0	203.8	264987.8	0.398E+05
Wearplate	600.0	10.0	31.0	60.0	186000.0	0.885E+04
Web	10.0	404.0	238.0	40.4	961520.0	0.845E+04
BasePlate	430.0	25.0	452.5	107.5	4864375.5	0.968E+05



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Totals	...	...	...	411.7	6276883.0	0.154E+06
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Distance to Centroid [C1]:

$$= AY / A$$

$$= 6276883 / 411.737$$

$$= 152.449 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 11559.596$$

$$= 2352.6282 \text{ Kgf}$$

Tension Stress, St = (Fh / As)	=	1.1098	N./mm ²
Allowed Stress, Sa = 0.6 * Yield Str	=	157.1972	N./mm ²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1000.0 - 535.0 \cdot \sin(120.0 / 2) / 1.0472$$

$$= 557.558 \text{ mm}$$

Bending Moment, M = Fh * d	=	1311.7538	Kg-m.
----------------------------	---	-----------	-------

Bending Stress, Sb = (M * C1 / I)	=	1.2743	N./mm ²
Allowed Stress, Sa = 2/3 * Yield Str	=	174.6635	N./mm ²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(11560 + 274)430.0 / (4 \cdot 1000.0 \cdot 174.663)}$$

$$= 14.638 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0 / 2) (533.0 + 28.0 + 10.0)$$

$$= 989.001 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 989.001 / (3 - 1)$$

$$= 494.501 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 494.5005 \cdot 430.0 / 2$$

$$= 1063.176 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= Q / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 11559.596 / (1000.0 \cdot 430.0)$$



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$$= 2.688 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 1063.2 \cdot 2.69$$

$$= 2858.113 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 39.5 + 24.725$$

$$= 64.225 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 2858.1 / 64.225$$

$$= 4.364 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	I _o
Rib+Web	10.0	405.0	...	40.5	...	0.554E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 405.0 - 10.0$$

$$= 395.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 64.225$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 405.0 / 2$$

$$= 202.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(5535.8 / 64.225)}$$

$$= 92.841 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000.0 - \cos(120.0/2) (533.0 + 28.0 + 10.0) - 25.0$$

$$= 689.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 262.0)}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 689.5 / 92.841$$

$$= 7.427$$



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Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 1884.2 / (2 \cdot 1000.0) \cdot 494.501 \cdot 689.5 / 2$$

$$= 160.609 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (7.4267 < 122.7394) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.43)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (7.43) / (8 \cdot 122.74) - (7.43)^3 / (8 \cdot 122.74^3))$$

$$= 154.8 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 4.36 / 154.8 + (160.61 \cdot 202.5 / 55358440) / 174.66$$

$$= 0.061$$

Check of Inside Ribs:**Inertia of Saddle, Inner Ribs - Axial Direction**

	B	D	Y	A	AY	I _o
Rib	10.0	395.0	0.0	39.5	0.0	0.554E+04
Web	494.5	10.0	0.0	49.5	0.0	4.12
Totals	...	...	...	89.0	...	0.554E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 88.95$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 405.0 / 2$$

$$= 202.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1000.0 - \cos(571.0/2) (25.0 + 0.0 + 0.0) - 0.0$$

$$= 404.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{5539.9 / 88.95}$$

$$= 78.918 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 404.0 / 78.918$$

$$= 5.119$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 1884.177 / (2 \cdot 1000.0)$$

$$= 0.942 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.942 \cdot 494.501 \cdot 404.0$$

$$= 188.212 \text{ Kg-m.}$$



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Bending Stress due to Transverse Force and Weight Load [σ_B , base, c]:

= Bending Moment / Section Modulus

= 188.212/273574.375

= 6.747 N./mm²

Compressive Allowable, $KL/r < C_c$ (5.1192 < 122.7394) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)

= (1-(5.12)²/(2 • 122.74²))262/

(5/3+3•(5.12)/(8• 122.74)-(5.12³)/(8•122.74³)

= 155.6 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C_1/I)/S_{ba}$

= 6.31/155.6 + (188.21 • 202.5/5539.876)/174.66

= 0.079

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: Tube inlet

From: 20

Pressure for Reinforcement Calculations	P	1000.000	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.00	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1060.00	mm
Design Length of Section	L	938.3333	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		450.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-182 F304L	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	453.0500	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		90.0000	mm
Flange Class		150	
Flange Grade		GR 2.3	



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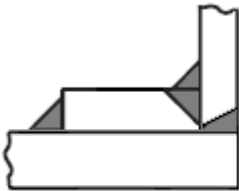
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The Pressure Design option was Design Pressure + static head.



Abutting/Set-on Nozzle With Pad

Reinforcement CALCULATION, Description: Tube inlet

ASME Code, Section VIII, Div. 1, 2021, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750 in.
Actual Thickness Used in Calculation	0.365 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000.0 \cdot 533.2 / (138 \cdot 1.0 - 0.6 \cdot 1000.0)$
 $= 3.8809 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1000.0 \cdot 136.525 / (118 \cdot 1.0 + 0.4 \cdot 1000.0)$
 $= 1.1532 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8429 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	509.0160	mm
Parallel to Vessel Wall, opening length	d	254.5080	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	9.877	4.170	NA
Area in Shell	A1	17.610	19.146	NA
Area in Nozzle Wall	A2	3.748	3.892	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.924	2.924	NA
Area in Element	A5	21.600	21.600	NA
TOTAL AREA AVAILABLE	Atot	45.882	47.561	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (254.508 \cdot 3.8809 \cdot 1.0 + 2 \cdot 9.271 \cdot 3.8809 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 9.877 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 254.508 (1.0 \cdot 10.8 - 1.0 \cdot 3.881) - 2 \cdot 9.271 \\
 &\quad (1.0 \cdot 10.8 - 1.0 \cdot 3.8809) \cdot (1 - 1.0) \\
 &= 17.610 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 27.0) \cdot (9.27 - 1.15) \cdot 0.8551 \\
 &= 3.748 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\
 &= 15.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 10.0^2 \cdot 1.0 \\
 &= 2.924 \text{ cm}^2
 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned}
 &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\
 &= (453.05 - 273.05) \cdot 12.0 \cdot 1.0 \\
 &= 21.600 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.1532 mm
Wall Thickness per UG16(b),	tr16b = 5.7000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.0809 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.0809 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.5982 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.7000 mm
Wall Thickness per table UG-45	tb3 = 8.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$



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= min[8.103, max(7.0809, 5.7)]
 = 7.0809 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(1.1532, 7.0809)
 = 7.0809 mm

Available Nozzle Neck Thickness = $0.875 \times 9.271 = 8.112$ mm → OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
 (see 319.4.4 and 302.3.5):**

Sustained	:	29.4,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	265.6 N./mm ²	Passed
Occasional	:	6.6,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	18.0,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 8.112, tr = 1.153, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \times E^* / (tg - c) = 0.142$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 8.112, tr = 1.153, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \times E^* / (tg - c) = 0.142$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 8.112, tr = 1.153, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \times E^* / (tg - c) = 0.142$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, c = 3.2 mm, E* = 1.0
 Thickness Ratio = $tr \times E^* / (tg - c) = 0.359$, Temp. Reduction = 71 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 8.112, tr = 1.153, c = 0.0 mm, E* = 1.0



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Thickness Ratio = $tr * E / (tg - c) = 0.142$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -48 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1000.00 / 1590.01 = 0.629$

Weld Size Calculations, Description: Tube inlet

Intermediate Calc. for nozzle/shell Welds Tmin 9.2710 mm
Intermediate Calc. for pad/shell Welds TminPad 10.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$10.6050 = 0.7 * Wo$ mm
Pad Weld	$5.4000 = 0.5 * TminPad$	$7.0700 = 0.7 * Wp$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= (Ar - A1) * Sv$
 $= (9.8773 - 17.6096) * 138$
 $= 0.00 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A2 + A5 + A4 - (Wi - Can / .707)^2 * fr2) * Sv$
 $= (3.7483 + 21.6 + 2.9239 - 0.0 * 0.86) * 138$
 $= 39783.47 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$= (\pi/2) * Dlo * Wo * 0.49 * Snw$
 $= (3.1416/2.0) * 273.05 * 15.0 * 0.49 * 118$
 $= 37931. \text{ Kg}$

Shear, Pad Element Weld [Spew]:

$= (\pi/2) * DP * WP * 0.49 * SEW$
 $= (3.1416/2.0) * 453.05 * 10.0 * 0.49 * 138$
 $= 49069. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:

$= (\pi/2) * Dlo * Wgpn * 0.74 * Seg$
 $= (3.1416/2) * 273.05 * 12.0 * 0.74 * 138$
 $= 53594. \text{ Kg}$

Shear, Nozzle Groove Weld [Sngw]:



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$$\begin{aligned}
 &= (\pi/2) \cdot (Dlr+Wgnva) \cdot (Wgnva-Can) \cdot 0.60 \cdot Sng \\
 &= (3.14/2) \cdot (254.508+9.0) \cdot (9.0-0.0) \cdot 0.6 \cdot 138 \\
 &= 31452. \text{ Kg}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNGW) = (49069 + 31452) = 80521 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 80520 Kg, must exceed W = 0 Kg or W1 = 39783 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2751 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 17.8858 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 281.8858 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 254.51 mm

Input Echo, WRC297 Item 1, Description: Tube inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Corrosion Allowance	3.2000	mm
Shell Diameter	1060.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

Using 2 * Yield for Discontinuity Stress Allowable (Div 2, 4.1.6.3), Sps.
 Make sure that material properties at this temperature are not time-dependent for Material: SA-516 70

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	273.050	mm
Nozzle Thickness	9.2710	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

Using 2 * Yield for Discontinuity Stress Allowable (Div 2, 4.1.6.3), Sps.
 Make sure that material properties at this temperature are not time-dependent for Material: SA-106 B

Diameter of Reinforcing Pad	453.050	mm
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Thickness of Reinforcing Pad 12.0000 mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1000.00	KPa.
Radial Load	P	-1912.11	Kgf
Circumferential Shear	Vc	1274.73	Kgf
Longitudinal Shear	Vl	1274.73	Kgf
Circumferential Moment	Mc	891.97	Kg-m.
Longitudinal Moment	Ml	891.97	Kg-m.
Torsional Moment	Mt	1274.24	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(544.6(14.0 - 3.2))}$$

$$= 526.134 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1089.200 mm
Nozzle Outside Diameter	(d) =	273.050 mm
Vessel Thickness used	(T) =	22.800 mm
Nozzle Thickness used	(t) =	9.271 mm
T / t	=	2.459
d / t	=	29.452

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.733$$

Nr/P	=	0.085
Mr/P	=	0.084
M0/P	=	0.061
N0/P	=	0.177
MrD/Mc	=	0.175
NrDL/Mc	=	0.112
M0d/Mc	=	0.148
N0DL/Mc	=	0.180
MrD/Ml	=	0.108
NrDL/Ml	=	0.077
M0D/Ml	=	0.087
N0DL/Ml	=	0.308

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-18	-18	18	18



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Outplane Bending (ML)	-32	32	32	-32
Normal Pressure Stress	22	23	22	23

Outplane Stress Summary	-8	30	93	3
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-40	40	40	-40
Inplane Pressure Stress	11	11	11	11

Inplane Stress Summary	-12	31	77	-38
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	5	5	3	3
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	16	37	94	43
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-54	54	54	-54
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	11	11	11	11

Outplane Stress Summary	-34	48	96	-39
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	18	-18	18	-18



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Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-64	64	64	-64
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	22	23	22	23

Inplane Stress Summary	-27	66	115	-48
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	36	67	117	51
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	17	15	17

Hoop Stress Summary	3	4	41	42
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	53	-53	53	-53
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-155	155	155	-155
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	-109	94	238	-181
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside



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Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11
Shear Stress Summary	14	14	8	8

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	116	96	238	224

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	17	15	17
Hoop Stress Summary	11	12	33	34

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	53	-53	53	-53
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-265	265	265	-265
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	-219	204	347	-291

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	11	11	11	11
Shear Stress Summary	8	8	14	14

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	231	204	348	327

Stress Computations at the Edge of the Reinforcing Pad:



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Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(538.6(14.0 - 3.2))}$$

$$= 524.736 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1077.200 mm
 Nozzle Outside Diameter (d) = 453.050 mm
 Vessel Thickness used (T) = 10.800 mm
 Nozzle Thickness used (t) = 99.271 mm
 T / t = 0.109
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.200$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.050
 Mr/P = 0.080
 M0/P = 0.023
 N0/P = 0.035
 MrD/Mc = 0.250
 NrDL/Mc = 0.101
 M0d/Mc = 0.075
 N0DL/Mc = 0.062
 MrD/Ml = 0.059
 NrDL/Ml = 0.056
 M0D/Ml = 0.018
 N0DL/Ml = 0.038

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	48	49	48	49
Outplane Stress Summary	52	44	100	21

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	76	-76	76	-76
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-58	58	58	-58
Inplane Pressure Stress	24	24	24	24



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Inplane Stress Summary	41	5	177	-94
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	5	5	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	54	45	177	115
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-74	74	74	-74
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	24	24	24	24

Outplane Stress Summary	-32	71	136	-56
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	76	-76	76	-76
Inplane Membrane (Mc)	-16	-16	16	16
Inplane Bending (Mc)	-248	248	248	-248
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	48	49	48	49

Inplane Stress Summary	-131	212	399	-250
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	1	1	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	131	212	399	251

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	22	23	22	23	22	23	22	23
Circ. Pl (SUS)	-12	-12	24	24	-3	-3	9	9
Circ. Q (SUS)	-19	19	45	-45	-46	46	82	-82
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-1	-1	7	7	-5	-5	17	17
Long. Q (SUS)	-22	22	58	-58	-41	41	67	-67
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	22.0	23.0	22.0	23.0	22.0	23.0	22.0	23.0
Pm+Pl (SUS)	11.0	11.6	46.0	47.0	19.1	20.1	31.3	32.2
Pm+Pl+Q (Total)	15.7	36.1	91.6	42.4	36.0	66.5	114.3	51.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	23.00	138.00	Passed
Pm+Pl (SUS)	47.03	207.00	Passed
Pm+Pl+Q (TOTAL)	114.30	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	48	49	48	49	48	49	48	49
Circ. Pl (SUS)	-1	-1	11	11	-8	-8	24	24
Circ. Q (SUS)	5	-5	39	-39	-172	172	324	-324
Long. Pm (SUS)	24	24	24	24	24	24	24	24
Long. Pl (SUS)	-1	-1	17	17	-5	-5	15	15



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Long. Q (SUS)	18	-18	134	-134	-52	52	96	-96
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	48.0	49.0	48.0	49.0	48.0	49.0	48.0	49.0
Pm+Pl (SUS)	47.0	48.0	59.1	60.1	40.0	41.0	72.0	73.0
Pm+Pl+Q (Total)	53.3	43.4	175.1	114.1	132.0	213.0	396.1	251.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	49.00	138.00	Passed
Pm+Pl (SUS)	73.03	207.00	Passed
Pm+Pl+Q (TOTAL)	396.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	17	15	17	15	17	15	17
Circ. Pl (SUS)	-12	-12	24	24	-5	-5	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-15	-15	19	19	-15	-15	19	19
Long. Q (SUS)	-102	102	208	-208	-212	212	318	-318
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	15.0	17.0	15.0	17.0	15.0	17.0	15.0	17.0
Pm+Pl (SUS)	12.5	14.3	39.7	41.6	19.0	20.9	33.2	35.0
Pm+Pl+Q (Total)	116.4	96.2	234.3	223.6	230.6	204.3	344.6	327.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	17.00	118.00	Passed
Pm+Pl (SUS)	41.58	177.00	Passed
Pm+Pl+Q (TOTAL)	344.63	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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Input, Nozzle Desc: Tube outlet

From: 20

Pressure for Reinforcement Calculations	P	1010.390	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.00	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1060.00	mm
Design Length of Section	L	938.3333	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		450.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	453.0500	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		90.0000	mm
Flange Class		150	
Flange Grade		GR 2.3	



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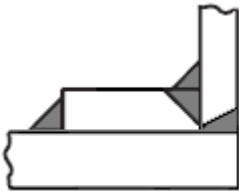
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The Pressure Design option was Design Pressure + static head.



Abutting/Set-on Nozzle With Pad

Reinforcement CALCULATION, Description: Tube outlet

ASME Code, Section VIII, Div. 1, 2021, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750 in.
Actual Thickness Used in Calculation	0.365 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1010.39 \cdot 533.2 / (138 \cdot 1.0 - 0.6 \cdot 1010.39)$
 $= 3.9214 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1010.39 \cdot 136.525 / (118 \cdot 1.0 + 0.4 \cdot 1010.39)$
 $= 1.1651 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8429 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	509.0160	mm
Parallel to Vessel Wall, opening length	d	254.5080	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	9.980	4.170	NA
Area in Shell	A1	17.506	19.146	NA
Area in Nozzle Wall	A2	3.743	3.892	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.924	2.924	NA
Area in Element	A5	21.600	21.600	NA
TOTAL AREA AVAILABLE	Atot	45.773	47.561	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (254.508 \cdot 3.9214 \cdot 1.0 + 2 \cdot 9.271 \cdot 3.9214 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 9.980 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 254.508 (1.0 \cdot 10.8 - 1.0 \cdot 3.921) - 2 \cdot 9.271 \\
 &\quad (1.0 \cdot 10.8 - 1.0 \cdot 3.9214) \cdot (1 - 1.0) \\
 &= 17.506 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 27.0) \cdot (9.27 - 1.17) \cdot 0.8551 \\
 &= 3.743 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\
 &= 15.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 10.0^2 \cdot 1.0 \\
 &= 2.924 \text{ cm}^2
 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned}
 &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\
 &= (453.05 - 273.05) \cdot 12.0 \cdot 1.0 \\
 &= 21.600 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.1651 mm
Wall Thickness per UG16(b),	tr16b = 5.7000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.1214 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.1214 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.5982 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.7000 mm
Wall Thickness per table UG-45	tb3 = 8.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$



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= min[8.103, max(7.1214, 5.7)]
 = 7.1214 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(1.1651, 7.1214)
 = 7.1214 mm

Available Nozzle Neck Thickness = 0.875 * 9.271 = 8.112 mm→OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
 (see 319.4.4 and 302.3.5):**

Sustained	:	29.5,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	265.5 N./mm ²	Passed
Occasional	:	6.7,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	18.0,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 8.112, tr = 1.165, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.144$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 8.112, tr = 1.165, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.144$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 8.112, tr = 1.165, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.144$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, c = 3.2 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.363$, Temp. Reduction = 68 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 8.112, tr = 1.165, c = 0.0 mm, E* = 1.0



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Thickness Ratio = $tr * E / (tg - c) = 0.144$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -48 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1010.39 / 1590.01 = 0.635$

Weld Size Calculations, Description: Tube outlet

Intermediate Calc. for nozzle/shell Welds Tmin 9.2710 mm
Intermediate Calc. for pad/shell Welds TminPad 10.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$10.6050 = 0.7 * Wo$ mm
Pad Weld	$5.4000 = 0.5 * TminPad$	$7.0700 = 0.7 * Wp$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= (Ar - A1) * Sv$
 $= (9.9804 - 17.5065) * 138$
 $= 0.00 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A2 + A5 + A4 - (Wi - Can / .707)^2 * fr2) * Sv$
 $= (3.7428 + 21.6 + 2.9239 - 0.0 * 0.86) * 138$
 $= 39775.71 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$= (\pi/2) * Dlo * Wo * 0.49 * Snw$
 $= (3.1416/2.0) * 273.05 * 15.0 * 0.49 * 118$
 $= 37931. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:

$= (\pi/2) * DP * WP * 0.49 * SEW$
 $= (3.1416/2.0) * 453.05 * 10.0 * 0.49 * 138$
 $= 49069. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:

$= (\pi/2) * Dlo * Wgpn * 0.74 * Seg$
 $= (3.1416/2) * 273.05 * 12.0 * 0.74 * 138$
 $= 53594. \text{ Kgf}$

Shear, Nozzle Groove Weld [Sngw]:



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$$\begin{aligned}
 &= (\pi/2) \cdot (Dlr+Wgnva) \cdot (Wgnva-Can) \cdot 0.60 \cdot Sng \\
 &= (3.14/2) \cdot (254.508+9.0) \cdot (9.0-0.0) \cdot 0.6 \cdot 138 \\
 &= 31452. \text{ Kg}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNGW) = (49069 + 31452) = 80521 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 80520 Kg, must exceed W = 0 Kg or W1 = 39775 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2751 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 17.8858 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 281.8858 mm

Input Echo, WRC297 Item 1, Description: Tube outlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Corrosion Allowance	3.2000	mm
Shell Diameter	1060.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

Using 2 * Yield for Discontinuity Stress Allowable (Div 2, 4.1.6.3), Sps.
 Make sure that material properties at this temperature are not
 time-dependent for Material: SA-516 70

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	273.050	mm
Nozzle Thickness	9.2710	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

Using 2 * Yield for Discontinuity Stress Allowable (Div 2, 4.1.6.3), Sps.
 Make sure that material properties at this temperature are not
 time-dependent for Material: SA-106 B

Diameter of Reinforcing Pad	453.050	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:



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These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	1010.39	KPa.
Radial Load	P	-1912.11	Kgf
Circumferential Shear	Vc	1274.73	Kgf
Longitudinal Shear	Vl	1274.73	Kgf
Circumferential Moment	Mc	891.97	Kg-m.
Longitudinal Moment	Ml	891.97	Kg-m.
Torsional Moment	Mt	1274.24	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(544.6(14.0 - 3.2))}$$

$$= 526.134 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1089.200 mm
Nozzle Outside Diameter	(d) =	273.050 mm
Vessel Thickness used	(T) =	22.800 mm
Nozzle Thickness used	(t) =	9.271 mm
T / t	=	2.459
d / t	=	29.452

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.733$$

Nr/P	=	0.085
Mr/P	=	0.084
M0/P	=	0.061
N0/P	=	0.177
MrD/Mc	=	0.175
NrDL/Mc	=	0.112
M0d/Mc	=	0.148
N0DL/Mc	=	0.180
MrD/Ml	=	0.108
NrDL/Ml	=	0.077
M0D/Ml	=	0.087
N0DL/Ml	=	0.308

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-18	-18	18	18
Outplane Bending (Ml)	-32	32	32	-32
Normal Pressure Stress	23	24	23	24
Outplane Stress Summary	-8	30	93	4



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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-40	40	40	-40
Inplane Pressure Stress	11	11	11	11
Inplane Stress Summary	-12	31	77	-38

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	16	37	94	43

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-54	54	54	-54
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	11	11	11	11
Outplane Stress Summary	-34	48	97	-38

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-64	64	64	-64
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0



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Inplane Pressure Stress	23	24	23	24
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Inplane Stress Summary	-27	66	115	-48
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	36	67	117	51
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	17	16	17

Hoop Stress Summary	3	4	41	42
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	53	-53	53	-53
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-155	155	155	-155
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-109	94	238	-181
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11



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Shear Stress Summary	14	14	8	8
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	116	97	238	224
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	17	16	17

Hoop Stress Summary	11	12	33	34
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	53	-53	53	-53
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-265	265	265	-265
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-219	204	348	-291
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	11	11	11	11

Shear Stress Summary	8	8	14	14
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	231	204	348	327
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{538.6(14.0 - 3.2)}$$



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= 524.736 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1077.200 mm
 Nozzle Outside Diameter (d) = 453.050 mm
 Vessel Thickness used (T) = 10.800 mm
 Nozzle Thickness used (t) = 99.271 mm
 T / t = 0.109
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.200$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.050
 Mr/P = 0.080
 M0/P = 0.023
 N0/P = 0.035
 MrD/Mc = 0.250
 NrDL/Mc = 0.101
 M0d/Mc = 0.075
 N0DL/Mc = 0.062
 MrD/Ml = 0.059
 NrDL/Ml = 0.056
 M0D/Ml = 0.018
 N0DL/Ml = 0.038

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	49	50	49	50
Outplane Stress Summary	53	45	101	22

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	76	-76	76	-76
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-58	58	58	-58
Inplane Pressure Stress	24	24	24	24
Inplane Stress Summary	41	5	177	-93

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			3	3	3	3
Shear Stress Summary			5	5	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress		55	46	177	116

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)		5	5	5	5
Outplane Bending (P)		22	-22	22	-22
Outplane Membrane (Mc)		-10	-10	10	10
Outplane Bending (Mc)		-74	74	74	-74
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		24	24	24	24
Outplane Stress Summary		-31	72	137	-56

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)		8	8	8	8
Inplane Bending (P)		76	-76	76	-76
Inplane Membrane (Mc)		-16	-16	16	16
Inplane Bending (Mc)		-248	248	248	-248
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		49	50	49	50
Inplane Stress Summary		-131	213	399	-250

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-1	-1	1	1
Torsional Shear (Mt)		3	3	3	3
Shear Stress Summary		1	1	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left	Left	Right	Right



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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	131	213	399	250

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	23	24	23	24	23	24	23	24
Circ. Pl (SUS)	-12	-12	24	24	-3	-3	9	9
Circ. Q (SUS)	-19	19	45	-45	-46	46	82	-82
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-1	-1	7	7	-5	-5	17	17
Long. Q (SUS)	-22	22	58	-58	-41	41	67	-67
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	23.0	24.0	23.0	24.0	23.0	24.0	23.0	24.0
Pm+Pl (SUS)	11.6	12.4	47.0	48.0	20.1	21.1	32.2	33.2
Pm+Pl+Q (Total)	15.4	36.5	92.5	43.4	35.9	67.4	115.2	51.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	138.00	Passed
Pm+Pl (SUS)	48.03	207.00	Passed
Pm+Pl+Q (TOTAL)	115.24	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	49	50	49	50	49	50	49	50
Circ. Pl (SUS)	-1	-1	11	11	-8	-8	24	24
Circ. Q (SUS)	5	-5	39	-39	-172	172	324	-324
Long. Pm (SUS)	24	24	24	24	24	24	24	24
Long. Pl (SUS)	-1	-1	17	17	-5	-5	15	15
Long. Q (SUS)	18	-18	134	-134	-52	52	96	-96
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1



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Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	49.0	50.0	49.0	50.0	49.0	50.0	49.0	50.0
Pm+Pl (SUS)	48.0	49.0	60.1	61.0	41.0	42.0	73.0	74.0
Pm+Pl+Q (Total)	54.2	44.4	175.1	115.1	131.0	214.0	397.1	250.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	50.00	138.00	Passed
Pm+Pl (SUS)	74.03	207.00	Passed
Pm+Pl+Q (TOTAL)	397.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	16	17	16	17	16	17	16	17
Circ. Pl (SUS)	-12	-12	24	24	-5	-5	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-15	-15	19	19	-15	-15	19	19
Long. Q (SUS)	-102	102	208	-208	-212	212	318	-318
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	16.0	17.0	16.0	17.0	16.0	17.0	16.0	17.0
Pm+Pl (SUS)	12.5	13.4	40.7	41.6	19.0	19.9	34.2	35.1
Pm+Pl+Q (Total)	116.4	97.1	235.3	222.6	230.6	205.3	345.6	326.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	17.00	118.00	Passed
Pm+Pl (SUS)	41.62	177.00	Passed
Pm+Pl+Q (TOTAL)	345.63	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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Input, Nozzle Desc: Shell outlet From: 50

Pressure for Reinforcement Calculations	P	6110.449	KPa.
Temperature for Internal Pressure	Temp	275	°C
Design External Pressure	Pext	103.00	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1066.00	mm
Design Length of Section	L	7538.8335	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		8104.70	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

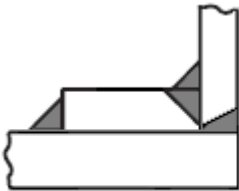
Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	15.0876	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	473.0500	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	22.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		100.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	



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The Pressure Design option was Design Pressure + static head.



Abutting/Set-on Nozzle With Pad

Reinforcement CALCULATION, Description: Shell outlet

ASME Code, Section VIII, Div. 1, 2021, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750 in.
Actual Thickness Used in Calculation	0.520 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6110.45 \cdot 535.0 / (138 \cdot 1.0 - 0.6 \cdot 6110.45)$
 $= 24.3375 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6110.45 \cdot 136.525 / (118 \cdot 1.0 + 0.4 \cdot 6110.45)$
 $= 6.9268 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9197 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	505.2934	mm
Parallel to Vessel Wall, opening length	d	252.6467	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		53.5041	mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137.0 / 138.0)$
 $= 0.993$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 0.993)$
 $= 0.855$



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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	61.488	9.924	NA
Area in Shell	A1	4.200	45.841	NA
Area in Nozzle Wall	A2	2.996	8.493	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	5.317	5.317	NA
Area in Element	A5	55.595	55.595	NA
TOTAL AREA AVAILABLE	Atot	68.108	115.246	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 449.2329 28.0000 mm
 Based on given Pad Diameter: 473.0500 24.6656 mm
 Based on the Estimated Diameter Limit: 503.7059 21.3874 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (252.6467 \cdot 24.3375 \cdot 1.0 + 2 \cdot 10.2016 \cdot 24.3375 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 61.488 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 252.647(1.0 \cdot 26.0 - 1.0 \cdot 24.337) - 2 \cdot 10.202$$

$$(1.0 \cdot 26.0 - 1.0 \cdot 24.3375) \cdot (1 - 1.0)$$

$$= 4.200 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 53.5) \cdot (10.2 - 6.93) \cdot 0.8551$$

$$= 2.996 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$$

$$= (1.) \cdot 0.86 + (0.0) \cdot 0.86 + 449.4456^2 \cdot 0.99$$

$$= 5.317 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (473.05 - 273.05) \cdot 28.0 \cdot 0.9928$$

$$= 55.595 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 9.9268 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 5.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 26.3375 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 26.3375 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.3995 \text{ mm}$



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Wall Thickness $tb2 = \max(trb2, tr16b) = 5.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 11.1026 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.103, \max(26.3375, 5.5)]$
 $= 11.1026 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(9.9268, 11.1026)$
 $= 11.1026 \text{ mm}$

Available Nozzle Neck Thickness = 13.2016 mm→OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	57.3,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	237.7 N./mm ²	Passed
Occasional	:	36.4,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	16.6,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 13.202$, $t_r = 6.927$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio $= t_r * E^* / (t_g - c) = 0.679$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-18 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 13.202$, $t_r = 6.927$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio $= t_r * E^* / (t_g - c) = 0.679$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-38 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 13.202$, $t_r = 6.927$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio $= t_r * E^* / (t_g - c) = 0.679$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-38 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 28.0$, $t_r = 24.337$, $c = 2.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio $= t_r * E^* / (t_g - c) = 0.936$, Temp. Reduction = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	2 °C
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Min Metal Temp. at Required thickness (UCS 66.1) -1 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.202, tr = 6.927, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.679$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -20 °C

Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -18 °C

Gov. MDMT of the Reinforcement Pad : -1 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -40 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $6110.45/10210.00 = 0.598$

Weld Size Calculations, Description: Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 10.2016 mm

Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	15.5540 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= (Ar - A1) • Sv

= (61.4879 - 4.2003) • 138

= 80612.62 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

= $(A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$

= (2.9965 + 55.5946 + 5.317 - 0.0 • 0.86) • 138

= 89928.70 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

= $(\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$

= (3.1416/2.0) • 273.05 • 10.0 • 0.49 • 118

= 25287. Kgf

Shear, Pad Element Weld [Spew]:

= $(\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$

= (3.1416/2.0) • 473.05 • 22.0 • 0.49 • 137

= 111901. Kgf



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Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 273.05 \cdot 28.0 \cdot 0.74 \cdot 137$
 $= 124148. \text{ Kg}$

Shear, Nozzle Groove Weld [Sngw]:
 $= (\pi/2) \cdot (Dlr+Wgnva) \cdot (Wgnva-Can) \cdot 0.60 \cdot Sng$
 $= (3.14/2) \cdot (252.647+15.088) \cdot (15.088-3.0) \cdot 0.6 \cdot 138$
 $= 42920. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNGW}) = (111901 + 42920) = 154820 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 154820 Kg, must exceed W = 80612 Kg or W1 = 89928 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6415 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 17.7817 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 345.7816 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 37.35 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 252.65 mm

Input Echo, WRC297 Item 1, Description: Shell outlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Corrosion Allowance	2.0000	mm
Shell Diameter	1066.000	mm
Shell Thickness	28.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

Using 2 * Yield for Discontinuity Stress Allowable (Div 2, 4.1.6.3), Sps.
 Make sure that material properties at this temperature are not time-dependent for Material: SA-516 70

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	273.050	mm
Nozzle Thickness	13.2016	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

Using 2 * Yield for Discontinuity Stress Allowable (Div 2, 4.1.6.3), Sps.
 Make sure that material properties at this temperature are not



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time-dependent for Material: SA-106 B

Diameter of Reinforcing Pad 473.050 mm
 Thickness of Reinforcing Pad 28.0000 mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	6110.45	KPa.
Radial Load	P	-1912.11	Kgf
Circumferential Shear	Vc	1274.73	Kgf
Longitudinal Shear	Vl	1274.73	Kgf
Circumferential Moment	Mc	891.97	Kg-m.
Longitudinal Moment	Ml	891.97	Kg-m.
Torsional Moment	Mt	1274.24	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(562.0(28.0 - 2.0))}$$

$$= 671.954 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1124.000 mm
Nozzle Outside Diameter	(d) =	273.050 mm
Vessel Thickness used	(T) =	54.000 mm
Nozzle Thickness used	(t) =	10.202 mm
T / t	=	5.293
d / t	=	26.765

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.108$$

Nr/P	=	0.068
Mr/P	=	0.048
M0/P	=	0.126
N0/P	=	0.311
MrD/Mc	=	0.089
NrDL/Mc	=	0.102
M0d/Mc	=	0.274
N0DL/Mc	=	0.200
MrD/Ml	=	0.081
NrDL/Ml	=	0.093
M0D/Ml	=	0.220
N0DL/Ml	=	0.408

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1



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Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	57	63	57	63

Outplane Stress Summary	45	70	83	50
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-5	5	5	-5
Inplane Pressure Stress	28	28	28	28

Inplane Stress Summary	24	31	37	23
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	45	71	83	50
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	28	28	28	28

Outplane Stress Summary	15	41	55	10
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right



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circumferential plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-5	5	5	-5
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	57	63	57	63
Inplane Stress Summary	52	67	66	57

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	52	67	67	57

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	87	94	87	94
Hoop Stress Summary	84	92	93	101

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	14	-14	14	-14
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-16	-16	16	16
Axial Bending (ML)	-63	63	63	-63
Axial Pressure Stress	43	43	43	43
Axial Stress Summary	-18	77	140	-15



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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	13	13	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	107	100	141	117

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	87	94	87	94
Hoop Stress Summary	87	94	91	98

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	14	-14	14	-14
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-69	69	69	-69
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	43	43	43	43
Axial Stress Summary	-24	83	146	-21

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10
Shear Stress Summary	7	7	13	13

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside



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Two * Max Shear Stress 112 98 149 123

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(548.0(28.0 - 2.0))}$
 $= 666.954 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1096.000 mm
 Nozzle Outside Diameter (d) = 473.050 mm
 Vessel Thickness used (T) = 26.000 mm
 Nozzle Thickness used (t) = 110.202 mm
 T / t = 0.236
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.802$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.060
 Mr/P = 0.112
 M0/P = 0.034
 N0/P = 0.068
 MrD/Mc = 0.271
 NrDL/Mc = 0.092
 M0d/Mc = 0.081
 N0DL/Mc = 0.085
 MrD/Ml = 0.089
 NrDL/Ml = 0.066
 M0D/Ml = 0.025
 N0DL/Ml = 0.083

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-2	-2	2	2
Outplane Bending (Ml)	-4	4	4	-4
Normal Pressure Stress	122	128	122	128
Outplane Stress Summary	123	126	136	123

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



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Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-14	14	14	-14
Inplane Pressure Stress	61	61	61	61
Inplane Stress Summary	65	57	98	31

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0	0
Outplane Shear (Vl)	0	0	0	0	0
Outplane Shear (Mt)	1	1	1	1	1
Shear Stress Summary	2	2	0	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	123	127	136	123	123

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1	1	1	1	1
Outplane Bending (P)	5	-5	5	-5	-5
Outplane Membrane (Mc)	-2	-2	2	2	2
Outplane Bending (Mc)	-13	13	13	-13	-13
Outplane Membrane (ML)	0	0	0	0	0
Outplane Bending (ML)	0	0	0	0	0
Normal Pressure Stress	61	61	61	61	61
Outplane Stress Summary	53	68	84	46	46

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	1	1	1	1	1
Inplane Bending (P)	18	-18	18	-18	-18
Inplane Membrane (Mc)	-2	-2	2	2	2
Inplane Bending (Mc)	-44	44	44	-44	-44
Inplane Membrane (ML)	0	0	0	0	0
Inplane Bending (ML)	0	0	0	0	0
Inplane Pressure Stress	122	128	122	128	128
Inplane Stress Summary	96	153	190	69	69

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0	0



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Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	96	153	190	70

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	57	63	57	63	57	63	57	63
Circ. Pl (SUS)	-3	-3	5	5	-1	-1	1	1
Circ. Q (SUS)	-10	10	18	-18	-4	4	6	-6
Long. Pm (SUS)	28	28	28	28	28	28	28	28
Long. Pl (SUS)	-1	-1	1	1	-1	-1	3	3
Long. Q (SUS)	-4	4	6	-6	-14	14	22	-22
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	57.0	63.0	57.0	63.0	57.0	63.0	57.0	63.0
Pm+Pl (SUS)	54.0	60.0	62.0	68.0	56.0	62.0	58.0	64.0
Pm+Pl+Q (Total)	44.0	70.0	80.0	50.0	52.0	66.0	64.1	58.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	63.00	138.00	Passed
Pm+Pl (SUS)	68.00	207.00	Passed
Pm+Pl+Q (TOTAL)	80.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	122	128	122	128	122	128	122	128
Circ. Pl (SUS)	-1	-1	3	3	-1	-1	3	3



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Circ. Q (SUS)	1	-1	9	-9	-26	26	62	-62
Long. Pm (SUS)	61	61	61	61	61	61	61	61
Long. Pl (SUS)	0	0	2	2	-1	-1	3	3
Long. Q (SUS)	4	-4	32	-32	-8	8	18	-18
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	122.0	128.0	122.0	128.0	122.0	128.0	122.0	128.0
Pm+Pl (SUS)	121.0	127.0	125.0	131.0	121.0	127.0	125.0	131.0
Pm+Pl+Q (Total)	122.0	126.0	134.0	122.0	95.0	153.0	187.0	69.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	128.00	138.00	Passed
Pm+Pl (SUS)	131.00	207.00	Passed
Pm+Pl+Q (TOTAL)	187.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	87	94	87	94	87	94	87	94
Circ. Pl (SUS)	-3	-3	5	5	-1	-1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	43	43	43	43	43	43	43	43
Long. Pl (SUS)	-14	-14	18	18	-14	-14	18	18
Long. Q (SUS)	-49	49	77	-77	-55	55	83	-83
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	87.0	94.0	87.0	94.0	87.0	94.0	87.0	94.0
Pm+Pl (SUS)	84.1	91.1	92.1	99.1	86.1	93.1	90.1	97.1
Pm+Pl+Q (Total)	106.7	98.1	139.4	116.1	113.1	97.7	146.5	121.4

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	94.00	118.00	Passed



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Pm+Pl (SUS)	99.10	177.00	Passed
Pm+Pl+Q (TOTAL)	146.55	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: Shell Inlet

From: 50

Pressure for Reinforcement Calculations	P	6100.000	KPa.
Temperature for Internal Pressure	Temp	275	°C
Design External Pressure	Pext	103.00	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1066.00	mm
Design Length of Section	L	7538.8335	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2604.70	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	15.0876	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	473.0500	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	22.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		100.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	



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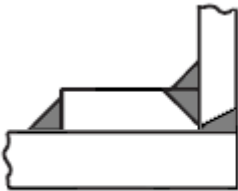
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The Pressure Design option was Design Pressure + static head.



Abutting/Set-on Nozzle With Pad

Reinforcement CALCULATION, Description: Shell Inlet

ASME Code, Section VIII, Div. 1, 2021, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750 in.
Actual Thickness Used in Calculation	0.520 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6100.0 \cdot 535.0 / (138 \cdot 1.0 - 0.6 \cdot 6100.0)$
 $= 24.2947 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6100.0 \cdot 136.525 / (118 \cdot 1.0 + 0.4 \cdot 6100.0)$
 $= 6.9152 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9197 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	505.2934	mm
Parallel to Vessel Wall, opening length	d	252.6467	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		53.5041	mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137.0 / 138.0)$
 $= 0.993$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 0.993)$
 $= 0.855$



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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	61.380	9.924	NA
Area in Shell	A1	4.308	45.841	NA
Area in Nozzle Wall	A2	3.007	8.493	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	5.317	5.317	NA
Area in Element	A5	55.595	55.595	NA
TOTAL AREA AVAILABLE	Atot	68.227	115.246	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 448.4177 28.0000 mm
 Based on given Pad Diameter: 473.0500 24.5515 mm
 Based on the Estimated Diameter Limit: 503.7059 21.2884 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (252.6467 \cdot 24.2947 \cdot 1.0 + 2 \cdot 10.2016 \cdot 24.2947 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 61.380 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 252.647(1.0 \cdot 26.0 - 1.0 \cdot 24.295) - 2 \cdot 10.202$$

$$(1.0 \cdot 26.0 - 1.0 \cdot 24.2947) \cdot (1 - 1.0)$$

$$= 4.308 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 53.5) \cdot (10.2 - 6.92) \cdot 0.8551$$

$$= 3.007 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$$

$$= (1.) \cdot 0.86 + (0.0) \cdot 0.86 + 449.4456^2 \cdot 0.99$$

$$= 5.317 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (473.05 - 273.05) \cdot 28.0 \cdot 0.9928$$

$$= 55.595 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 9.9152 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 5.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 26.2947 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 26.2947 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.3995 \text{ mm}$



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Wall Thickness tb2 = max(trb2, tr16b) = 5.5000 mm
 Wall Thickness per table UG-45 tb3 = 11.1026 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[11.103, max(26.2947, 5.5)]
 = 11.1026 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(9.9152, 11.1026)
 = 11.1026 mm

Available Nozzle Neck Thickness = 13.2016 mm→OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	57.3,	Allowable :	118.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	237.7 N./mm ²	Passed
Occasional :	36.3,	Allowable :	156.9 N./mm ²	Passed
Shear :	16.6,	Allowable :	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 13.202, tr = 6.915, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.678$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-18 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 13.202, tr = 6.915, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.678$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-38 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 13.202, tr = 6.915, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.678$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-38 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 28.0, tr = 24.295, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.934$, Temp. Reduction = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	2 °C
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Min Metal Temp. at Required thickness (UCS 66.1) -1 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.202, tr = 6.915, c = 3.0 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.678, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -20 °C

Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -18 °C

Gov. MDMT of the Reinforcement Pad : -1 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -40 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 6100.00/10210.00 = 0.597

Weld Size Calculations, Description: Shell Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 10.2016 mm

Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	15.5540 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= (Ar - A1) • Sv

= (61.3799 - 4.3083) • 138

= 80308.66 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv

= (3.0071 + 55.5946 + 5.317 - 0.0 • 0.86) • 138

= 89943.63 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

= (π/2) • Dlo • Wo • 0.49 • Snw

= (3.1416/2.0) • 273.05 • 10.0 • 0.49 • 118

= 25287. Kgf

Shear, Pad Element Weld [Spew]:

= (π/2) • DP • WP • 0.49 • SEW

= (3.1416/2.0) • 473.05 • 22.0 • 0.49 • 137

= 111901. Kgf



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Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 273.05 \cdot 28.0 \cdot 0.74 \cdot 137$
 $= 124148. \text{ Kg}$

Shear, Nozzle Groove Weld [Sngw]:
 $= (\pi/2) \cdot (Dlr+Wgnva) \cdot (Wgnva-Can) \cdot 0.60 \cdot Sng$
 $= (3.14/2) \cdot (252.647+15.088) \cdot (15.088-3.0) \cdot 0.6 \cdot 138$
 $= 42920. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNGW}) = (111901 + 42920) = 154820 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 154820 Kg, must exceed W = 80308 Kg or W1 = 89943 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6415 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 17.7817 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 345.7816 mm

Input Echo, WRC297 Item 1, Description: Shell Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Corrosion Allowance	2.0000	mm
Shell Diameter	1066.000	mm
Shell Thickness	28.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

Using 2 * Yield for Discontinuity Stress Allowable (Div 2, 4.1.6.3), Sps.
 Make sure that material properties at this temperature are not
 time-dependent for Material: SA-516 70

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	273.050	mm
Nozzle Thickness	13.2016	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

Using 2 * Yield for Discontinuity Stress Allowable (Div 2, 4.1.6.3), Sps.
 Make sure that material properties at this temperature are not
 time-dependent for Material: SA-106 B

Diameter of Reinforcing Pad	473.050	mm
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Thickness of Reinforcing Pad 28.0000 mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	6100.00	KPa.
Radial Load	P	-1912.11	Kgf
Circumferential Shear	Vc	1274.73	Kgf
Longitudinal Shear	Vl	1274.73	Kgf
Circumferential Moment	Mc	891.97	Kg-m.
Longitudinal Moment	Ml	891.97	Kg-m.
Torsional Moment	Mt	1274.24	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(562.0(28.0 - 2.0))}$$

$$= 671.954 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1124.000 mm
Nozzle Outside Diameter	(d) =	273.050 mm
Vessel Thickness used	(T) =	54.000 mm
Nozzle Thickness used	(t) =	10.202 mm
T / t	=	5.293
d / t	=	26.765

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.108$$

Nr/P	=	0.068
Mr/P	=	0.048
M0/P	=	0.126
N0/P	=	0.311
MrD/Mc	=	0.089
NrDL/Mc	=	0.102
M0d/Mc	=	0.274
N0DL/Mc	=	0.200
MrD/Ml	=	0.081
NrDL/Ml	=	0.093
M0D/Ml	=	0.220
N0DL/Ml	=	0.408

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4



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Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	57	63	57	63
Outplane Stress Summary	45	70	83	50

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-5	5	5	-5
Inplane Pressure Stress	28	28	28	28
Inplane Stress Summary	24	31	37	23

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	45	70	83	50

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	28	28	28	28
Outplane Stress Summary	15	41	55	10

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1



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Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-5	5	5	-5
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	57	63	57	63
Inplane Stress Summary	52	66	66	57

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	52	67	67	57

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	87	94	87	94
Hoop Stress Summary	84	91	93	100

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	14	-14	14	-14
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-16	-16	16	16
Axial Bending (ML)	-63	63	63	-63
Axial Pressure Stress	43	43	43	43
Axial Stress Summary	-18	77	140	-15

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside



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Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	13	13	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	106	99	141	117

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	87	94	87	94
Hoop Stress Summary	86	94	91	98

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	14	-14	14	-14
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-69	69	69	-69
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	43	43	43	43
Axial Stress Summary	-24	83	146	-21

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10
Shear Stress Summary	7	7	13	13

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	112	98	149	123

Stress Computations at the Edge of the Reinforcing Pad:



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Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(548.0(28.0 - 2.0))}$$

$$= 666.954 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1096.000 mm
Nozzle Outside Diameter	(d) =	473.050 mm
Vessel Thickness used	(T) =	26.000 mm
Nozzle Thickness used	(t) =	110.202 mm
T / t	=	0.236
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.802$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.060
Mr/P	=	0.112
M0/P	=	0.034
N0/P	=	0.068
MrD/Mc	=	0.271
NrDL/Mc	=	0.092
M0d/Mc	=	0.081
N0DL/Mc	=	0.085
MrD/Ml	=	0.089
NrDL/Ml	=	0.066
M0D/Ml	=	0.025
N0DL/Ml	=	0.083

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-4	4	4	-4
Normal Pressure Stress	122	128	122	128
Outplane Stress Summary	123	126	136	122

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-14	14	14	-14
Inplane Pressure Stress	61	61	61	61



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Inplane Stress Summary	65	57	97	31
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	0	0
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	123	126	136	122
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-13	13	13	-13
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	61	61	61	61

Outplane Stress Summary	53	68	84	46
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-44	44	44	-44
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	122	128	122	128

Inplane Stress Summary	95	153	189	69
------------------------	----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
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 Nozzle Calcs.: Shell Inlet

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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	95	153	189	69

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	57	63	57	63	57	63	57	63
Circ. Pl (SUS)	-3	-3	5	5	-1	-1	1	1
Circ. Q (SUS)	-10	10	18	-18	-4	4	6	-6
Long. Pm (SUS)	28	28	28	28	28	28	28	28
Long. Pl (SUS)	-1	-1	1	1	-1	-1	3	3
Long. Q (SUS)	-4	4	6	-6	-14	14	22	-22
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	57.0	63.0	57.0	63.0	57.0	63.0	57.0	63.0
Pm+Pl (SUS)	54.0	60.0	62.0	68.0	56.0	62.0	58.0	64.0
Pm+Pl+Q (Total)	44.0	70.0	80.0	50.0	52.0	66.0	64.1	58.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	63.00	138.00	Passed
Pm+Pl (SUS)	68.00	207.00	Passed
Pm+Pl+Q (TOTAL)	80.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	122	128	122	128	122	128	122	128
Circ. Pl (SUS)	-1	-1	3	3	-1	-1	3	3
Circ. Q (SUS)	1	-1	9	-9	-26	26	62	-62
Long. Pm (SUS)	61	61	61	61	61	61	61	61
Long. Pl (SUS)	0	0	2	2	-1	-1	3	3



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Long. Q (SUS)	4	-4	32	-32	-8	8	18	-18
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	122.0	128.0	122.0	128.0	122.0	128.0	122.0	128.0
Pm+Pl (SUS)	121.0	127.0	125.0	131.0	121.0	127.0	125.0	131.0
Pm+Pl+Q (Total)	122.0	126.0	134.0	122.0	95.0	153.0	187.0	69.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	128.00	138.00	Passed
Pm+Pl (SUS)	131.00	207.00	Passed
Pm+Pl+Q (TOTAL)	187.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	87	94	87	94	87	94	87	94
Circ. Pl (SUS)	-3	-3	5	5	-1	-1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	43	43	43	43	43	43	43	43
Long. Pl (SUS)	-14	-14	18	18	-14	-14	18	18
Long. Q (SUS)	-49	49	77	-77	-55	55	83	-83
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	87.0	94.0	87.0	94.0	87.0	94.0	87.0	94.0
Pm+Pl (SUS)	84.1	91.1	92.1	99.1	86.1	93.1	90.1	97.1
Pm+Pl+Q (Total)	106.7	98.1	139.4	116.1	113.1	97.7	146.5	121.4

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	94.00	118.00	Passed
Pm+Pl (SUS)	99.10	177.00	Passed
Pm+Pl+Q (TOTAL)	146.55	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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 Nozzle Calcs.: Vent

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Input, Nozzle Desc: Vent**From: 50**

Pressure for Reinforcement Calculations	P	6100.000	KPa.
Temperature for Internal Pressure	Temp	275	°C
Design External Pressure	Pext	103.00	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1066.00	mm
Design Length of Section	L	7538.8335	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3604.70	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	200.3250	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		70.0000	mm
Flange Class		600	



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 Nozzle Calcs.: Vent

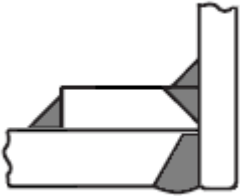
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Flange Grade

GR 1.1

The Pressure Design option was Design Pressure + static head.



Insert/Set-in Nozzle With Pad, no Inside projection

Reinforcement CALCULATION, Description: Vent

ASME Code, Section VIII, Div. 1, 2021, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.503 in.
Actual Thickness Used in Calculation	0.436 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6100.0 \cdot 535.0 / (138 \cdot 1.0 - 0.6 \cdot 6100.0)$
 $= 24.2947 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6100.0 \cdot 22.09 / (118 \cdot 1.0 - 0.6 \cdot 6100.0)$
 $= 1.1785 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3478 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	112.3250	mm
Parallel to Vessel Wall	Rn+tn+t	56.1625	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		36.1860	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Taking a UG-36(c)(3)(a) exemption for nozzle: Vent.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 4.1785	mm
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 26.2947	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.2947	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3995	mm



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 FileName : 174-C_BEU_Dia 1000-AK
 Nozzle Calcs.: Vent

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Wall Thickness tb2 = max(trb2, tr16b) = 5.5000 mm
 Wall Thickness per table UG-45 tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[6.42, max(26.2947, 5.5)]
 = 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(4.1785, 6.42)
 = 6.4200 mm

Available Nozzle Neck Thickness = 0.875 * 11.074 = 9.690 mm—OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.69, tr = 1.178, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.176$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.69, tr = 1.178, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.176$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.69, tr = 1.178, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.176$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 16.0, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.934$, Temp. Reduction = 4 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-18 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.69, tr = 1.178, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.176$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



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 Nozzle Calcs.: Vent

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Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -40 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 6100.00/10210.00 = 0.597

Weld Size Calculations, Description: Vent

Intermediate Calc. for nozzle/shell Welds Tmin 8.0744 mm
 Intermediate Calc. for pad/shell Welds TminPad 16.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6521 = 0.7 * tmin.	14.1400 = 0.7 * Wo mm
Pad Weld	8.0000 = 0.5 * TminPad	8.4840 = 0.7 * Wp mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6506 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.8541 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 278.8541 mm

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 Nozzle Calcs.: Drain

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Input, Nozzle Desc: Drain**From: 50**

Pressure for Reinforcement Calculations	P	6110.449	KPa.
Temperature for Internal Pressure	Temp	275	°C
Design External Pressure	Pext	103.00	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1066.00	mm
Design Length of Section	L	7538.8335	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4604.70	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	200.3250	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	15.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		70.0000	mm
Flange Class		600	



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 Nozzle Calcs.: Drain

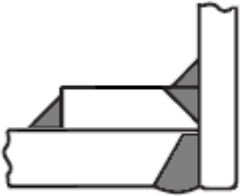
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Flange Grade

GR 1.1

The Pressure Design option was Design Pressure + static head.



Insert/Set-in Nozzle With Pad, no Inside projection

Reinforcement CALCULATION, Description: Drain

ASME Code, Section VIII, Div. 1, 2021, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.503 in.
 Actual Thickness Used in Calculation 0.436 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6110.45 \cdot 535.0 / (138 \cdot 1.0 - 0.6 \cdot 6110.45)$
 $= 24.3375 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6110.45 \cdot 22.09 / (118 \cdot 1.0 - 0.6 \cdot 6110.45)$
 $= 1.1806 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3478 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	112.3250	mm
Parallel to Vessel Wall	Rn+tn+t	56.1625	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		36.1860	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Taking a UG-36(c)(3)(a) exemption for nozzle: Drain.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 4.1806	mm
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 26.3375	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.3375	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3995	mm



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 Nozzle Calcs.: Drain

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Wall Thickness tb2 = max(trb2, tr16b) = 5.5000 mm
 Wall Thickness per table UG-45 tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[6.42, max(26.3375, 5.5)]
 = 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(4.1806, 6.42)
 = 6.4200 mm

Available Nozzle Neck Thickness = 0.875 * 11.074 = 9.690 mm—OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.69, tr = 1.181, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.176$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.69, tr = 1.181, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.176$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.69, tr = 1.181, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.176$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 16.0, c = 2.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.936$, Temp. Reduction = 4 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-18 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.69, tr = 1.181, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.176$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



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 FileName : 174-C_BEU_Dia 1000-AK
 Nozzle Calcs.: Drain

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Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -40 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $6110.45/10210.00 = 0.598$

Weld Size Calculations, Description: Drain

Intermediate Calc. for nozzle/shell Welds Tmin 8.0744 mm
 Intermediate Calc. for pad/shell Welds TminPad 16.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6521 = 0.7 * t_{min}$	$14.1400 = 0.7 * W_o$ mm
Pad Weld	$8.0000 = 0.5 * T_{minPad}$	$10.6050 = 0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6516 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.8541 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 278.8541 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 37.35 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 56.16 mm

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Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	6100.00	KPa.
Shell Maximum Operating Pressure	Psox,max	6100.00	KPa.
Shell Minimum Operating Pressure	Psox,min	0.00	KPa.
Shell Thickness	ts	28.0000	mm
Shell Internal Corrosion Allowance	cas	2.0000	mm
Shell External Corrosion Allowance	caext	0.0000	mm
Inside Diameter of Shell	Ds	1066.000	mm
Shell Temperature for Internal Pressure	Ts	275.00	°C
Shell Material		SA-516 70	

Using 2 * Yield for Discontinuity Stress Allowable (UG-23(e)), Sps.
 Make sure that material properties at this temperature are not
 time-dependent for Material: SA-516 70

Shell Material UNS Number		K02700	
Shell Allowable Stress at Temperature	Ss	137.01	N./mm ²
Shell Allowable Stress at Ambient		138.01	N./mm ²

Channel Description: Channel shell

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	1000.00	KPa.
Channel Maximum Operating Pressure	Ptox,max	1000.00	KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00	KPa.
Channel Thickness	tc	14.0000	mm
Channel Corrosion Allowance	cac	3.2000	mm
Inside Diameter of Channel	Dc	1060.000	mm
Channel Design Temperature	TEMPC	82.00	°C
Channel Material		SA-516 70	

Using 2 * Yield for Discontinuity Stress Allowable (UG-23(e)), Sps.
 Make sure that material properties at this temperature are not
 time-dependent for Material: SA-516 70

Channel Material UNS Number		K02700	
Channel Allowable Stress at Temperature	Sc	138.01	N./mm ²
Channel Allowable Stress at Ambient		138.01	N./mm ²

Tube Data:

Number of Tube Holes	Nt	682	
Tube Wall Thickness	et	2.7700	mm
Tube Outside Diameter	D	25.4000	mm
Total Straight Tube Length	Lt	7100.00	mm
Straight Tube Length (bet. inner tubsht faces)	L	6825.00	mm
Design Temperature of the Tubes		275.00	°C
Tube Material		SA-213 TP304L	
Tube Material UNS Number		S30403	
Is this a Welded Tube		No	
Tube Material Specification used		Smls. tube	
Tube Allowable Stress at Temperature		100.35	N./mm ²
Tube Allowable Stress At Ambient		115.00	N./mm ²
Tube Yield Stress At design Temperature	Syt	111.00	N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500	mm
Tube Layout Pattern		Triangular	
Fillet Weld Leg	af	2.4000	mm



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Groove Weld Leg	ag	2.4000	mm
Tube-Tubesheet Joint Weld Type		Full Strength	
Method for Tube-Tubesheet Jt. Allow.		UW-20	
Is the Tube Inset		No	
Radius to Outermost Tube Hole Center	ro	509.427	mm
Largest Center-to-Center Tube Distance	U1	76.2000	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	8.0000	mm
Area of Tube Layout	Ap	6160.6	cm ²
Perimeter of Tube Layout	Cp	3009.9	mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature	T	275.00	°C
Tubesheet Material		SA-965 F304L	
Tubesheet Material UNS Number		S30403	
Tubesheet Allowable Stress at Temperature	S	74.15	N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	115.00	N./mm ²
Thickness of Tubesheet	h	275.0000	mm
Tubesheet Corr. Allowance (Shell side)	Cats	13.0000	mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000	mm
Tubesheet Outside Diameter	A	1380.000	mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		YES	
Is the Tubesheet Flat on Tubeside?		NO	
Step 1 Diameter on the Tubeside	dt1	1119.00	mm
Step 1 Depth on the Tubeside	ht1	8.00	mm
Step 2 Diameter on the Tubeside	dt2	1047.00	mm
Step 2 Depth on the Tubeside	ht2	8.00	mm
Is the Tubesheet Flat on Shellside?		NO	
Calculated Tubesheet Diameter as per UHX-10(b)		1380.00	mm

Note: Tubesheet diameter is now: 1380.000 mm per UHX-10(b).

Dimension G for the Channel Side	Gc	1105.151	mm
Area of the Untubed Lanes	AL	700.1	cm ²

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	NO
Run Multiple Load Cases	YES

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both	
Flange Outside Diameter	A	1380.000	mm
Flange Inside Diameter	B	1056.000	mm
Flange Face Outside Diameter	Fod	1143.000	mm
Flange Face Inside Diameter	Fid	1056.000	mm
Gasket Outside Diameter	Go	1118.000	mm
Gasket Inside Diameter	Gi	1092.000	mm
Small end Hub thk.	g0	26.0000	mm
Large end Hub thk.	g1	36.0000	mm
Gasket Factor,	m	3.00	
Gasket Design Seating Stress	y	68.95	N./mm ²
Flange Facing Sketch	Code Sketch 1a		
Column for Gasket Seating	Code Column II		
Gasket Thickness	tg	6.3500	mm



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Full face Gasket Flange Option	Program Selects	
Length of Partition Gasket	lp	1050.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²
Partition Gasket Facing Sketch	Code	Sketch 1a
Partition Gasket Column for Gasket Seating	Code	Column II

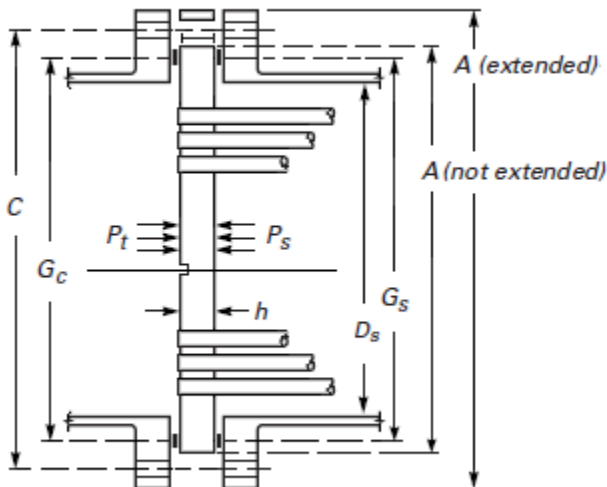
Bolting Information:

Diameter of Bolt Circle	C	1270.000 mm
Nominal Bolt Diameter	dB	47.6250 mm
Type of Thread Series	TEMA	Thread Series
Number of Bolts	n	36
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Alternate Flange Operating Bolt Load, Wm1	699550.75 Kgf
Alternate Flange Seating Bolt Load, Wm2	193738.56 Kgf
Alternate Flange Design Bolt Load, W	938521.75 Kgf

Tubesheet Integral with	None
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 259.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Tubesheet Gasketed With Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX, 2021

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	275.0 °C	0.18674E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%



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Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	275.0 °C	0.17745E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Material Group G

Elastic Mod. at Design Temperature	275.0 °C	0.17745E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(1000.0 \cdot (25.4/2 - 0.0)) / (100.35 \cdot 1.0 + 0.4 \cdot 1000.0)$

= $0.1261 + 0.0000 = 0.1261$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	HA-3	at	275.00 °C
Elastic Modulus for Material			172442880.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.7700	25.40	7334.43	9.17	50.0000	0.0130823	54.05

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 8270.4014$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.2947	25.40	7334.43	11.07	50.0000	0.0089782	50.65

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 50.6494) / (3 \cdot 11.0688) = 6100.8013$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.2947	mm
Allowable Internal Pressure at Corroded thickness	23978.19	KPa.
Required Internal Design Pressure	1000.00	KPa.
Allowable External Pressure at Corroded thickness	8270.40	KPa.
Required External Design Pressure	6100.00	KPa.
Required Thickness due to Shell Side pressure	2.2947	mm

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2021

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000	mm
Basic Gasket Width,	$b0 = N / 2$	6.500	mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422	mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	1105.156	mm

Required Tubesheet Flanged Extension thk. per UHX-9 (operating) [hropr]:

= $\sqrt{(1.9 \cdot WM1 \cdot hg / (S \cdot G))}$

= $\sqrt{(1.9 \cdot 114676.36 \cdot 82.422 / (74.15 \cdot 1105.151))}$



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= 46.359 mm

Required Tubesheet Flanged Extension thk. per UHX-9 (seating) [hrseat]:

$$= \sqrt{\frac{1.9 \cdot W \cdot h_g}{(S_a \cdot G)}} \\
= \sqrt{\frac{1.9 \cdot 938521 \cdot 82.422}{(115.0 \cdot 1105.151)}} \\
= 106.495 \text{ mm}$$

Gasket Reaction Diameter (Gc or Gs as applicable): 1105.156 mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2021, Original Thickness :

UHX-12.5.1 Step 1:

Compute the Equivalent Outer Tube Limit Circle Diameter [Do]:

$$= 2 \cdot r_o + dt \\
= 2 \cdot 509.427 + 25.4 = 1044.254 \text{ mm}$$

Determine the Basic Ligament Efficiency for Shear [μ]:

$$= (p - dt) / p \\
= (31.75 - 25.4) / 31.75 = 0.2$$

UHX-12.5.2 Step 2:

Compute the Ratio [ps]:

$$= G_s / D_o \text{ (Configurations d, e, f)} \\
= 1105.1555 / 1044.254 = 1.0583$$

Compute the Ratio [pc]:

$$= G_c / D_o \text{ (Configurations d)} \\
= 1105.1509 / 1044.254 = 1.0583$$

Moment on Tubesheet due to Pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot P_s - (pc - 1) \cdot (pc^2 + 1) \cdot P_t] \\
= 1044.254^2 / 16 \cdot [(1.058 - 1) \cdot (1.058^2 + 1) \cdot 0.0 - \\
(1.058 - 1) \cdot (1.058^2 + 1) \cdot 1000.0] \\
= -8426039.0000 \text{ KPa} \cdot \text{mm}^2$$

UHX-12.5.3 Step 3, Determination of Effective Elastic Properties :

Compute the Ratio [p]:

$$= l_{tx} / h = 0.0 / 259.0 = 0.0 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Compute the Effective Tube Hole Diameter [d*]:

$$= \max(dt - 2tt(E_t T / E)(S_t / S)(p), dt - 2tt) \\
= \max(25.4 - 2 \cdot 2.77 (.17745E+09 / .17745E+09) \cdot \\
(100 / 74)(0.0), 25.4 - 2 \cdot 2.77) \\
= 25.4000 \text{ mm}$$

Compute the Effective Tube Pitch [p*]:

$$= p / \sqrt{1 - 4 \cdot \min(A_l, 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) } \\
= 31.75 / \sqrt{1 - 4 \cdot \min(700.09, 4 \cdot 1044.254 \cdot 31.75) / (\pi \cdot 1044.254^2) } \\
= 33.1331 \text{ mm}$$

Compute the Effective Ligament Efficiency for Bending [μ*]:

$$= (p^* - d^*) / p^* = (33.1331 - 25.4) / 33.1331 = 0.23339$$

E*/E and ν* for Triangular pattern from Fig. UHX-11.3.



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$h/p = 8.157480$; $\mu^* = 0.233394$; $E^* = 34745236$ KPa.
 $E^*/E = 0.195798$; $\nu^* = 0.414711$

Note: As h/p (8.157) is > 2 , data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

UHX-12.5.5 Step 5:

Diameter ratio [K]:
 $= A/D_o = 1380.0/1044.254 = 1.3215$

Coefficient [F]:
 $= (1 - \nu^*)/E^* \cdot (E \cdot \ln(K))$
 $= (1 - 0.41)/34745236 \cdot (0.17745E+09 \cdot \ln(1.32))$
 $= 0.8333$

UHX-12.5.6 Step 6:

Moment Acting on Unperforated Tubesheet Rim [M*]:
 $= M_{ts} + W^* \cdot (G_c - G_s)/(2 \cdot \pi \cdot D_o)$
 $= -8426039.0 + 114676.4 \cdot (1105.151 - 1105.151)/(2 \cdot \pi \cdot 1044.254)$
 $= -8426039.0000$ KPa. $\cdot$ mm²

UHX-12.5.7 Step 7:

Maximum Bending Moment acting on Periphery of Tubesheet [Mp]:
 $= ((M^*) - D_o^2/32 \cdot F(P_s - P_t)) / (1 + F)$
 $= ((-8426039.0) - 1044.254^2/32 \cdot 0.833 (0.0 - 1000.0)) / (1 + 0.83)$
 $= 10893726.0000$ KPa. $\cdot$ mm²

Maximum Bending Moment acting on Center of Tubesheet [Mo]:
 $= M_p + D_o^2/64 \cdot (3 + \nu^*)(P_s - P_t)$
 $= 10893726 + 1044.254^2/64 \cdot (3 + 0.415)(0.0 - 1000.0)$
 $= -47287964.0000$ KPa. $\cdot$ mm²

Maximum Bending Moment acting on Tubesheet [M]:
 $= \max(\text{abs}(M_p), \text{abs}(M_o))$
 $= \max(\text{abs}(10893726), \text{abs}(-0.5E+08))$
 $= 47287964.0000$ KPa. $\cdot$ mm²

UHX-12.5.8 Step 8:

Tubesheet Bending Stress at Original Thickness:
 $= 6 \cdot M / (\mu^* (h - h_g)^2)$
 $= 6 \cdot 47287964 / (0.2334 (259.0 - 5.0)^2)$
 $= 18.8439$ N./mm²

The Allowable Tubesheet Bending Stress [σ_{All}]:
 $= 2 \cdot S = 2 \cdot 74.15 = 148.3$ N./mm²

Tubesheet Bending Stress at Final Thickness [σ]:
 $= 6 \cdot M / (\mu^* (h - h_g)^2)$
 $= 6 \cdot 47287964 / (0.2334 (95.7256 - 5.0)^2)$
 $= 147.6991$ N./mm²

Required Tubesheet Thickness, for Bending Stress [HreqB]:
 $= h + C_{ats} + C_{atc} = 95.7256 + 13.0 + 3.0 = 111.7256$ mm

Required Tubesheet Thickness for Given Loadings (includes CA) [Hreq]:



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$$= \max(\text{HreqB}, \text{HreqS}) = \max(111.7256, 33.2988) = 111.7256 \text{ mm}$$

UHX-12.5.9 Step 9:

$$\text{abs}(P_s - P_t) = \text{abs}(0.0 - 1000.0) = 1000.0 \text{ KPa.}$$

Shear Stress check [rlimit]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S, 0.533 \cdot S_y) \\ &= 4 \cdot 0.2 \cdot 259.0 / 1044.25 \cdot \min(0.8 \cdot 74.15, 0.533 \cdot 111.001) \\ &= 11.739 \text{ N/mm}^2 \end{aligned}$$

[Ps - Pt] is ≤ the Shear Stress limit (rlimit). Therefore, the required thickness for shear is for information only.

Minimum Required Thickness for Shear [HreqS]:

$$\begin{aligned} &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S, 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot \text{abs}(P_s - P_t) + C_{at} + C_{atc} \\ &= 1 / (4 \cdot 0.2 \cdot 59.163) \cdot (4 \cdot 6160.6 / 3009.9) \cdot \text{abs}((0.0 - 1000.0)) + 16.0 \\ &= 33.2988 \text{ mm} \end{aligned}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (25.4 - 2.77) 2.77 \\ &= 1.969 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S_a \\ &= \pi \cdot 2.77 (25.4 - 2.77) 100.351 \\ &= 2015.143 \text{ Kg} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ &= 0.85 \cdot \pi \cdot 2.4 (25.4 + 0.67 \cdot 2.4) 74.15 \\ &= 1308.7521 \text{ Kg} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= S_a / S_w \text{ where } S_w = \min(S_a \text{ and } S_t) \\ &= 100.351 / 74.15 \\ &= 1.353 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\ &= 0.55 \cdot \pi \cdot 2.4 (25.4 + 0.67 \cdot 2.4) 74.15 \\ &= 846.8395 \text{ Kg} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right) \\ &= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1.353 \cdot 1.0)} - 0.75 \cdot 25.4 \right) \\ &= 4.499 \text{ mm} \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 2.2497 mm
 Minimum Required Groove Weld Leg [agr]: 2.2497 mm



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Tube-Tubesheet Jt allowable, 2015.14 is $\geq$ tube strength 2015.14 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	18.8 $\leq$	148.3 N./mm ²	Ok
Tubesheet shear stress	4.0 $\leq$	59.2 N./mm ²	Ok

Thickness results for loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	111.726	275.000	Ok
Tube-Tubesheet Fillet Weld Leg :	2.250	2.400	Ok
Tube-Tubesheet Groove Weld Leg :	2.250	2.400	Ok

Detailed Results for load Case D2 corr. (Psd,max + Ptd,min)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2021

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000	mm
Basic Gasket Width,	$b_0 = N / 2$	6.500	mm
Effective Gasket Width,	$b = \sqrt{b_0} * 2.5$	6.422	mm
Gasket Reaction Diameter,	$G = G_o - 2.0 * b$	1105.156	mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2021, Original Thickness :

UHX-12.5.1 Step 1:

Compute the Equivalent Outer Tube Limit Circle Diameter [Do]:
 $= 2 * r_o + d_t$
 $= 2 * 509.427 + 25.4 = 1044.254 \text{ mm}$

Determine the Basic Ligament Efficiency for Shear [μ]:
 $= (p - d_t) / p$
 $= (31.75 - 25.4) / 31.75 = 0.2$

UHX-12.5.2 Step 2:

Compute the Ratio [ρ_s]:
 $= G_s / D_o$ (Configurations d, e, f)
 $= 1105.1555 / 1044.254 = 1.0583$

Compute the Ratio [ρ_c]:
 $= G_c / D_o$ (Configurations d)
 $= 1105.1509 / 1044.254 = 1.0583$

Moment on Tubesheet due to Pressures (Ps, Pt) [Mts]:



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$$\begin{aligned}
 &= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt] \\
 &= 1044.254^2/16 \cdot [(1.058 - 1) \cdot (1.058^2 + 1) \cdot 6100.0 - \\
 &\quad (1.058 - 1) \cdot (1.058^2 + 1) \cdot 0.0] \\
 &= 51403064.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

UHX-12.5.3 Step 3, Determination of Effective Elastic Properties :

Compute the Ratio [p]:
 $= ltx/h = 0.0/259.0 = 0.0$ (must be $0 \leq p \leq 1$)

Compute the Effective Tube Hole Diameter [d*]:
 $= \max(dt - 2tt(EtT/E)(StT/S)(p), dt - 2tt)$
 $= \max(25.4 - 2 \cdot 2.77 (.17745E+09 / .17745E+09) \cdot$
 $\quad (100/74) \cdot (0.0) , 25.4 - 2 \cdot 2.77)$
 $= 25.4000 \text{ mm}$

Compute the Effective Tube Pitch [p*]:
 $= p / \sqrt{ (1 - 4 \cdot \min(Al, 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$
 $= 31.75 / \sqrt{ (1 - 4 \cdot \min(700.09, 4 \cdot 1044.254 \cdot 31.75) / (\pi \cdot 1044.254^2)) }$
 $= 33.1331 \text{ mm}$

Compute the Effective Ligament Efficiency for Bending [μ^*]:
 $= (p^* - d^*)/p^* = (33.1331 - 25.4) / 33.1331 = 0.23339$

E*/E and v* for Triangular pattern from Fig. UHX-11.3.
 $h/p = 8.157480$; $\mu^* = 0.233394$; $E^* = 34745236 \text{ KPa}$.
 $E^*/E = 0.195798$; $v^* = 0.414711$

Note: As h/p (8.157) is > 2, data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

UHX-12.5.5 Step 5:

Diameter ratio [K]:
 $= A/Do = 1380.0/1044.254 = 1.3215$

Coefficient [F]:
 $= (1 - v^*)/E^* \cdot (E \cdot \ln(K))$
 $= (1 - 0.41) / 34745236 \cdot (0.17745E+09 \cdot \ln(1.32))$
 $= 0.8333$

UHX-12.5.6 Step 6:

Moment Acting on Unperforated Tubesheet Rim [M*]:
 $= Mts + W^* \cdot (Gc - Gs) / (2 \cdot \pi \cdot Do)$
 $= 51403064 + 699525 \cdot (1105.151 - 1105.151) / (2 \cdot \pi \cdot 1044.254)$
 $= 51403064.0000 \text{ KPa} \cdot \text{mm}^2$

UHX-12.5.7 Step 7:

Maximum Bending Moment acting on Periphery of Tubesheet [Mp]:
 $= ((M^*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$
 $= ((51403064) - 1044.254^2/32 \cdot 0.833 (6100.0 - 0.0)) / (1 + 0.83)$
 $= -66449428.0000 \text{ KPa} \cdot \text{mm}^2$

Maximum Bending Moment acting on Center of Tubesheet [Mo]:
 $= Mp + Do^2/64 \cdot (3 + v^*)(Ps - Pt)$
 $= -0.7E+08 + 1044.254^2/64 \cdot (3 + 0.415) (6100.0 - 0.0)$
 $= 288458880.0000 \text{ KPa} \cdot \text{mm}^2$



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Maximum Bending Moment acting on Tubesheet [M]:
 $= \max(\text{abs}(M_p), \text{abs}(M_o))$
 $= \max(\text{abs}(-0.7\text{E}+08), \text{abs}(.28846\text{E}+09))$
 $= 288458880.0000 \text{ KPa} \cdot \text{mm}^2$

UHX-12.5.8 Step 8:

Tubesheet Bending Stress at Original Thickness:
 $= 6 \cdot M / (\mu \cdot (h - h_g)^2)$
 $= 6 \cdot 0.28846\text{E}+09 / (0.2334 (259.0 - 5.0)^2)$
 $= 114.9484 \text{ N./mm}^2$

The Allowable Tubesheet Bending Stress [σ_{All}]:
 $= 2 \cdot S = 2 \cdot 74.15 = 148.3 \text{ N./mm}^2$

Tubesheet Bending Stress at Final Thickness [σ]:
 $= 6 \cdot M / (\mu \cdot (h - h_g)^2)$
 $= 6 \cdot 0.28846\text{E}+09 / (0.2334 (228.6349 - 5.0)^2)$
 $= 148.2830 \text{ N./mm}^2$

Required Tubesheet Thickness, for Bending Stress [HreqB]:
 $= h + C_{ats} + C_{atc} = 228.6349 + 13.0 + 3.0 = 244.6349 \text{ mm}$

Required Tubesheet Thickness for Given Loadings (includes CA) [Hreq]:
 $= \max(H_{reqB}, H_{reqS}) = \max(244.6349, 121.5228) = 244.6349 \text{ mm}$

UHX-12.5.9 Step 9:

$\text{abs}(P_s - P_t) = \text{abs}(6100.0 - 0.0) = 6100.0 \text{ KPa}$.

Shear Stress check [τ_{limit}]:
 $= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S, 0.533 \cdot S_y)$
 $= 4 \cdot 0.2 \cdot 259.0 / 1044.25 \cdot \min(0.8 \cdot 74.15, 0.533 \cdot 111.001)$
 $= 11.739 \text{ N./mm}^2$

[$P_s - P_t$] is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum Required Thickness for Shear [HreqS]:
 $= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S, 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot \text{abs}(P_s - P_t) + C_{ats} + C_{atc}$
 $= 1 / (4 \cdot 0.2 \cdot 59.163) \cdot (4 \cdot 6160.6 / 3009.9) \cdot \text{abs}((6100.0 - 0.0)) + 16.0$
 $= 121.5228 \text{ mm}$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [At]:
 $= \pi (d_o - t) t$
 $= \pi (25.4 - 2.77) 2.77$
 $= 1.969 \text{ cm}^2$

Tube Weld Size Results per UW-20:

Tube Strength [Ft]:
 $= \pi \cdot t (d_o - t) S_a$
 $= \pi \cdot 2.77 (25.4 - 2.77) 100.351$
 $= 2015.143 \text{ Kg}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$



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$$= 0.85 \cdot \pi \cdot 2.4 (25.4 + 0.67 \cdot 2.4) 74.15$$

$$= 1308.7521 \text{ Kg}$$

Weld Strength Factor [fw]:
 $= S_a / S_w$ where $S_w = \min(S_a \text{ and } S_t)$
 $= 100.351 / 74.15$
 $= 1.353$

Design Strength Ratio [fd]:
 $= 1$ for full strength welds

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not $> F_t$)
 $= 0.55 \cdot \pi \cdot 2.4 (25.4 + 0.67 \cdot 2.4) 74.15$
 $= 846.8395 \text{ Kg}$

Minimum Length of Weld [ar]:
 $= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$
 $= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1.353 \cdot 1.0)} \right) -$
 $0.75 \cdot 25.4$
 $= 4.499 \text{ mm}$

Minimum Required Fillet Weld Leg [afr]: 2.2497 mm
 Minimum Required Groove Weld Leg [agr]: 2.2497 mm

Tube-Tubesheet Jt allowable, 2015.14 is $\geq$ tube strength 2015.14 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D2 corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	114.9 $\leq$	148.3 N./mm ²	Ok
Tubesheet shear stress	24.1 $\leq$	59.2 N./mm ²	Ok

Thickness results for loadcase D2 corr. (Psd,max + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	244.635	275.000	Ok
Tube-Tubesheet Fillet Weld Leg :	2.250	2.400	Ok
Tube-Tubesheet Groove Weld Leg :	2.250	2.400	Ok

Detailed Results for load Case D3 corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2021

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000	mm
Basic Gasket Width,	$b_0 = N / 2$	6.500	mm
Effective Gasket Width,	$b = \sqrt{b_0} \cdot 2.5$	6.422	mm
Gasket Reaction Diameter,	$G = G_o - 2.0 \cdot b$	1105.156	mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:



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Per Edition 2021, Original Thickness :

UHX-12.5.1 Step 1:

Compute the Equivalent Outer Tube Limit Circle Diameter [Do]:
 $= 2 \cdot ro + dt$
 $= 2 \cdot 509.427 + 25.4 = 1044.254 \text{ mm}$

Determine the Basic Ligament Efficiency for Shear [μ]:
 $= (p - dt) / p$
 $= (31.75 - 25.4) / 31.75 = 0.2$

UHX-12.5.2 Step 2:

Compute the Ratio [ρ_s]:
 $= G_s / Do$ (Configurations d, e, f)
 $= 1105.1555 / 1044.254 = 1.0583$

Compute the Ratio [ρ_c]:
 $= G_c / Do$ (Configurations d)
 $= 1105.1509 / 1044.254 = 1.0583$

Moment on Tubesheet due to Pressures (Ps, Pt) [Mts]:
 $= Do^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (pc - 1) \cdot (pc^2 + 1) \cdot Pt]$
 $= 1044.254^2 / 16 \cdot [(1.058 - 1) \cdot (1.058^2 + 1) \cdot 6100.0 - (1.058 - 1) \cdot (1.058^2 + 1) \cdot 1000.0]$
 $= 42977024.0000 \text{ KPa} \cdot \text{mm}^2$

UHX-12.5.3 Step 3, Determination of Effective Elastic Properties :

Compute the Ratio [ρ]:
 $= ltx/h = 0.0 / 259.0 = 0.0$ (must be $0 \leq \rho \leq 1$)

Compute the Effective Tube Hole Diameter [d^*]:
 $= \max(dt - 2 \cdot tt(Et/E) (StT/S) (\rho), dt - 2 \cdot tt)$
 $= \max(25.4 - 2 \cdot 2.77 (.17745E+09 / .17745E+09) \cdot (100/74) \cdot (0.0), 25.4 - 2 \cdot 2.77)$
 $= 25.4000 \text{ mm}$

Compute the Effective Tube Pitch [p^*]:
 $= p / \sqrt{ 1 - 4 \cdot \min(Al, 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$
 $= 31.75 / \sqrt{ 1 - 4 \cdot \min(700.09, 4 \cdot 1044.254 \cdot 31.75) / (\pi \cdot 1044.254^2) }$
 $= 33.1331 \text{ mm}$

Compute the Effective Ligament Efficiency for Bending [μ^*]:
 $= (p^* - d^*) / p^* = (33.1331 - 25.4) / 33.1331 = 0.23339$

E^*/E and v^* for Triangular pattern from Fig. UHX-11.3.
 $h/p = 8.157480$; $\mu^* = 0.233394$; $E^* = 34745236 \text{ KPa}$.
 $E^*/E = 0.195798$; $v^* = 0.414711$

Note: As h/p (8.157) is > 2 , data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

UHX-12.5.5 Step 5:

Diameter ratio [K]:



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$$= A/Do = 1380.0/1044.254 = 1.3215$$

Coefficient [F]:

$$= (1 - v^*)/E^* \cdot (E \cdot \ln(K)) \\
= (1 - 0.41)/34745236 \cdot (0.17745E+09 \cdot \ln(1.32)) \\
= 0.8333$$

UHX-12.5.6 Step 6:

Moment Acting on Unperforated Tubesheet Rim [M*]:

$$= Mts + W^* \cdot (Gc - Gs)/(2 \cdot \pi \cdot Do) \\
= 42977024 + 699550 \cdot (1105.151 - 1105.151)/(2 \cdot \pi \cdot 1044.254) \\
= 42977024.0000 \text{ KPa} \cdot \text{mm}^2$$

Note: W is the maximum of the bolt loads between the shell and channel sides.*

UHX-12.5.7 Step 7:

Maximum Bending Moment acting on Periphery of Tubesheet [Mp]:

$$= ((M^*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F) \\
= ((42977024) - 1044.254^2/32 \cdot 0.833 (6100.0 - 1000.0)) / (1 + 0.83) \\
= -55555704.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum Bending Moment acting on Center of Tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + v^*)(Ps - Pt) \\
= -0.6E+08 + 1044.254^2/64 \cdot (3 + 0.415)(6100.0 - 1000.0) \\
= 241170928.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum Bending Moment acting on Tubesheet [M]:

$$= \max(\text{abs}(Mp), \text{abs}(Mo)) \\
= \max(\text{abs}(-0.6E+08), \text{abs}(2.4117E+09)) \\
= 241170928.0000 \text{ KPa} \cdot \text{mm}^2$$

UHX-12.5.8 Step 8:

Tubesheet Bending Stress at Original Thickness:

$$= 6 \cdot M / (\mu^* (h - hg)^2) \\
= 6 \cdot 0.24117E+09 / (0.2334 (259.0 - 5.0)^2) \\
= 96.1046 \text{ N./mm}^2$$

The Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S = 2 \cdot 74.15 = 148.3 \text{ N./mm}^2$$

Tubesheet Bending Stress at Final Thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - hg)^2) \\
= 6 \cdot 0.24117E+09 / (0.2334 (209.496 - 5.0)^2) \\
= 148.2660 \text{ N./mm}^2$$

Required Tubesheet Thickness, for Bending Stress [HreqB]:

$$= h + Cats + Catc = 209.496 + 13.0 + 3.0 = 225.496 \text{ mm}$$

Required Tubesheet Thickness for Given Loadings (includes CA) [Hreq]:

$$= \max(HreqB, HreqS) = \max(225.496, 104.224) = 225.496 \text{ mm}$$

UHX-12.5.9 Step 9:

$$\text{abs}(Ps - Pt) = \text{abs}(6100.0 - 1000.0) = 5100.0 \text{ KPa.}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h/Do \cdot \min(0.8 \cdot S, 0.533 \cdot Sy)$$



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$$= 4 \cdot 0.2 \cdot 259.0/1044.25 \cdot \min(0.8 \cdot 74.15, 0.533 \cdot 111.001)$$

$$= 11.739 \text{ N./mm}^2$$

[Ps - Pt] is ≤ the Shear Stress limit (τlimit). Therefore, the required thickness for shear is for information only.

Minimum Required Thickness for Shear [HreqS]:

$$= 1/(4 \cdot \mu \cdot \min(0.8 \cdot S, 0.533 \cdot S_y)) \cdot (4 \cdot A_p/C_p) \cdot \text{abs}(P_s - P_t) + C_{atc} + C_{atc}$$

$$= 1/(4 \cdot 0.2 \cdot 59.163) \cdot (4 \cdot 6160.6/3009.9) \cdot \text{abs}((6100.0 - 1000.0)) + 16.0$$

$$= 104.2240 \text{ mm}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.77) 2.77$$

$$= 1.969 \text{ cm}^2$$

Tube Weld Size Results per UW-20:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S_a$$

$$= \pi \cdot 2.77 (25.4 - 2.77) 100.351$$

$$= 2015.143 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.4 (25.4 + 0.67 \cdot 2.4) 74.15$$

$$= 1308.7521 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S_a / S_w \text{ where } S_w = \min(S_a \text{ and } S_t)$$

$$= 100.351/74.15$$

$$= 1.353$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.4 (25.4 + 0.67 \cdot 2.4) 74.15$$

$$= 846.8395 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)}) - 0.75 \cdot d_o$$

$$= 2(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1.353 \cdot 1.0)}) - 0.75 \cdot 25.4$$

$$= 4.499 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 2.2497 mm
 Minimum Required Groove Weld Leg [agr]: 2.2497 mm

Tube-Tubesheet Jt allowable, 2015.14 is >= tube strength 2015.14 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
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Tubesheet bend. stress	96.1	<=	148.3 N./mm ²	Ok
Tubesheet shear stress	20.2	<=	59.2 N./mm ²	Ok

Thickness results for loadcase D3 corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	225.496	275.000	Ok
Tube-Tubesheet Fillet Weld Leg :	2.250	2.400	Ok
Tube-Tubesheet Groove Weld Leg :	2.250	2.400	Ok

U-Tube Tubesheet results per ASME UHX-12 2021

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc	98.545	106.495	17 148 4 59	Ps+Pt D1	Ok
D2uc	231.668	...	104 148 23 59	Ps+Pt D2	Ok
D3uc	212.546	...	87 148 19 59	Ps+Pt D3	Ok
D1c	111.726	106.495	19 148 4 59	Ps+Pt-c D1	Ok
D2c	244.635	...	115 148 24 59	Ps+Pt-c D2	Ok
D3c	225.496	...	96 148 20 59	Ps+Pt-c D3	Ok
Max:	244.6349	106.495 mm	0.775	0.407 (Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	244.635	275.000	Ok
Tubesheet Thickness Flanged Extension :	106.495	259.000	Ok
Tube Thickness :	2.295	2.770	Ok
Tube-Tubesheet Fillet Weld Leg :	2.250	2.400	Ok
Tube-Tubesheet Groove Weld Leg :	2.250	2.400	Ok



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Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	7869.97	1.000	7869.91	1.000
Tubesheet Shear Stress	14972.11	1.000	14972.11	1.000
Tube Pressure Stress	23978.18	1.000	8270.39	1.000
Tubesheet Extension Stress	8184.08	...	No Calc	No Calc
Minimum MAWP	7869.97		7869.91	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	13486.68	1.000	13486.68	1.000
Tubesheet Shear Stress	24347.54	1.000	24347.54	1.000
Tube Pressure Stress	27478.06	1.000	14312.24	1.000
Tubesheet Extension Stress	8184.08	...	No Calc	No Calc
Minimum MAPnc	8184.08		13486.68	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

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